

Department of Computer Science

COSC 336 — Introduction to Software Engineering

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Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

Phase 4 & 5 :

Group Number: 2

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1. Context Diagram

1.1 External Entities and Data Exchanges

The system has nine human roles and one external service. “Input” means data sent to the system; “output” means data sent from the system. Every human role also signs in through Process 1 (sign-in credentials in; access decision out). These two flows apply to all nine roles and are not drawn, to keep the diagrams readable.

Table 1.1:

Entitles and data exchanges

Entity	Inputs to the system	Outputs from the system
Department Representative	Surplus publication request, Resource request, Receipt confirmation, Receipt discrepancy report	Publication result, Request status, Transfer progress, Notification
Requester	Search criteria, Resource request, Request change or cancellation, Match decision, Assistant question	Search results, Match recommendations, Request status, Assistant answer, Notification
Asset Custodian	Asset record, Classification request, Classification decision, Handover arrangements, Release record, Inspection finding, Availability confirmation, Life-cycle outcome, Suggestion review decision	Classification suggestion, Validation message, Asset identifier and status, Transfer assignment, Sustainable-action suggestion, Notification
University Approver	Approval decision, Reservation instruction, Repair authorization, Life-cycle approval, Report approval	Review package, Report draft for review, Audit history, Notification
Procurement Officer	Procurement note, Purchase estimate, Report request	Operational report
Finance Officer	Cost and value record, Report request	Financial report
Maintenance Staff	Defect report, Repair progress record	Repair assignment, Asset history view, Notification
Sustainability Officer	Calculation input and assumption, Report request	Sustainability report, Sustainable-action suggestion
System Administrator	Account and role data, Department record	Account confirmation, Audit and technical log

External AI Service	Classification result, Match result, Generated text result	Classification input, Matching input, Text-generation input

Note: for the External AI Service the roles are reversed. The system sends it only permitted fields (Section 6.4 of this document) and receives results back. Value, cost and personal contact data are never sent. Role mapping: the Phase 3 "University Approver" is the business approval role; Procurement and Finance responsibilities are separate actors here (D1).

1.2 Context Diagram

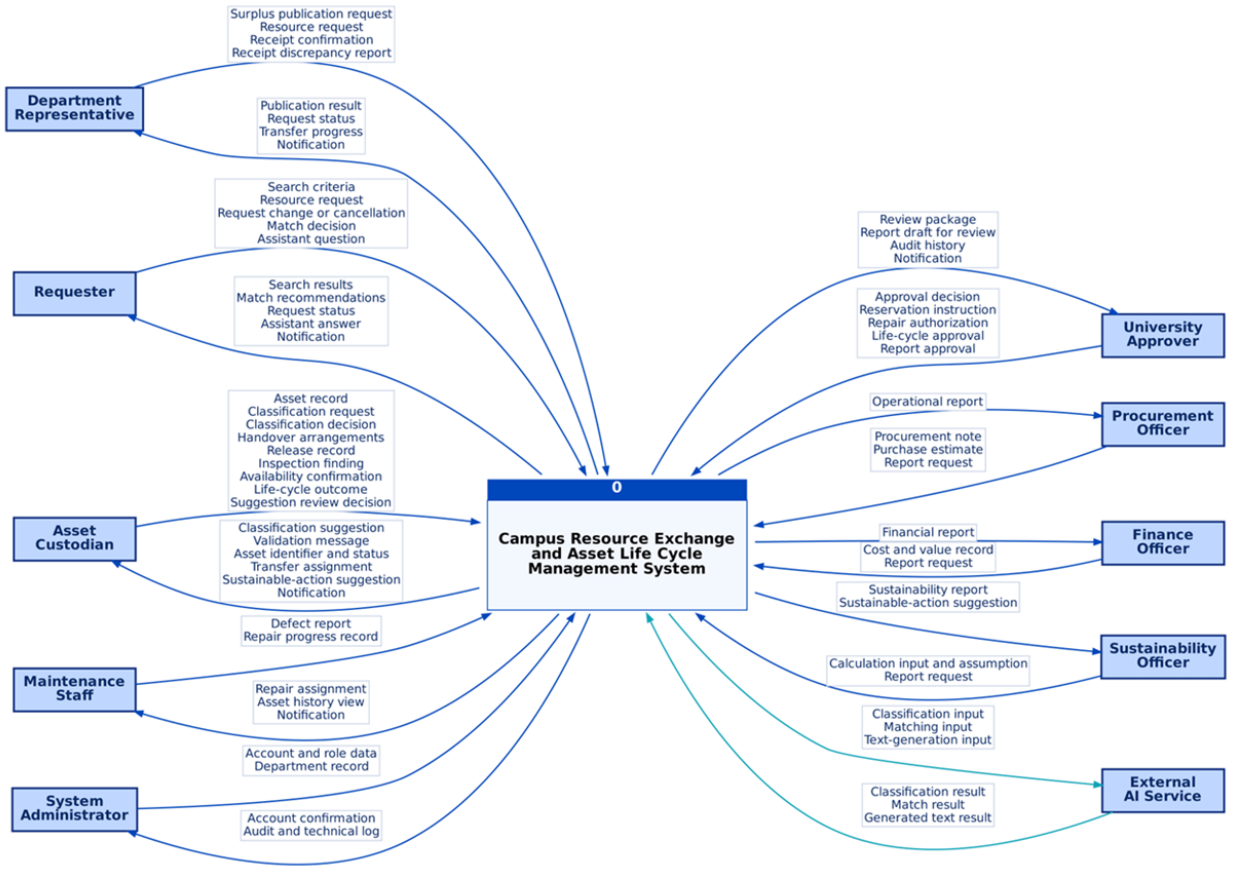


Figure 1.1: Context diagram (Process 0). Blue arrows are data flows, teal arrows go to or from the External AI Service; each arrow lists its flow names. Sign-in credentials and access decision (all nine roles) are not drawn.

2. Data-Flow Diagrams

2.1 Level 1 DFD

Seven processes (1.0 to 7.0) and nine data stores (D1 to D9). To keep every flow name readable on A4, the Level 1 diagram is drawn on three sheets (Figures 2.1a to 2.1c). Each sheet shows the processes of that sheet with every entity and store they use; an entity or store that appears on more than one sheet is the same item. Process 7.0 delivers notifications and audit history, so no store sends data directly to an entity. Permission checks against D1 are a convention and are not drawn. Table 2.1 lists every flow in text.

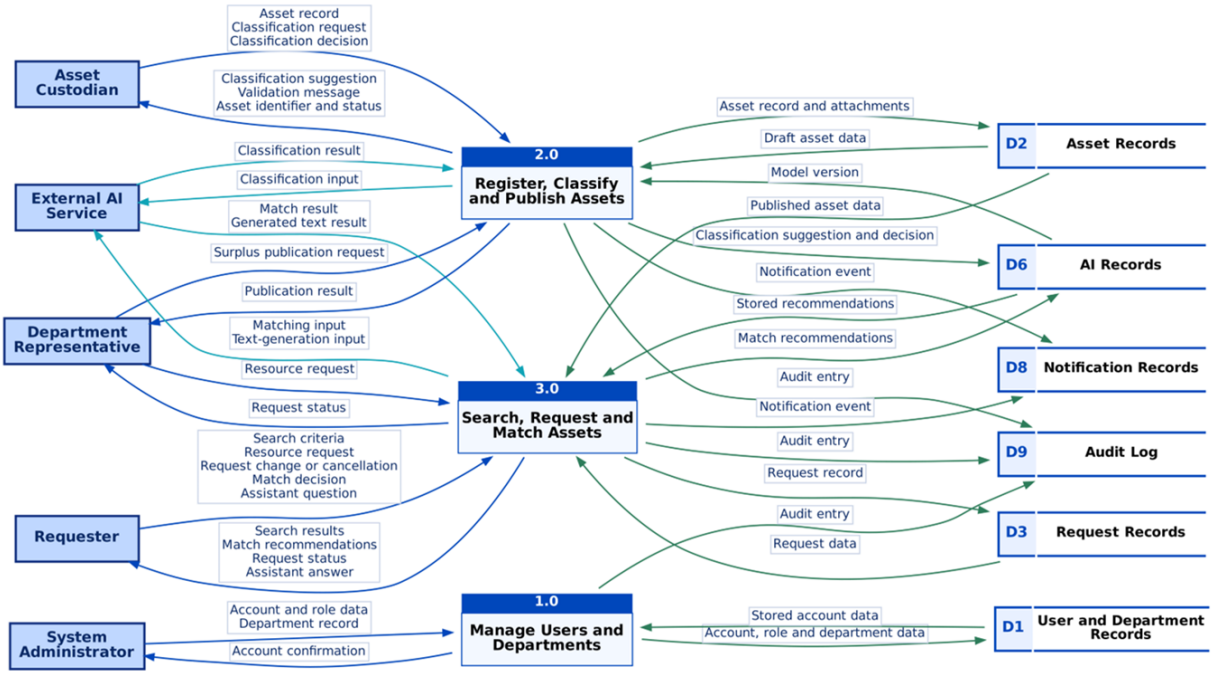


Figure 2.1a: Level 1 DFD, sheet 1 of 3 (processes 1.0, 2.0 and 3.0).

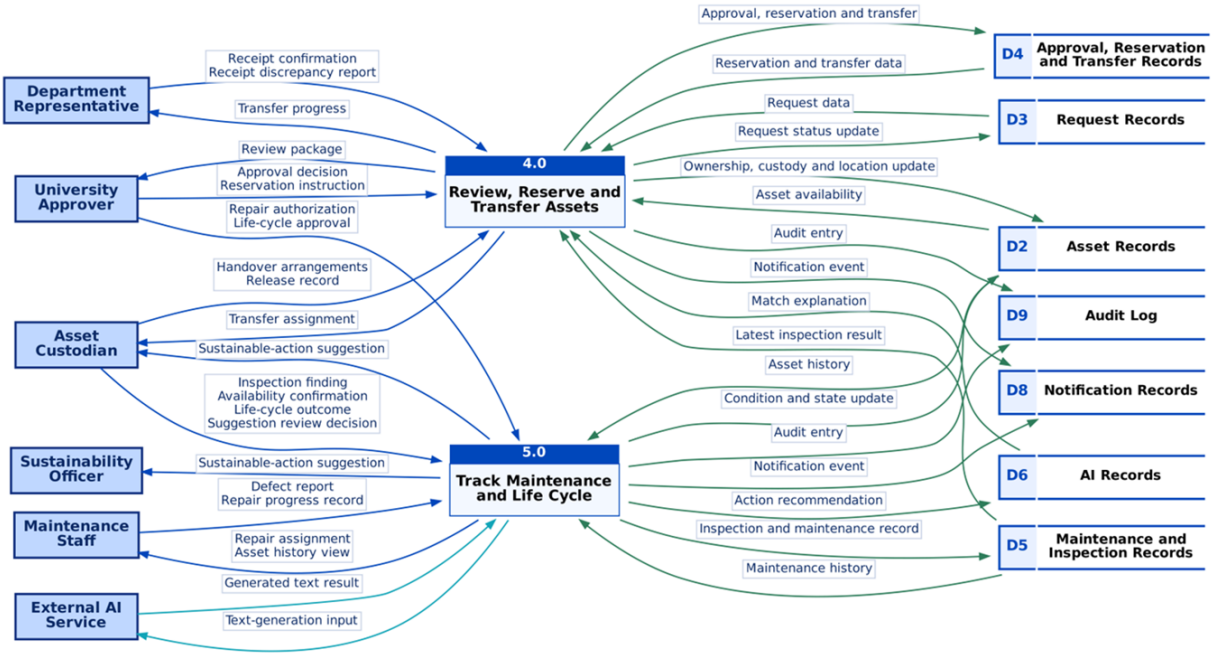


Figure 2.1b: Level 1 DFD, sheet 2 of 3 (processes 4.0 and 5.0).

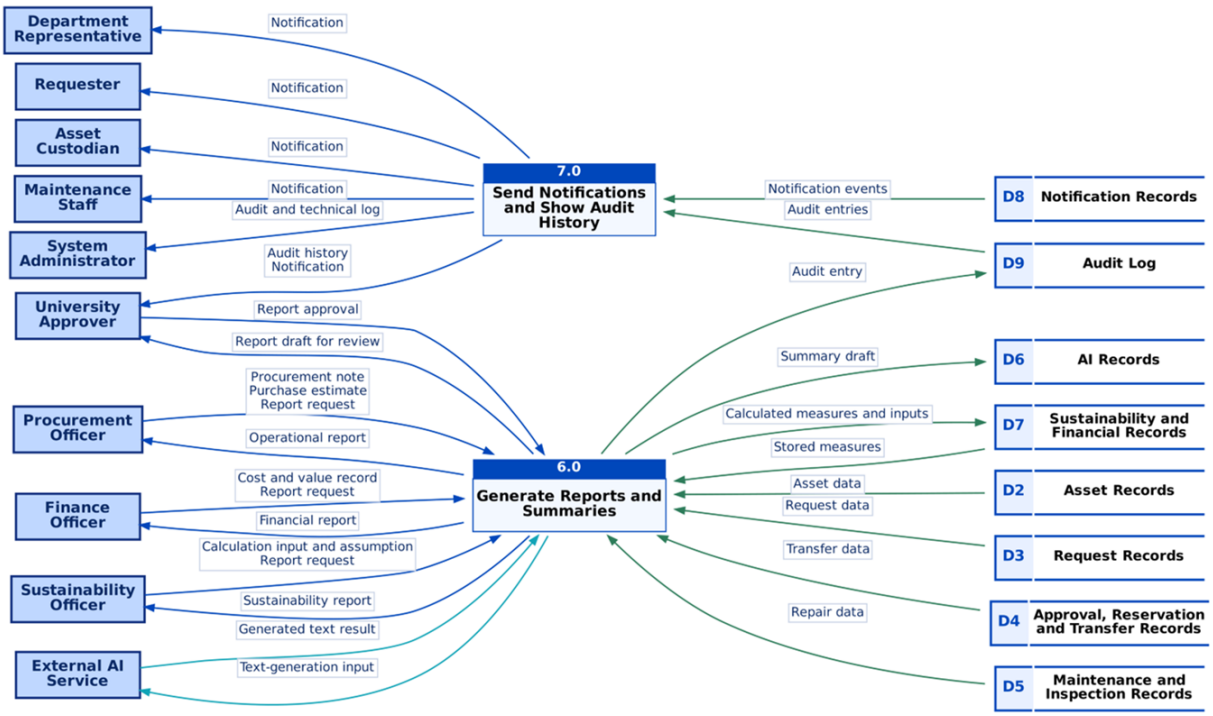


Figure 2.1c: Level 1 DFD, sheet 3 of 3 (processes 6.0 and 7.0). Blue = entity flows, teal = AI service flows, green = process–store flows.

Table 2.1: Level 1 processes and store flows

Process	Name	Phase 3 requirements
1.0	Manage Users and Departments	FR-01–FR-05
2.0	Register, Classify and Publish Assets	FR-06–FR-11, FR-39, FR-40, FR-47, FR-48, FR-52
3.0	Search, Request and Match Assets	FR-12–FR-17, FR-41–FR-48, FR-50, FR-52
4.0	Review, Reserve and Transfer Assets	FR-18–FR-26
5.0	Track Maintenance and Life Cycle	FR-27–FR-32, FR-49
6.0	Generate Reports and Summaries	FR-35–FR-38, FR-51, FR-52
7.0	Send Notifications and Show Audit History	FR-33, FR-34

From	To	Data flow
1.0 Manage Users and Departments	D1 User and Department Records	Account, role and department data
D1 User and Department Records	1.0 Manage Users and Departments	Stored account data
1.0 Manage Users and Departments	D9 Audit Log	Audit entry
2.0 Register, Classify and Publish Assets	D2 Asset Records	Asset record and attachments
D2 Asset Records	2.0 Register, Classify and Publish Assets	Draft asset data
2.0 Register, Classify and Publish Assets	D6 AI Records	Classification suggestion and decision
D6 AI Records	2.0 Register, Classify and Publish Assets	Model version
2.0 Register, Classify and Publish Assets	D9 Audit Log	Audit entry
2.0 Register, Classify and Publish Assets	D8 Notification Records	Notification event
D2 Asset Records	3.0 Search, Request and Match Assets	Published asset data
3.0 Search, Request and Match Assets	D3 Request Records	Request record

D3 Request Records	3.0 Search, Request and Match Assets	Request data
3.0 Search, Request and Match Assets	D6 AI Records	Match recommendations
D6 AI Records	3.0 Search, Request and Match Assets	Stored recommendations
3.0 Search, Request and Match Assets	D8 Notification Records	Notification event
3.0 Search, Request and Match Assets	D9 Audit Log	Audit entry
D3 Request Records	4.0 Review, Reserve and Transfer Assets	Request data
D2 Asset Records	4.0 Review, Reserve and Transfer Assets	Asset availability
D6 AI Records	4.0 Review, Reserve and Transfer Assets	Match explanation
D5 Maintenance and Inspection Records	4.0 Review, Reserve and Transfer Assets	Latest inspection result
4.0 Review, Reserve and Transfer Assets	D3 Request Records	Request status update
4.0 Review, Reserve and Transfer Assets	D4 Approval, Reservation and Transfer Records	Approval, reservation and transfer
D4 Approval, Reservation and Transfer Records	4.0 Review, Reserve and Transfer Assets	Reservation and transfer data
4.0 Review, Reserve and Transfer Assets	D2 Asset Records	Ownership, custody and location update
4.0 Review, Reserve and Transfer Assets	D8 Notification Records	Notification event
4.0 Review, Reserve and Transfer Assets	D9 Audit Log	Audit entry
5.0 Track Maintenance and Life Cycle	D5 Maintenance and Inspection Records	Inspection and maintenance record
D5 Maintenance and Inspection Records	5.0 Track Maintenance and Life Cycle	Maintenance history
5.0 Track Maintenance and Life Cycle	D2 Asset Records	Condition and state update
D2 Asset Records	5.0 Track Maintenance and Life Cycle	Asset history

5.0 Track Maintenance and Life Cycle	D6 AI Records	Action recommendation
5.0 Track Maintenance and Life Cycle	D8 Notification Records	Notification event
5.0 Track Maintenance and Life Cycle	D9 Audit Log	Audit entry
D2 Asset Records	6.0 Generate Reports and Summaries	Asset data
D3 Request Records	6.0 Generate Reports and Summaries	Request data
D4 Approval, Reservation and Transfer Records	6.0 Generate Reports and Summaries	Transfer data
D5 Maintenance and Inspection Records	6.0 Generate Reports and Summaries	Repair data
D7 Sustainability and Financial Records	6.0 Generate Reports and Summaries	Stored measures
6.0 Generate Reports and Summaries	D7 Sustainability and Financial Records	Calculated measures and inputs
6.0 Generate Reports and Summaries	D6 AI Records	Summary draft
6.0 Generate Reports and Summaries	D9 Audit Log	Audit entry
D8 Notification Records	7.0 Send Notifications and Show Audit History	Notification events
D9 Audit Log	7.0 Send Notifications and Show Audit History	Audit entries

Checks made: every process has at least one input and one output; no data moves store-to-store or between a store and an external entity without a process; “Audit entry” and “Notification event” flows are written by the process that performs the action.

2.2 Level 2 DFDs

Every Level 1 process is expanded into a Level 2 diagram. Process 2.0 was expanded first because it is the riskiest AI process; processes 1.0 and 3.0 to 7.0 use the same style and rules. In each diagram the dashed boundary is the parent process, blue arrows are entity flows, teal arrows go to or from the External AI Service and green arrows go to or from data stores. Only the parent's Level 1 flows cross the boundary, with the same names and directions (Tables 2.3c to 2.3h). Each figure is followed by a table that maps every sub-process to the Phase 3 requirements listed for its parent in Table 2.1, so all of them are covered.

2.2.1 Process 1.0 — Manage Users and Departments

Process 1.0 checks every sign-in and lets the System Administrator maintain accounts, roles and departments. Account and department data are validated separately and saved by one sub-process, so every change to D1 produces one audit entry. Sign-in credentials and Access decision are the two flows shared by all nine roles (C69, C70); they are drawn here once, against an "All Users" entity, because this is the process that handles them.

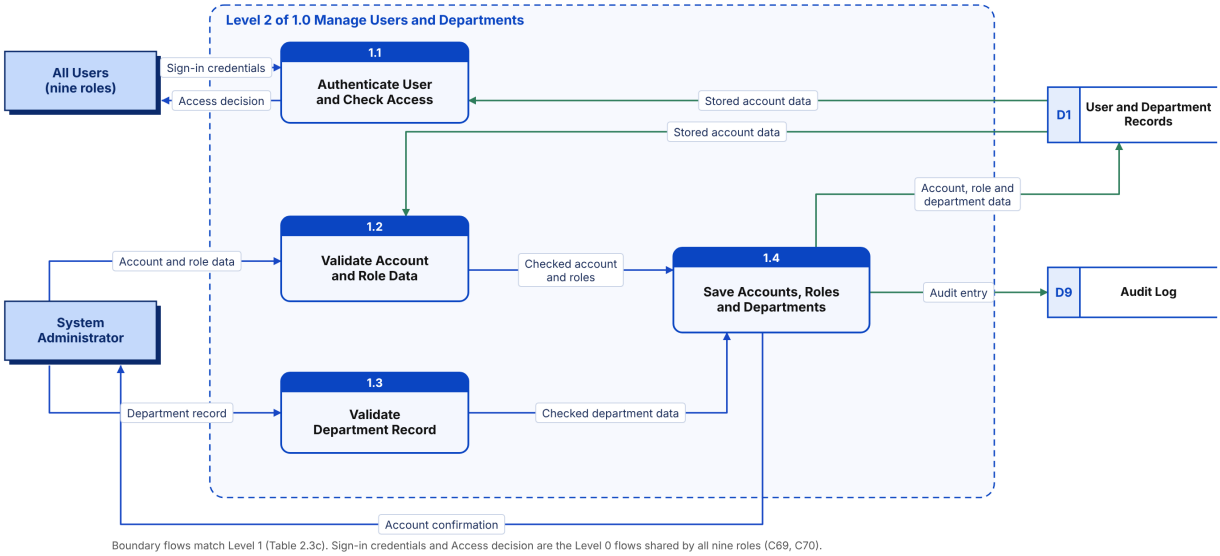


Figure 2.2a: Level 2 DFD of Process 1.0 — Manage Users and Departments.

Table 2.2a: Sub-processes of 1.0 and the requirements they cover

Sub-process	Name	What it does	Phase 3 requirements
1.1	Authenticate User and Check Access	Checks sign-in credentials against stored account data and returns the access decision for the user's role.	FR-01

Sub-process	Name	What it does	Phase 3 requirements
1.2	Validate Account and Role Data	Checks new or changed accounts and role assignments (unique e-mail, valid role, active status).	FR-02, FR-03
1.3	Validate Department Record	Checks new or changed department records (unique code, active flag).	FR-04
1.4	Save Accounts, Roles and Departments	Writes checked data to D1, records the audit entry and confirms the change to the administrator.	FR-02–FR-05

2.2.2 Process 2.0 — Register, Classify and Publish Assets

Why Process 2 was expanded: it is the most complex AI-related process in the system. It validates data, sends only permitted fields to the AI service, handles failure and low confidence, records the human decision with the model version, and publishes surplus assets. It is also the process shown in sequence diagram 8.3.1. The Level 2 diagram keeps all boundary flows of Process 2 (Table 2.3b).

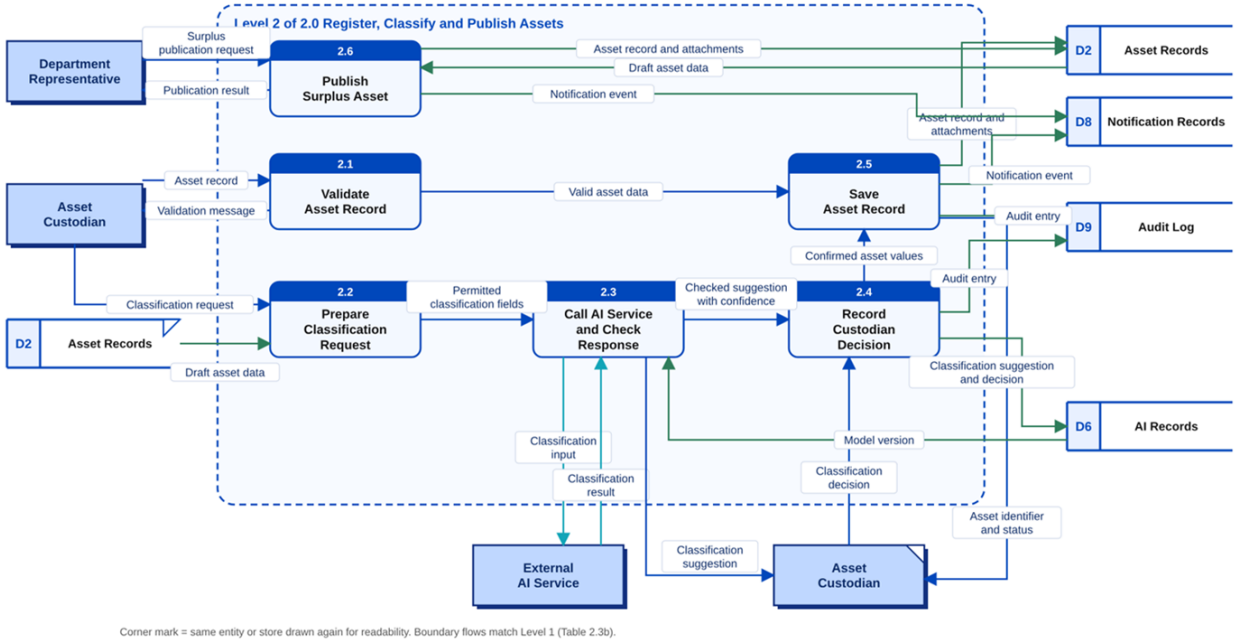


Figure 2.2b: Level 2 DFD of Process 2.0 — Register, Classify and Publish Assets. The dashed boundary is crossed only by the Level 1 flows of Process 2.0 (Table 2.3b).

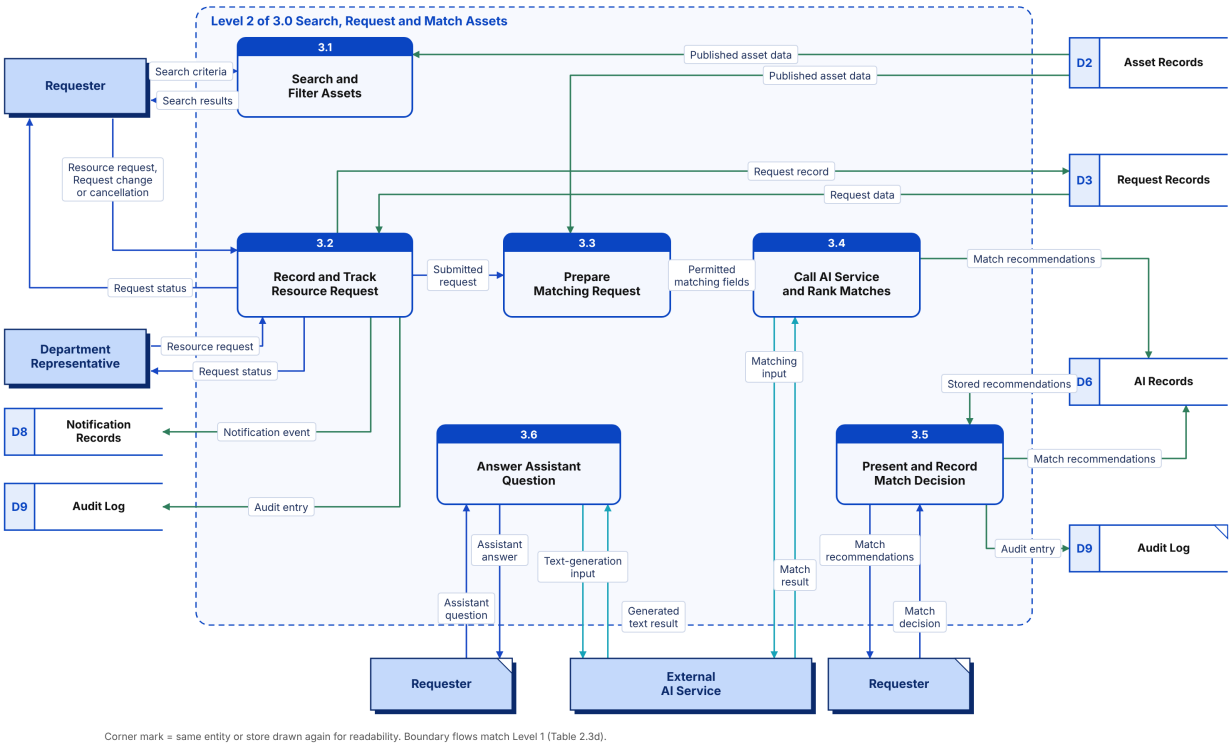
When the AI service times out, is unavailable or returns an invalid result, sub-process 2.3 returns a “Classification suggestion” that contains an “AI unavailable” status, and the custodian enters the category manually. When confidence is low, the suggestion is shown with a warning and nothing is pre-selected.

Table 2.2b: Sub-processes of 2.0 and the requirements they cover

Sub-process	Name	What it does	Phase 3 requirements
2.1	Validate Asset Record	Checks required fields, quantity, dates and attachment types; returns a validation message.	FR-06, FR-07, FR-08
2.2	Prepare Classification Request	Reads the draft and keeps only permitted fields (name, description, specifications, photos).	FR-39, FR-40
2.3	Call AI Service and Check Response	Calls the AI service, checks the result and confidence, and handles timeout or invalid output.	FR-39, FR-47
2.4	Record Custodian Decision	Stores the suggestion with the accept, edit or reject decision and the model version.	FR-48, FR-52
2.5	Save Asset Record	Saves the confirmed asset, issues its identifier and status, and writes audit and notification entries.	FR-09, FR-11
2.6	Publish Surplus Asset	Publishes a registered asset as surplus when the Department Representative requests it.	FR-10

2.2.3 Process 3.0 — Search, Request and Match Assets

Process 3.0 has two AI calls: ranking matches for a submitted request (3.3 and 3.4) and answering the natural-language assistant (3.6). Matches are stored first and then shown to the Requester by 3.5, which also records the Requester's decision, so a recommendation is never acted on without a person. If the AI service is unavailable, 3.4 stores no recommendation and the request continues normally (keyword search in 3.1 stays available).



Corner mark = same entity or store drawn again for readability. Boundary flows match Level 1 (Table 2.3d).

Figure 2.2c: Level 2 DFD of Process 3.0 — Search, Request and Match Assets.

Table 2.2c: Sub-processes of 3.0 and the requirements they cover

Sub-process	Name	What it does	Phase 3 requirements
3.1	Search and Filter Assets	Searches published assets by keyword, category, condition and location, and returns the results.	FR-12, FR-13, FR-14
3.2	Record and Track Resource Request	Records new, changed or cancelled requests from the Requester or Department Representative and returns their status.	FR-15, FR-16, FR-17
3.3	Prepare Matching Request	Combines the submitted request with published asset data and keeps only permitted matching fields.	FR-41, FR-42
3.4	Call AI Service and Rank Matches	Sends the matching input, checks the result, ranks assets by score and confidence and stores the recommendations.	FR-43, FR-44, FR-45, FR-46, FR-47
3.5	Present and Record Match Decision	Shows the stored recommendations with reasons and records the Requester's decision with the model version.	FR-48, FR-52

Sub-process	Name	What it does	Phase 3 requirements
3.6	Answer Assistant Question	Sends the question to the AI service and returns the generated answer, with a fallback message when the service fails.	FR-50

2.2.4 Process 4.0 — Review, Reserve and Transfer Assets

Process 4.0 follows the approval chain: the University Approver receives one review package (4.1), decides (4.2) and instructs the reservation (4.3). Release by the sending Asset Custodian (4.4) and receipt by the receiving Department Representative (4.5) stay separate, and ownership, custody and location change in D2 only after receipt is confirmed.

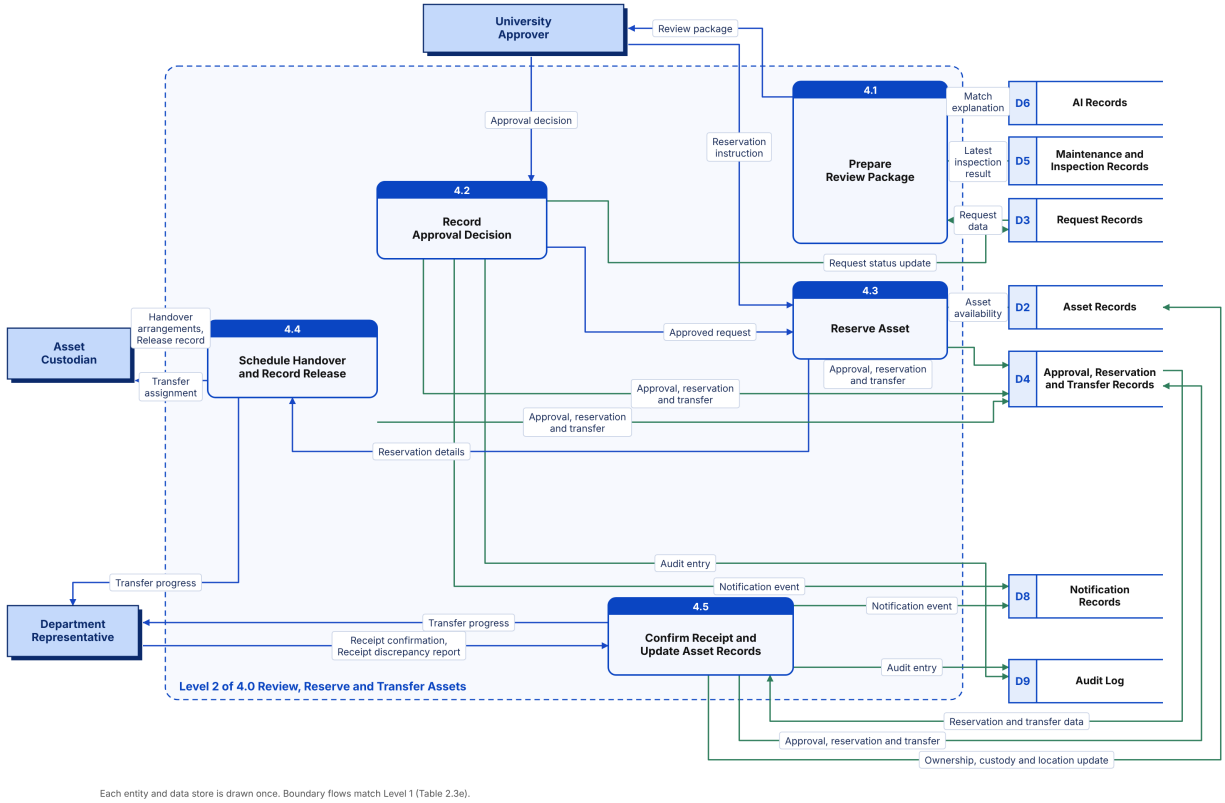


Figure 2.2d: Level 2 DFD of Process 4.0 — Review, Reserve and Transfer Assets.

Table 2.2d:

Sub-processes of 4.0 and the requirements they cover

Sub-process	**Name**	**What it does**	**Subprocess: Phase 3 requirements**
4.1	Prepare Review Package	Combines the request, the match explanation and the latest inspection result for the Approver.	FR-20
4.2	Record Approval Decision	Records approve, reject or return-for-information with the reason, and updates the request status.	FR-20, FR-21
4.3	Reserve Asset	Checks availability again and reserves the approved quantity.	FR-18,FR-19
4.4	Schedule Handover and Record Release	Assigns the handover to the custodian, records the release and reports transfer progress.	FR-22, FR-23
4.5	Confirm Receipt and Update Asset Records	Records receipt or a discrepancy, then updates ownership, custody and location in D2.	FR-24, FR-25, FR-26

2.2.5 Process 5.0 — Track Maintenance and Life Cycle

Process 5.0 covers the repair loop (5.1 to 5.3) and the life-cycle decision (5.4 and 5.5). The AI service only drafts a sustainable-action suggestion; the Asset Custodian reviews it and the University Approver approves the life-cycle outcome before the asset state changes in D2.

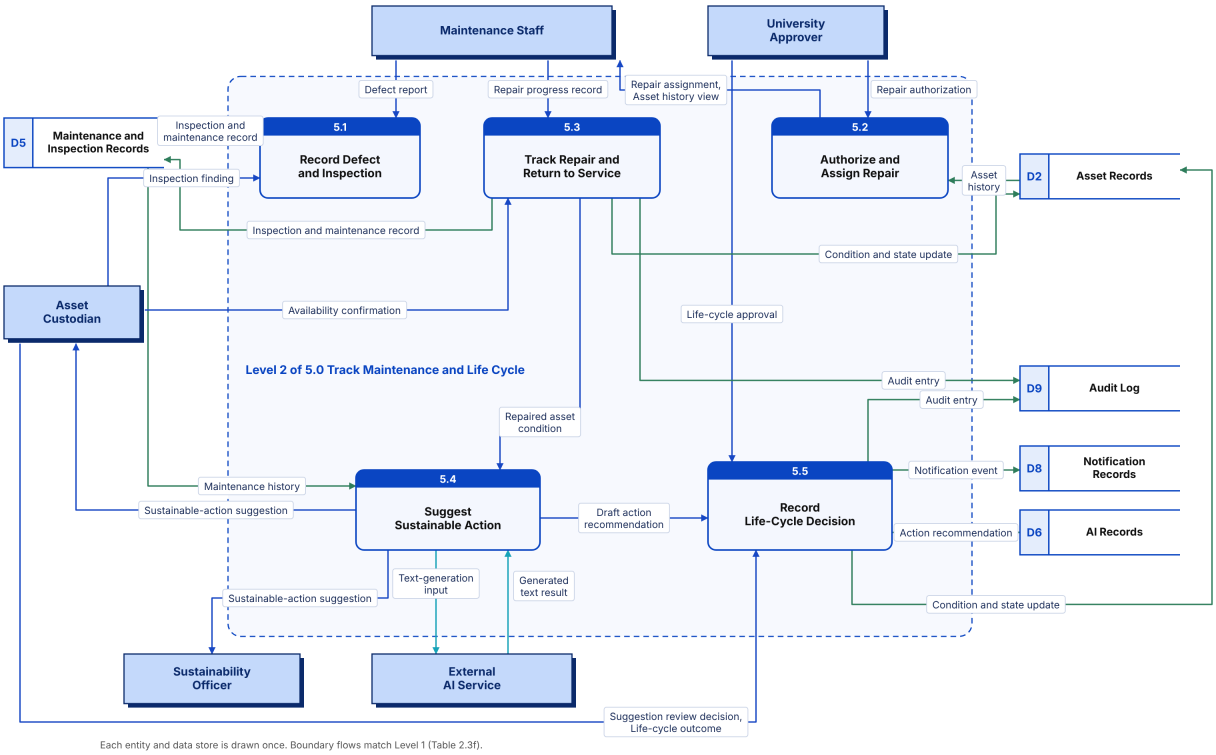


Figure 2.2e: Level 2 DFD of Process 5.0 — Track Maintenance and Life Cycle.

Table 2.2e: Sub-processes of 5.0 and the requirements they cover

Sub-process	Name	What it does	Phase 3 requirements
5.1	Record Defect and Inspection	Records defect reports from Maintenance Staff and inspection findings from the Asset Custodian.	FR-27, FR-28
5.2	Authorize and Assign Repair	Uses the asset history, receives the repair authorization and assigns the repair to Maintenance Staff.	FR-29
5.3	Track Repair and Return to Service	Records repair progress and cost, and returns the asset to service after availability is confirmed.	FR-30, FR-31
5.4	Suggest Sustainable Action	Uses the maintenance history and asks the AI service to draft a reuse, refurbish, donate, recycle or retire suggestion.	FR-49
5.5	Record Life-Cycle Decision	Records the reviewed suggestion, the outcome and the approval, then updates the asset state.	FR-32

2.2.6 Process 6.0 — Generate Reports and Summaries

Process 6.0 calculates every reported figure itself (6.2) before the AI service is asked for a summary (6.3), so the AI text can only describe calculated values. Reports are issued to the three officers only after the University Approver approves them (6.4).

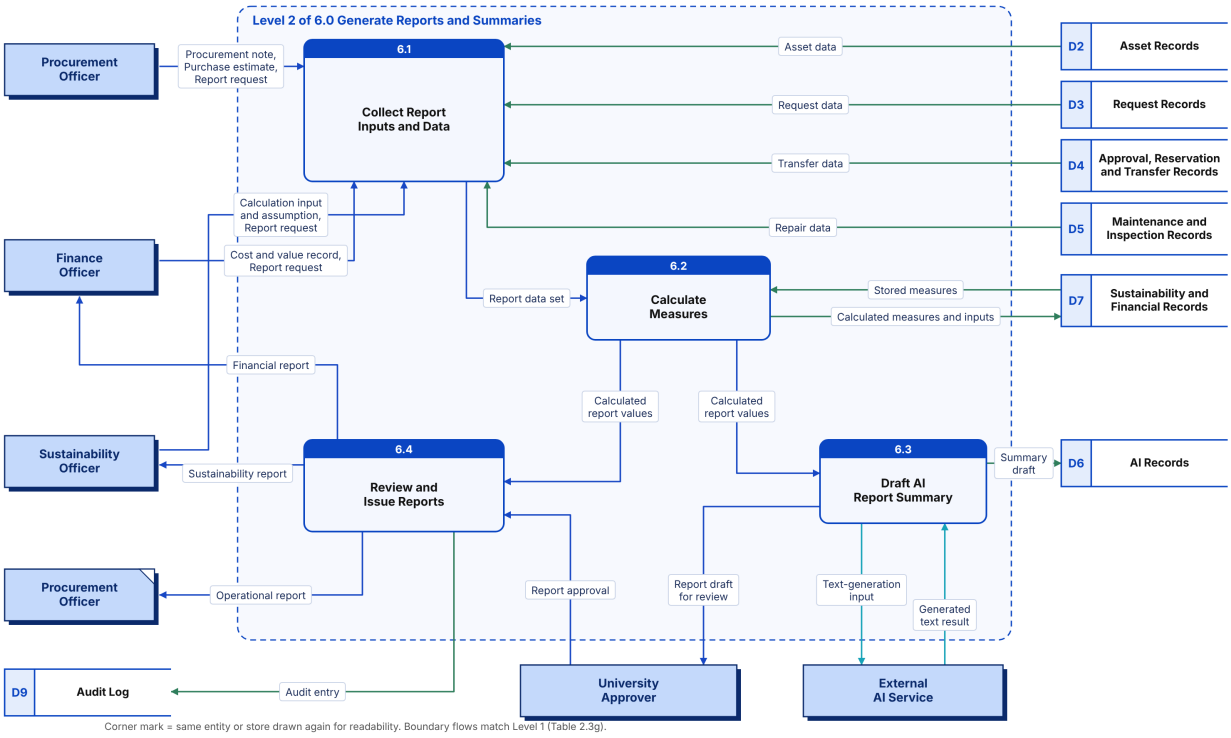


Figure 2.2f: Level 2 DFD of Process 6.0 — Generate Reports and Summaries.

Table 2.2f: Sub-processes of 6.0 and the requirements they cover

Sub-process	Name	What it does	Phase 3 requirements
6.1	Collect Report Inputs and Data	Receives report requests and officer inputs and reads asset, request, transfer and repair data.	FR-35
6.2	Calculate Measures	Calculates operational, financial and sustainability measures; marks missing inputs as incomplete.	FR-36, FR-37
6.3	Draft AI Report Summary	Asks the AI service for a summary of the calculated values and sends the draft for review.	FR-51

Sub-process	Name	What it does	Phase 3 requirements
6.4	Review and Issue Reports	Records the approval decision and issues the operational, financial and sustainability reports.	FR-38, FR-52

2.2.7 Process 7.0 — Send Notifications and Show Audit History

Process 7.0 is the only process that sends data from D8 and D9 to people. Notification events are addressed by role before delivery, and audit entries are filtered by the viewer's permission, so the University Approver sees business audit history and the System Administrator sees the audit and technical log.

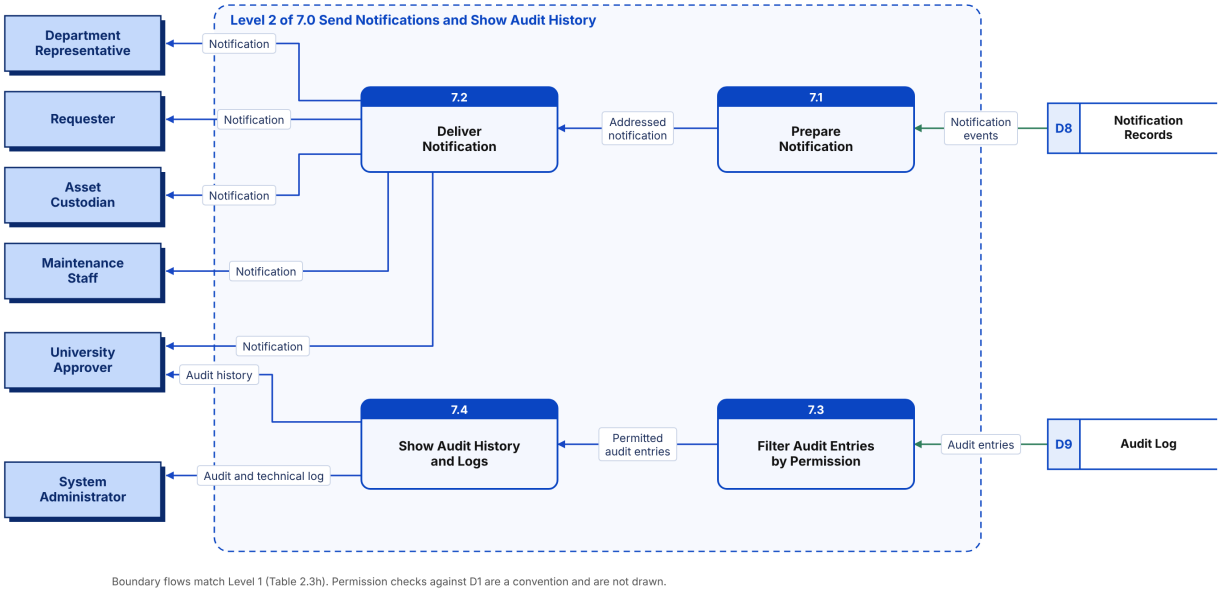


Figure 2.2g: Level 2 DFD of Process 7.0 — Send Notifications and Show Audit History.

Table 2.2g: Sub-processes of 7.0 and the requirements they cover

Sub-process	Name	What it does	Phase 3 requirements
7.1	Prepare Notification	Reads notification events and works out the recipients and message for each role.	FR-33
7.2	Deliver Notification	Delivers the in-app notification to each recipient.	FR-33
7.3	Filter Audit Entries by Permission	Reads audit entries and keeps only those the viewer's role may see.	FR-34

Sub-process	Name	What it does	Phase 3 requirements
7.4	Show Audit History and Logs	Shows audit history to the Approver and the audit and technical log to the administrator.	FR-34

2.3 Balancing Table

Table 2.3a maps every context flow to Level 1. “Sign-in credentials” and “Access decision” are common to all nine roles and map to Process 1.

ID	Context flow	Entity	Direction	Level 1 process	Preserved ?
C01	Surplus publication request	Department Representative	Entity → 0	2.0	Yes
C02	Resource request	Department Representative	Entity → 0	3.0	Yes
C03	Receipt confirmation	Department Representative	Entity → 0	4.0	Yes
C04	Receipt discrepancy report	Department Representative	Entity → 0	4.0	Yes
C05	Publication result	Department Representative	0 → Entity	2.0	Yes
C06	Request status	Department Representative	0 → Entity	3.0	Yes
C07	Transfer progress	Department Representative	0 → Entity	4.0	Yes
C08	Notification	Department Representative	0 → Entity	7.0	Yes
C09	Search criteria	Requester	Entity → 0	3.0	Yes
C10	Resource request	Requester	Entity → 0	3.0	Yes

C11	Request change or cancellation	Requester	Entity → 0	3.0	Yes
C12	Match decision	Requester	Entity → 0	3.0	Yes
C13	Assistant question	Requester	Entity → 0	3.0	Yes
C14	Search results	Requester	0 → Entity	3.0	Yes
C15	Match recommendations	Requester	0 → Entity	3.0	Yes
C16	Request status	Requester	0 → Entity	3.0	Yes
C17	Assistant answer	Requester	0 → Entity	3.0	Yes
C18	Notification	Requester	0 → Entity	7.0	Yes
C19	Asset record	Asset Custodian	Entity → 0	2.0	Yes
C20	Classification request	Asset Custodian	Entity → 0	2.0	Yes
C21	Classification decision	Asset Custodian	Entity → 0	2.0	Yes
C22	Handover arrangements	Asset Custodian	Entity → 0	4.0	Yes
C23	Release record	Asset Custodian	Entity → 0	4.0	Yes
C24	Inspection finding	Asset Custodian	Entity → 0	5.0	Yes
C25	Availability confirmation	Asset Custodian	Entity → 0	5.0	Yes
C26	Life-cycle outcome	Asset Custodian	Entity → 0	5.0	Yes
C27	Suggestion review decision	Asset Custodian	Entity → 0	5.0	Yes
C28	Classification suggestion	Asset Custodian	0 → Entity	2.0	Yes
C29	Validation message	Asset Custodian	0 → Entity	2.0	Yes
C30	Asset identifier and status	Asset Custodian	0 → Entity	2.0	Yes
C31	Transfer assignment	Asset Custodian	0 → Entity	4.0	Yes
C32	Sustainable-action suggestion	Asset Custodian	0 → Entity	5.0	Yes
C33	Notification	Asset Custodian	0 → Entity	7.0	Yes
C34	Approval decision	University Approver	Entity → 0	4.0	Yes
C35	Reservation instruction	University Approver	Entity → 0	4.0	Yes

C36	Repair authorization	University Approver	Entity → 0	5.0	Yes
C37	Life-cycle approval	University Approver	Entity → 0	5.0	Yes
C38	Report approval	University Approver	Entity → 0	6.0	Yes
C39	Review package	University Approver	0 → Entity	4.0	Yes
C40	Report draft for review	University Approver	0 → Entity	6.0	Yes
C41	Audit history	University Approver	0 → Entity	7.0	Yes
C42	Notification	University Approver	0 → Entity	7.0	Yes
C43	Procurement note	Procurement Officer	Entity → 0	6.0	Yes
C44	Purchase estimate	Procurement Officer	Entity → 0	6.0	Yes
C45	Report request	Procurement Officer	Entity → 0	6.0	Yes
C46	Operational report	Procurement Officer	0 → Entity	6.0	Yes
C47	Cost and value record	Finance Officer	Entity → 0	6.0	Yes
C48	Report request	Finance Officer	Entity → 0	6.0	Yes
C49	Financial report	Finance Officer	0 → Entity	6.0	Yes
C50	Defect report	Maintenance Staff	Entity → 0	5.0	Yes
C51	Repair progress record	Maintenance Staff	Entity → 0	5.0	Yes
C52	Repair assignment	Maintenance Staff	0 → Entity	5.0	Yes
C53	Asset history view	Maintenance Staff	0 → Entity	5.0	Yes
C54	Notification	Maintenance Staff	0 → Entity	7.0	Yes
C55	Calculation input and assumption	Sustainability Officer	Entity → 0	6.0	Yes

C56	Report request	Sustainability Officer	Entity → 0	6.0	Yes
C57	Sustainability report	Sustainability Officer	0 → Entity	6.0	Yes
C58	Sustainable-action suggestion	Sustainability Officer	0 → Entity	5.0	Yes
C59	Account and role data	System Administrator	Entity → 0	1.0	Yes
C60	Department record	System Administrator	Entity → 0	1.0	Yes
C61	Account confirmation	System Administrator	0 → Entity	1.0	Yes
C62	Audit and technical log	System Administrator	0 → Entity	7.0	Yes
C63	Classification input	External AI Service	0 → Entity	2.0	Yes
C64	Matching input	External AI Service	0 → Entity	3.0	Yes
C65	Text-generation input	External AI Service	0 → Entity	3.0, 5.0, 6.0	Yes
C66	Classification result	External AI Service	Entity → 0	2.0	Yes
C67	Match result	External AI Service	Entity → 0	3.0	Yes
C68	Generated text result	External AI Service	Entity → 0	3.0, 5.0, 6.0	Yes
C69	Sign-in credentials	All nine human roles	Entity → 0	1.0	Yes (not drawn)
C70	Access decision	All nine human roles	0 → Entity	1.0	Yes (not drawn)

Table 2.3b: Level 1 → Level 2 (Process 2). The sub-process column shows where each parent boundary flow lands.

Parent flow (Level 1)	**Level 1 direction**	**Level 2 location**
Asset record	AC → 2.0	AC → 2.1
Validation message	2.0 → AC	2.1 → AC
Classification request	AC → 2.0	AC → 2.2

Classification input	2.0 → AI	2.3 → AI
Classification result	AI → 2.0	AI → 2.3
Classification suggestion	2.0 → AC	2.3 → AC
Classification decision	AC → 2.0	AC → 2.4
Asset identifier and status	2.0 → AC	2.5 → AC
Surplus publication request	DR → 2.0	DR → 2.6
Publication result	2.0 → DR	2.6 → DR
Draft asset data	D2 → 2.0	D2 → 2.2 and D2 → 2.6
Asset record and attachments	2.0 → D2	2.5 → D2 and 2.6 → D2
Classification suggestion and decision	2.0 → D6	2.4 → D6
Model version	D6 → 2.0	D6 → 2.3
Audit entry	2.0 → D9	2.4 → D9 and 2.5 → D9
Notification event	2.0 → D8	2.5 → D8 and 2.6 → D8

Internal Level 2 flows (not visible at Level 1): Permitted classification fields (2.2 → 2.3), Checked suggestion with confidence (2.3 → 2.4), Confirmed asset values (2.4 → 2.5), Valid asset data (2.1 → 2.5).

Tables 2.3c to 2.3h check the other six Level 2 diagrams in the same way. Abbreviations: AC Asset Custodian, AI External AI Service, DR Department Representative, FO Finance Officer, MS Maintenance Staff, PO Procurement Officer, RQ Requester, SA System Administrator, SO Sustainability Officer, UA University Approver, Users all nine roles. Every Level 1 flow of each process appears once at Level 2 with the same name and direction; flows that split are listed with each sub-process that sends or receives them.

Table 2.3c: Level 1 → Level 2 (Process 1).

Parent flow (Level 1)	**Level 1 direction**	**Level 2 location**
Sign-in credentials	Users → 1.0	Users → 1.1
Access decision	1.0 → Users	1.1 → Users
Account and role data	SA → 1.0	SA → 1.2
Department record	SA → 1.0	SA → 1.3
Account confirmation	1.0 → SA	1.4 → SA

Parent flow (Level 1)	Level 1 direction	Level 2 location
Stored account data	D1 → 1.0	D1 → 1.1 and D1 → 1.2
Account, role and department data	1.0 → D1	1.4 → D1
Audit entry	1.0 → D9	1.4 → D9

Internal Level 2 flows (not visible at Level 1): Checked account and roles (1.2 → 1.4), Checked department data (1.3 → 1.4).

Table 2.3d: Level 1 → Level 2 (Process 3).

Parent flow (Level 1)	Level 1 direction	Level 2 location
Search criteria	RQ → 3.0	RQ → 3.1
Search results	3.0 → RQ	3.1 → RQ
Resource request	RQ → 3.0	RQ → 3.2
Resource request	DR → 3.0	DR → 3.2
Request change or cancellation	RQ → 3.0	RQ → 3.2
Request status	3.0 → RQ	3.2 → RQ
Request status	3.0 → DR	3.2 → DR
Matching input	3.0 → AI	3.4 → AI
Match result	AI → 3.0	AI → 3.4
Match recommendations	3.0 → RQ	3.5 → RQ
Match decision	RQ → 3.0	RQ → 3.5
Assistant question	RQ → 3.0	RQ → 3.6
Assistant answer	3.0 → RQ	3.6 → RQ
Text-generation input	3.0 → AI	3.6 → AI
Generated text result	AI → 3.0	AI → 3.6
Published asset data	D2 → 3.0	D2 → 3.1 and D2 → 3.3
Request record	3.0 → D3	3.2 → D3

Parent flow (Level 1)	Level 1 direction	Level 2 location
Request data	D3 → 3.0	D3 → 3.2
Match recommendations	3.0 → D6	3.4 → D6 and 3.5 → D6
Stored recommendations	D6 → 3.0	D6 → 3.5
Notification event	3.0 → D8	3.2 → D8
Audit entry	3.0 → D9	3.2 → D9 and 3.5 → D9

Internal Level 2 flows (not visible at Level 1): Submitted request (3.2 → 3.3), Permitted matching fields (3.3 → 3.4).

Table 2.3e: Level 1 → Level 2 (Process 4).

Parent flow (Level 1)	Level 1 direction	Level 2 location
Approval decision	UA → 4.0	UA → 4.2
Reservation instruction	UA → 4.0	UA → 4.3
Review package	4.0 → UA	4.1 → UA
Handover arrangements	AC → 4.0	AC → 4.4
Release record	AC → 4.0	AC → 4.4
Transfer assignment	4.0 → AC	4.4 → AC
Receipt confirmation	DR → 4.0	DR → 4.5
Receipt discrepancy report	DR → 4.0	DR → 4.5
Transfer progress	4.0 → DR	4.4 → DR and 4.5 → DR
Request data	D3 → 4.0	D3 → 4.1
Asset availability	D2 → 4.0	D2 → 4.3
Match explanation	D6 → 4.0	D6 → 4.1
Latest inspection result	D5 → 4.0	D5 → 4.1
Request status update	4.0 → D3	4.2 → D3
Approval, reservation and transfer	4.0 → D4	4.2, 4.3, 4.4 and 4.5 → D4

Parent flow (Level 1)	Level 1 direction	Level 2 location
Reservation and transfer data	D4 → 4.0	D4 → 4.5
Ownership, custody and location update	4.0 → D2	4.5 → D2
Notification event	4.0 → D8	4.2 → D8 and 4.5 → D8
Audit entry	4.0 → D9	4.2 → D9 and 4.5 → D9

Internal Level 2 flows (not visible at Level 1): Approved request (4.2 → 4.3), Reservation details (4.3 → 4.4).

Table 2.3f: Level 1 → Level 2 (Process 5).

Parent flow (Level 1)	Level 1 direction	Level 2 location
Defect report	MS → 5.0	MS → 5.1
Repair progress record	MS → 5.0	MS → 5.3
Repair assignment	5.0 → MS	5.2 → MS
Asset history view	5.0 → MS	5.2 → MS
Inspection finding	AC → 5.0	AC → 5.1
Availability confirmation	AC → 5.0	AC → 5.3
Life-cycle outcome	AC → 5.0	AC → 5.5
Suggestion review decision	AC → 5.0	AC → 5.5
Sustainable-action suggestion	5.0 → AC	5.4 → AC
Sustainable-action suggestion	5.0 → SO	5.4 → SO
Repair authorization	UA → 5.0	UA → 5.2
Life-cycle approval	UA → 5.0	UA → 5.5
Text-generation input	5.0 → AI	5.4 → AI
Generated text result	AI → 5.0	AI → 5.4
Inspection and maintenance record	5.0 → D5	5.1 → D5 and 5.3 → D5
Maintenance history	D5 → 5.0	D5 → 5.4

Parent flow (Level 1)	Level 1 direction	Level 2 location
Condition and state update	5.0 → D2	5.3 → D2 and 5.5 → D2
Asset history	D2 → 5.0	D2 → 5.2
Action recommendation	5.0 → D6	5.5 → D6
Notification event	5.0 → D8	5.5 → D8
Audit entry	5.0 → D9	5.3 → D9 and 5.5 → D9

Internal Level 2 flows (not visible at Level 1): Repaired asset condition (5.3 → 5.4), Draft action recommendation (5.4 → 5.5).

Table 2.3g: Level 1 → Level 2 (Process 6).

Parent flow (Level 1)	Level 1 direction	Level 2 location
Procurement note	PO → 6.0	PO → 6.1
Purchase estimate	PO → 6.0	PO → 6.1
Report request	PO → 6.0	PO → 6.1
Operational report	6.0 → PO	6.4 → PO
Cost and value record	FO → 6.0	FO → 6.1
Report request	FO → 6.0	FO → 6.1
Financial report	6.0 → FO	6.4 → FO
Calculation input and assumption	SO → 6.0	SO → 6.1
Report request	SO → 6.0	SO → 6.1
Sustainability report	6.0 → SO	6.4 → SO
Report draft for review	6.0 → UA	6.3 → UA
Report approval	UA → 6.0	UA → 6.4
Text-generation input	6.0 → AI	6.3 → AI
Generated text result	AI → 6.0	AI → 6.3
Asset data	D2 → 6.0	D2 → 6.1

Parent flow (Level 1)	Level 1 direction	Level 2 location
Request data	D3 → 6.0	D3 → 6.1
Transfer data	D4 → 6.0	D4 → 6.1
Repair data	D5 → 6.0	D5 → 6.1
Stored measures	D7 → 6.0	D7 → 6.2
Calculated measures and inputs	6.0 → D7	6.2 → D7
Summary draft	6.0 → D6	6.3 → D6
Audit entry	6.0 → D9	6.4 → D9

Internal Level 2 flows (not visible at Level 1): Report data set (6.1 → 6.2), Calculated report values (6.2 → 6.3 and 6.2 → 6.4).

Table 2.3h: Level 1 → Level 2 (Process 7).

Parent flow (Level 1)	Level 1 direction	Level 2 location
Notification events	D8 → 7.0	D8 → 7.1
Audit entries	D9 → 7.0	D9 → 7.3
Notification	7.0 → DR, RQ, AC, MS, UA	7.2 → DR, RQ, AC, MS, UA
Audit history	7.0 → UA	7.4 → UA
Audit and technical log	7.0 → SA	7.4 → SA

Internal Level 2 flows (not visible at Level 1): Addressed notification (7.1 → 7.2), Permitted audit entries (7.3 → 7.4).

2.4 Data-Store Descriptions

Store ID	Name	Main data	Processes that read / write it
D1	User and Department Records	Accounts, roles, role assignments, departments, access status	1.0 (write, read); all processes read permissions

D2	Asset Records	Assets, categories, quantities, states, attachments, asset history events	2.0 (write, read); 3.0 (read); 4.0 (read, write ownership/custody/location after receipt); 5.0 (read, write condition and state); 6.0 (read)
D3	Request Records	Resource requests and request status history	3.0 (write, read); 4.0 (read, write status); 6.0 (read)
D4	Approval, Reservation and Transfer Records	Approvals, reservations, transfers, receipt records, discrepancies	4.0 (write, read); 6.0 (read)
D5	Maintenance and Inspection Records	Defect reports, inspections, maintenance records, repair costs	5.0 (write, read); 4.0 (read inspection result before approval/release); 6.0 (read)
D6	AI Records	Model versions, classification suggestions, match recommendations, action recommendations, report summaries, embeddings	2.0, 3.0, 5.0, 6.0 (write, read)
D7	Sustainability and Financial Records	Sustainability impacts, emission factors, costs, assumptions, calculated measures	6.0 (write, read)
D8	Notification Records	Notification events and read status	2.0, 3.0, 4.0, 5.0 (write events); 7.0 (read, mark delivered)
D9	Audit Log	Who/what/when, affected record, before and after values, AI version, reasons	All processes 1.0–6.0 (write); 7.0 (read)

3. System Architecture

This section defines how the prototype is divided into software areas and how those areas communicate. The architecture continues the technical approach selected in Phase 2: a web application using Next.js and TypeScript, Supabase services, PostgreSQL with pgvector, and an external AI provider. The main aim is to keep the core resource-exchange workflows reliable even when the AI service is unavailable.

3.1 Architecture Decision

The team compared two realistic options before selecting the prototype architecture. Both could support the requirements, but they do not require the same development and integration effort.

Table 3.1

Architecture Options and Decision

Option	Advantages	Limitations	Decision
Layered modular monolith with external-service adapters	Fits a three-member team; simpler deployment and testing; consistent transactions; manual functions continue if AI fails.	Internal modules cannot be deployed or scaled separately. Clear boundaries are needed to avoid tight coupling.	Selected
Service-oriented architecture	Services may be deployed and scaled separately, and some failures can be isolated.	More APIs, security, deployment, monitoring and integration testing; difficult cross-service transactions for reservations and transfers.	Not selected for the semester prototype

The selected option is a layered modular monolith with external-service adapters. It matches FR-01–FR-53, NFR-Q06 and NFR-AI10 while remaining practical for the semester schedule. The accepted trade-off is reduced independent scaling. The team will manage this by keeping clear interfaces between presentation, workflow, data, security, reporting and AI modules. The browser will not access the database or external AI provider directly.

3.2 Architecture View

Figure 3.1 illustrates the proposed Circular Campus Resource Exchange System using a layered modular-monolith architecture. The system consists of four horizontal layers: Presentation,

Application, AI and Analytics, and Data. Security is represented as a cross-cutting concern, while external integrations are placed outside the system boundary. Reporting and Notifications are implemented as application modules. The AI Gateway coordinates AI-assisted functions and controls communication with the external AI provider. Core business workflows remain available when AI services are unavailable. This figure is also referenced in Section 8.6 as the UML Component Diagram.

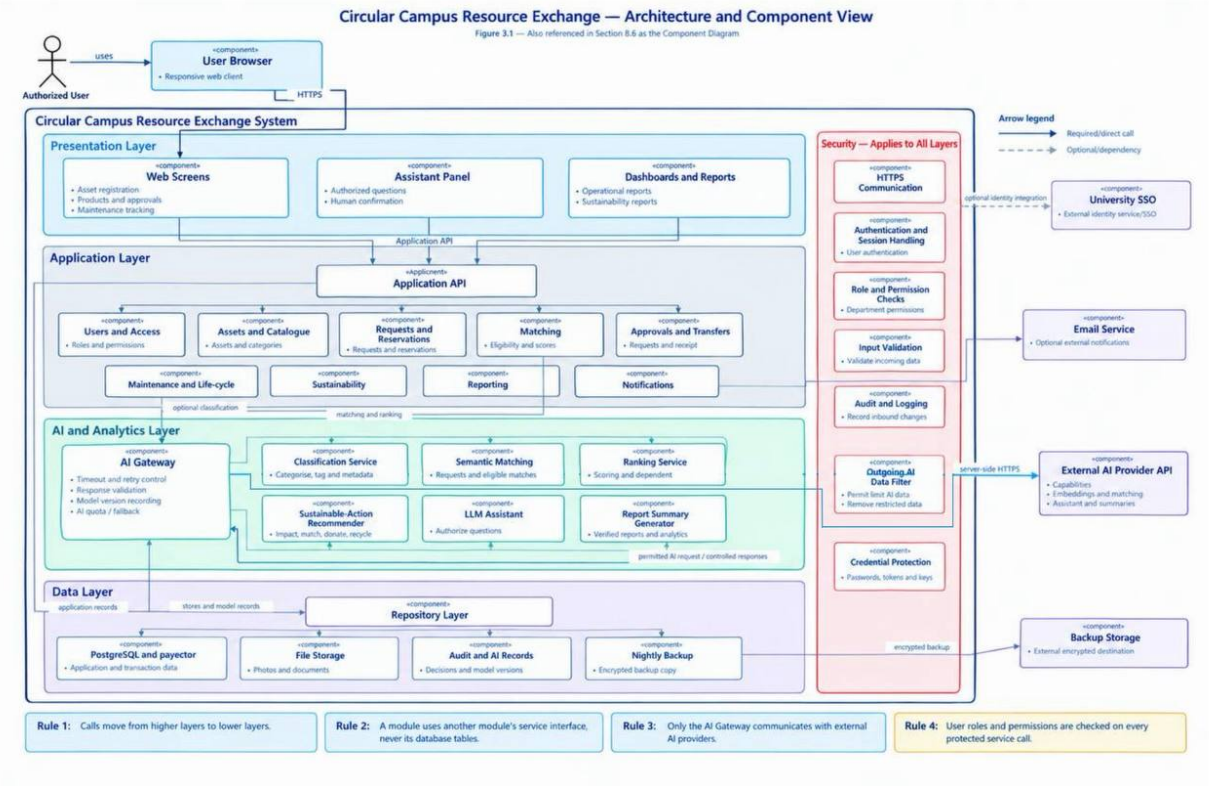


Figure 3.1

UML Component View of the Proposed System

Table 3.2

Architecture Areas and Main Responsibilities

Area	Main responsibility	Examples
Presentation	Displays role-appropriate screens and sends requests to the application API.	Responsive web UI, dashboards, forms, assistant panel

Application	Applies business rules and coordinates the core workflows.	Assets, requests, reservations, approvals, transfers, maintenance
AI and analytics	Coordinates advisory AI functions and changes provider results into the system format.	Classification, semantic matching, ranking, sustainable actions, assistant, summaries
Data	Stores application, file, audit and AI-monitoring records.	PostgreSQL, pgvector, object storage, audit/model records
Integration	Keeps provider-specific code outside the business modules.	External AI adapter; future university identity adapter
Notifications	Creates and stores messages for authorized recipients.	Request, approval, transfer, receipt and maintenance notifications
Security	Checks sessions, permissions, data validation and outgoing AI fields.	Authentication, role guard, secrets, field allow-list
Reporting	Calculates verified reports and optionally requests an AI draft summary.	Operational, financial and sustainability reporting

3.3 Responsibilities and Communication

The browser sends HTTPS requests to the application API. Before a protected request reaches a business module, the system checks the session, role, department access and permission for the selected record. Application services use the repository layer for database operations. Uploaded photographs and documents are stored separately, while their identifiers remain linked to the asset record.

Every AI-assisted request passes through the AI Gateway, which applies permission checks, outgoing-data filtering, response validation, timeout handling and model-version recording. The gateway coordinates communication with the external AI provider through the controlled server-side integration path. Provider-specific credentials are never exposed to the browser.

The reporting module calculates verified values first and requests an AI summary only after the report is complete. Notifications are created after the related business action succeeds; a notification failure does not undo the business record.

Table 3.3

Communication and AI-Failure Behaviour

Function	Normal communication	When AI is unavailable

Asset classification	Asset module → AI gateway → classification service	Custodian uses the normal category list and manual fields.
Matching and ranking	Request module → eligibility rules → AI gateway → ranking	Keyword search, filters and rule-based eligibility remain available; no AI rank is shown.
Sustainable action	Life-cycle module → queued AI job	Authorized user follows inspection, safety and life-cycle checklists.
Assistant	Web UI → application API → AI gateway	Normal forms, search, record pages and help pages remain available.
Report summary	Reporting module → optional queued summary	The standard verified report remains available without AI text.
Approval and transfer	Application workflow modules only	The workflow continues because AI never authorizes, releases or receives an asset.

3.4 Traditional and AI-Assisted Functions

Fixed rules control permission, availability, quantity, safety, approval and calculations. AI is used only where it can help interpret descriptions or prepare a draft. It cannot replace the rule checks or the authorized user's decision.

Table 3.4

Traditional and AI-Assisted Functions

Area	**Traditional function**	**AI-assisted function**	**Final control / fallback**
Search	Keywords and filters use stored fields.	Semantic search compares meaning and may find assets with different wording.	Keyword search and filters continue if AI is down.
Matching	Eligibility rules remove unavailable, unsafe or unsuitable assets.	Eligible assets receive a compatibility score, rank and explanation.	A human reviews the list; normal approval is still required.
Reporting	Verified formulas calculate operational,	AI prepares an optional draft written summary.	An authorized user accepts, edits or rejects the draft;

	financial and sustainability values.		standard reports continue.
Classification	Custodian selects a category, tags and materials manually.	AI suggests values and flags missing information.	Custodian confirms or edits the suggestion before saving.
Life-cycle action	Staff use inspection, safety, cost and policy rules.	AI suggests repair, transfer, donation, recycling or another suitable action.	Controlled action remains a human decision and requires the normal approval.

4. AI-Service Integration Design

The five AI functions have different inputs, waiting times and failure effects, so the design does not call them in the same way. Every call passes through the AI Gateway. The gateway checks permissions, applies an outgoing-field allow-list, validates the response, enforces the time limit and records the model/service version. AI output remains advisory and cannot approve a request, reserve an asset, release university property or change an asset's life-cycle state.

Table 4.1

Calling Patterns Used by the AI Services

Pattern	Meaning	Used for
Ask and wait	The user starts the action and waits for a short response.	Classification, current ranking and the conversational assistant
Compute ahead	The system prepares a reusable result before the user asks for it.	Asset and request semantic representations
Queue and notify	The system accepts the job, lets the user continue and sends an in-app notice when ready.	Sustainable-action review and report-summary drafting

For interactive classification, matching and recommendation calls, the prototype keeps the Phase 3 target: at least 90% should finish within 15 seconds in the test environment, with a hard stop at 30 seconds. One retry is allowed only for a temporary error and only inside the same 30-second user wait. A background job may start one later attempt without keeping the user waiting.

Table 4.2

Controls Applied to Every AI Service

Control	Design rule
Access	Check the user's role, department and record access before collecting AI inputs.
Privacy	Send only documented fields. Exclude passwords, tokens and unrelated personal, financial or confidential data.
Validation	Reject a response that does not match the expected schema or required value ranges.
Human control	Label the output as a suggestion or draft and keep the normal approval and safety rules.
Traceability	Store related record IDs, model/service version, time, output, confidence status and human decision.

Failure safety	An AI error must not delete, duplicate or partly update an asset, request, recommendation or report

4.1 Asset Classification

Classification helps an Asset Custodian complete a draft asset record. The suggestion is shown beside the original fields, and nothing is saved until the custodian accepts or edits it.

Design item	Specified design
Inputs / outputs	Inputs: asset ID, name, description, specifications, condition and approved image/document content where needed. Outputs: suggested category, subcategory, material, tags, clearer description, missing fields, reason, confidence status and model version.
Calling pattern and reason	Ask and wait. The custodian is already editing the asset and can continue manually.
Time limit / retry	Target 15 seconds; hard timeout 30 seconds; one temporary-error retry inside the total limit.
AI-down fallback	Keep the draft unchanged and show the manual category, material, tag and description fields.
Low confidence	Below 0.70, missing mandatory information or confidence not supplied. Show a warning and do not preselect the suggestion.
Explanation / record	Show the description words or specifications used. Record asset ID, model/template version, suggestion, confidence status, time and user decision.

Source: *FR-39, FR-40, FR-46–FR-48, FR-52–FR-53; BR-AI-01, BR-AI-03–BR-AI-06 and NFR-AI01–NFR-AI02*

4.2 Semantic Matching

Semantic matching compares the meaning of a request with published asset descriptions after visibility and eligibility checks. It supports normal search but does not replace keyword filters.

Design item	Specified design
Inputs / outputs	Inputs: request description, specifications, purpose, preferred condition, date and location; approved text/specifications for eligible assets. Outputs: candidate IDs, similarity values, matched evidence,

	missing-data warnings, confidence status and model version.
Calling pattern and reason	Compute ahead. Create an asset representation when the asset is published or materially updated, and a request representation at submission. Reuse them during comparison.
Time limit / retry	Each background call stops after 30 seconds and may be retried once later. Opening recommendations follows the 15-second AI target; structured keyword/filter search keeps the 3-second NFR-02 target.
AI-down fallback	Use keywords, filters and rule checks, clearly labelled as rule-based rather than semantic results.
Low confidence	Similarity below 0.70, missing required request information or unavailable confidence is marked Weak match. No top-pick claim is made.
Explanation / record	Show matching request terms, specifications or purpose. Record request/asset IDs, similarity, confidence status, model version and time.

Source: *FR-41–FR-48, FR-52–FR-53; NFR-02, NFR-05, NFR-AI03–NFR-AI08 and NFR-AI10*

4.3 Recommendation Ranking

Ranking only considers assets that pass availability, quantity, status, safety and mandatory-specification checks. The result orders candidates; it does not approve any request or transfer.

Design item	**Specified design**
Inputs / outputs	Inputs: request, eligible asset IDs, normalized factor values, missing-data flags and semantic similarity. Outputs: compatibility score, rank, factor contributions, explanation, evidence coverage, confidence status, warnings and version.
Calling pattern and reason	Ask and wait. Ranking is refreshed when the requester opens recommendations because availability may have changed.

Time limit / retry	Target 15 seconds; hard timeout 30 seconds; one temporary-error retry inside the limit. Eligibility is checked again before display.
AI-down fallback	Show eligible keyword/rule results without an AI rank. The user can continue through normal review.
Low confidence	Show Weak recommendations when the top compatibility is under 70/100, confidence is below 0.70 or not supplied, or evidence coverage is below 80%.
Explanation / record	Show positive/negative factor contributions, missing fields, scoring version, model version and confidence separately. Store rank and the user's decision.

Source: *FR-42–FR-48, FR-52–FR-53; BR-AI-01–BR-AI-04 and NFR-AI03–NFR-AI08*

4.4 Sustainable-Action Recommendations

This service suggests possible next actions such as continued use, transfer, repair, refurbishment, donation, recycling, component recovery or disposal. It cannot remove a safety restriction or replace university approval.

Design item	**Specified design**
Inputs / outputs	Inputs: condition, age, inspection/safety result, history, expected end-of-life, value, repair/transfer cost, material and approved factors. Outputs: ordered options, reasons, required follow-up, missing data, confidence status, assumptions and version.
Calling pattern and reason	Queue and notify. The analysis uses more history and cost data, so the user can continue while it runs.
Time limit / retry	Queue receipt within 3 seconds; AI call target 15 seconds and hard stop 30 seconds. One failed temporary job may be queued for one later attempt.
AI-down fallback	Show the normal inspection, safety, cost and life-cycle checklist. Do not create a new AI suggestion for that cycle.
Low confidence	Below 0.70, confidence not supplied, or missing safety/inspection data. Required safety data blocks the action and requests inspection.
Explanation / record	Show condition, history, cost, safety and sustainability factors. Record inputs,

	assumptions, version, confidence status, decision and reason.

Source: *FR-46–FR-49, FR-52–FR-53; BR-AI-01–BR-AI-04, BR-AI-08–BR-AI-10 and NFR-SA03–NFR-SA04.*

4.5 LLM Assistant and AI-Generated Report Summaries

The assistant and report summarizer may use the same provider, but they are separate services because their permissions, input records, validation and failure effects are different.

4.5.1 Conversational Assistant

Design item	Specified design
Inputs / outputs	Inputs: user ID/role, question and authorized record references. Outputs: answer, links, source references, clarification request, confirmation-required flag and model version.
Calling pattern and reason	Ask and wait because the user expects a direct reply.
Time limit / retry	Normal design target 8 seconds; progress message if longer; hard timeout 30 seconds; one temporary retry inside the limit.
AI-down fallback	Show Assistant unavailable and link to search, help pages and normal forms.
Low confidence	Below 0.70 or not supplied: show Answer needs verification and the related record/help source. Unverified text is not official information.
Human control / record	A prepared request must be shown for confirmation, then pass normal validation. Record model version, permitted references, answer reference, time and access result.

4.5.2 AI-Generated Report Summaries

Design item	Specified design
Inputs / outputs	Inputs: report ID, period, verified values/units, assumptions, source references, filters and missing-data notices. Outputs: draft summary, metric references, limitations, model version and time.
Calling pattern and reason	Queue and notify. The normal report is generated first, so the optional draft does not delay it.

Time limit / retry	Standard report retains the 5-second NFR-03 target. Each AI attempt stops after 30 seconds; one temporary failure may be retried once in the queue.
AI-down fallback	Keep the verified standard report and omit the AI summary.
Low confidence / validation	If provider confidence is unavailable, store Not supplied. Any number mismatch, unsupported claim or missing limitation blocks approval.
Human control / record	Authorized user accepts, edits or rejects the draft. Store report/metric references, model/template version, decision and approved final text.

Source: *Assistant: FR-50, FR-52–FR-53. Report summary: FR-35–FR-38, FR-51–FR-53, BR-AI-07–BR-AI-09 and NFR-AI09.*

5. Human Approval Points

5.1 Human-Approval Swimlane Overview

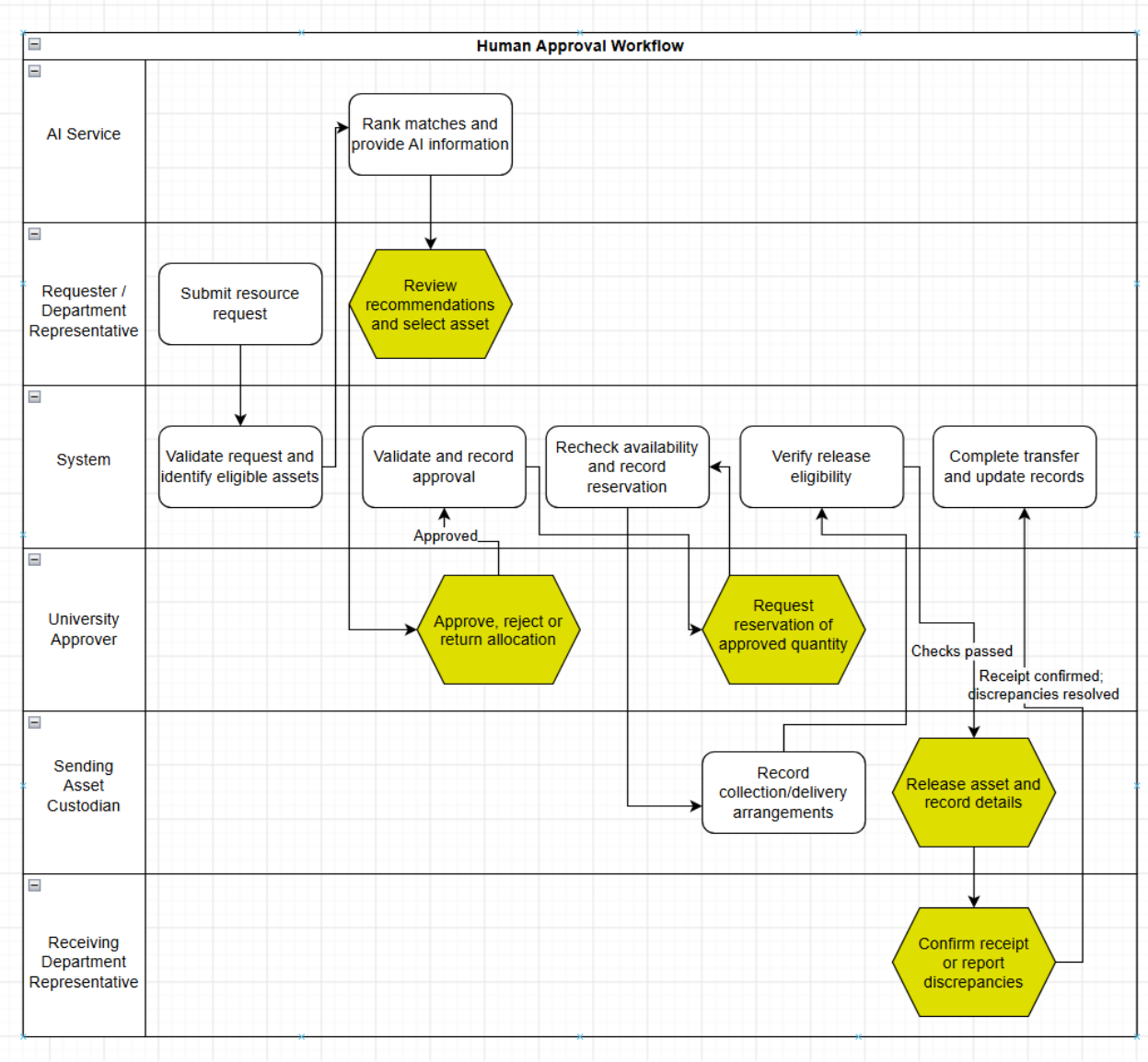


Figure 5.1: Human Approval Workflow

This stage allows the AI to provide a score, reason, and confidence level. Where AI is not available or appropriate, the user can perform a manual search and select relevant assets. Approval is still required.

Approval ensures you cannot approve yourself and prevents inappropriate/ineligible allocations. Reservations will only allow you to reserve the correct amount of approved items; they also have a time limit. Safety/Inspection restrictions and duplicate releases are prevented.
Receipt will record how much you received, what condition it was in, and where it went. The item will be in a Registered/Unpublished status or Unavailable status depending on any restrictions.
The flow shows the successful route through the process; any request that is rejected or returned will not go through. Any yellow hexagons are points where a person will need to review/approve.

5.2 Human Approval and Review Points

This is achieved by distinguishing between suggestions provided by AI, human approval, and logging of the execution of the action. AI can recommend classification, matches, actions, or summary reports, but the authorized person will still be accountable for reviewing them. The system validates the permissions and workflow prerequisites before accepting the decision.

Each logged decision will indicate the user who made the decision, time and date of the decision, and the associated record. Any AI decision will identify the output and the model/service used.

Decision point	**Responsible role**	**Information presented**	**Reason for human control**	**Recorded outcome**
Review an AI classification	Asset Custodian	Asset details, suggested category/tags, explanation and confidence or missing-information warning	Incorrect classification can affect search and matching	Accepted, corrected or rejected suggestion; confirmed values and override reason where applicable

Review a recommended match	Requester or authorized Department Representative	Requested specifications, available quantity, asset condition, match score, explanation and uncertainty	The requester must judge whether the proposed asset meets the intended need	Selected or rejected recommendation and any override reason; selection does not constitute approval
Approve, reject or return a request	University Approver	Request, proposed allocation, quantity, availability, condition, safety restrictions and any AI explanation	Allocation and transfer require authorized approval; users cannot approve their own requests	Decision, reason, approver and timestamp
Reserve an approved allocation	University Approver	Approved request, selected asset, eligible quantity and reservation expiry	The allocation must follow approval and avoid conflicting reservations	Linked reservation, reserved quantity and expiry; remaining eligible units stay available
Release an asset	Sending Asset Custodian	Valid approval, allocated quantity, handover arrangements, condition and safety/inspection information	Authorization alone does not prove that physical release occurred	Released asset and quantity, releasing user and release date/time
Confirm receipt	Receiving Department Representative	Transfer record, delivered quantity, condition and destination	Ownership and custody changes require confirmation of delivery	Receipt details and discrepancies; completion only after discrepancies are resolved

Authorize a controlled repair or life-cycle action	University Approver within delegated authority	Asset condition, inspection/history, proposed action, available cost information and any AI recommendation	Repair, refurbishment, donation, recycling, retirement and disposal can have financial or irreversible consequences	Approval or rejection and reason; staff record actual execution separately
Clear an asset for return to use	Authorized inspection staff and Asset Custodian	Inspection findings, maintenance outcome, resulting condition and remaining restrictions	Repair completion does not automatically establish that an asset is safe and available	Authorized condition assessment followed by the Custodian's availability confirmation
Confirm an assistant-prepared request	Requester or authorized Department Representative	Complete draft request, including specifications, quantity, purpose and required date	AI may misunderstand the user's intended request	User confirmation followed by permission/field validation and standard submission
Review an AI-generated report summary	Authorized report reviewer	Draft summary, underlying calculated report, reporting period, sources and assumptions	Generated statements must agree with verified records before official use	Accepted, edited or rejected summary and the reviewed version

6. Security, Data Storage and Recovery

6.1 Authentication and Access Boundaries

Users will need to authenticate to gain access to any secure functionality. The server will verify users role, department and if the user has permission to access the requested record. This is done regardless of whether a user wants access directly to an endpoint or has modified the record ID. When a user logs out their session will be invalidated.

The interface will interact with the application server. The application server controls access to the database and external AI calls. The AI services will only have access to the correct fields required to complete their task and cannot directly authorise requests or make changes to business records. Authorisation and handing over are outlined in section 5.

Related requirements: FR-01–FR-03; NFR-SE01–NFR-SE03 and NFR-SE10.

6.2 Role-Based Permissions

Permissions are denied unless explicitly granted. A user holding multiple roles still cannot approve their own request.

Role	Permitted access and actions	Main restriction
Requester	Search published assets; submit, track and update their own requests where the status permits.	Cannot publish assets or approve requests.
Department Representative	Manage authorized departmental records and requests; confirm receipt for their department when designated.	Cannot authorize their own request or an unapproved transfer.
Asset Custodian	Register and update assigned assets; publish eligible surplus assets; arrange and record approved releases; record authorized inspections.	Cannot approve requests or release assets without valid approval.
University Approver / Business Administrator	Review requests; approve, reject or return them; reserve approved quantities; authorize controlled lifecycle actions within delegated authority.	Cannot self-approve or act outside delegated authority.

Procurement Officer	View authorized procurement information and record or verify assigned purchasing information.	Cannot approve transfers without a separately assigned approval role.
Finance Officer	View authorized financial records; record or verify assigned costs and review financial reports.	Cannot change unrelated asset or approval records.
Maintenance Staff	Record assigned inspections, repairs, progress, costs and resulting condition.	Maintenance completion does not independently authorize publication or transfer.
Sustainability Officer	Review sustainability reports and verify authorized calculation inputs and assumptions.	Cannot authorize transfers or disposal without a separately assigned approval role.
System Administrator	Manage accounts, approved role assignments, configuration, technical logs and recovery operations.	Technical administration does not automatically grant business approval authority.
AI Service	Produce recommendations and summaries using approved inputs.	Cannot approve, reserve, release, confirm receipt or independently modify business records.

Procurement and Finance are separated to fit the list of roles for the project. They are grouped together under Procurement and Finance Officer in the SRS. The University Approver reflects the business authority. This role is different from the System Administrator.

Related Requirements: Sec 2 of SRS, FR-01 to FR-06, BR-08.

6.3 Audit Records

All major changes will be logged within the system. This includes changes to assets, requests, reservations, approvals, transfers and permissions. Within the log it will show:

- Responsible user or service identifier.
- Action and affected record identifier.
- Timestamp.

- Previous and new values or status, where applicable.
- Decision or correction reason, where required.

If the decision was made using AI then the output, model/version, related records and if they were accepted, edited or rejected by the user will be logged. If there has been an override then the reason will be shown.

Only authorised staff can view audit trails. Application users cannot make any changes/deletions. Passwords, tokens and API keys are not stored.

Related Requirements: FR-34, FR-48, FR-52. NFR-SE06, NFR-AI08.

6.4 Data Storage & Security

The database stores all related data relating to users, departments, assets, requests, reservations, approvals and transfers. It also stores historical and reporting data. Any related changes will be processed using transactions to ensure that if something goes wrong during the process then the transfer/reservation will not be partially completed. Quantity and reservation changes are done atomically to avoid double allocations.

Passwords are stored as salted hashes. All network activity is done over HTTPS. Secrets are not stored in the source code or on GitHub. The server will check that mandatory fields, types and allowed values are correct before making any changes. Demo data will be used.

External AI requests will only have a set list of fields that can be sent. Search and reporting functionality will still be available if the AI services are down.

Related requirements: NFR-SE04–NFR-SE10; NFR-Q04–NFR-Q05; NFR-AI10.

6.5 Backup and Recovery

The following is the proposed prototype backup design:

Item	Design
Backup contents	Database records, audit history and any stored supporting documents.

Frequency	Daily during active development and before database migrations.
Location	An access-restricted team backup folder separate from the running application, with encryption at rest. Secrets are stored separately.
Responsibility	The System Administrator role controls backup and restoration.
Recovery procedure	Restore the latest valid backup into a separate environment, verify it, then resume application use.
Restore test	Restore a backup and verify login, record counts, linked asset/request/transfer records, statuses and audit history. Record the result.

A failed save rolls back the affected transaction. Backup restoration is used for wider data loss or corruption; it does not replace transaction handling.

Related requirements: NFR-Q05 and NFR-SE06–NFR-SE07.

7. User-Interface Mock-ups

7.1 Asset Registration

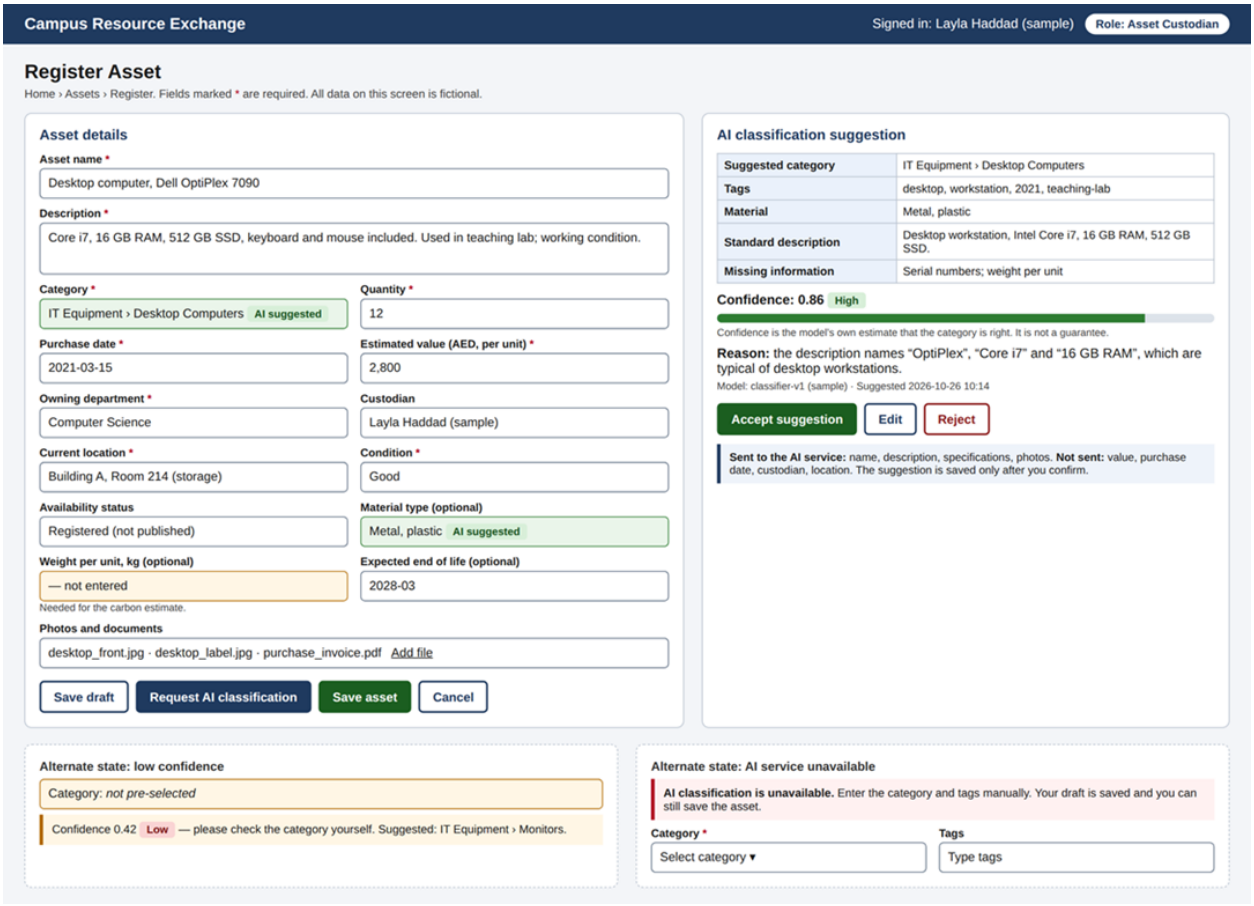


Figure 7.1: Asset Registration (role: Asset Custodian). All data is fictional.

Screen element	What it shows / does	Phase 3 link
Role banner	Signed-in user and role (Asset Custodian). Only roles allowed to register assets see this screen.	FR-01, FR-07
Asset details form	Required fields: name, description, category, quantity, purchase date, estimated value, owning department, location, condition. Custodian and availability status are shown. Optional: material type, weight per unit, expected end of life (needed for carbon and asset-life calculations). Photos and documents can be attached.	FR-07, FR-08

Request AI classification	Sends only name, description, specifications and photos. Value, purchase date, custodian and location are not sent. The page states this.	FR-39, FR-40
AI suggestion panel	Suggested category, tags, material, standard description, missing information, confidence (number, label and bar), plain-language reason, and model version.	FR-39, FR-48, FR-52
Human decision	Accept suggestion, Edit, or Reject. The suggestion is stored with the decision and model version; nothing is saved from the AI automatically. “Save asset” stays under the custodian’s control.	FR-48, FR-52
Alternate: low confidence	Category is not pre-selected, a warning is shown, and the custodian chooses the category.	FR-48
Alternate: AI unavailable	A message explains the service is unavailable; manual category and tags are shown; the draft is saved and the asset can still be registered.	FR-47

Accessibility: every field has a visible label; required fields use text “*” not colour alone; status is shown with text (“High”, “Low”) as well as colour; text contrast is at least 4.5:1; all actions are buttons reachable by keyboard.

7.2 Resource-Request Submission

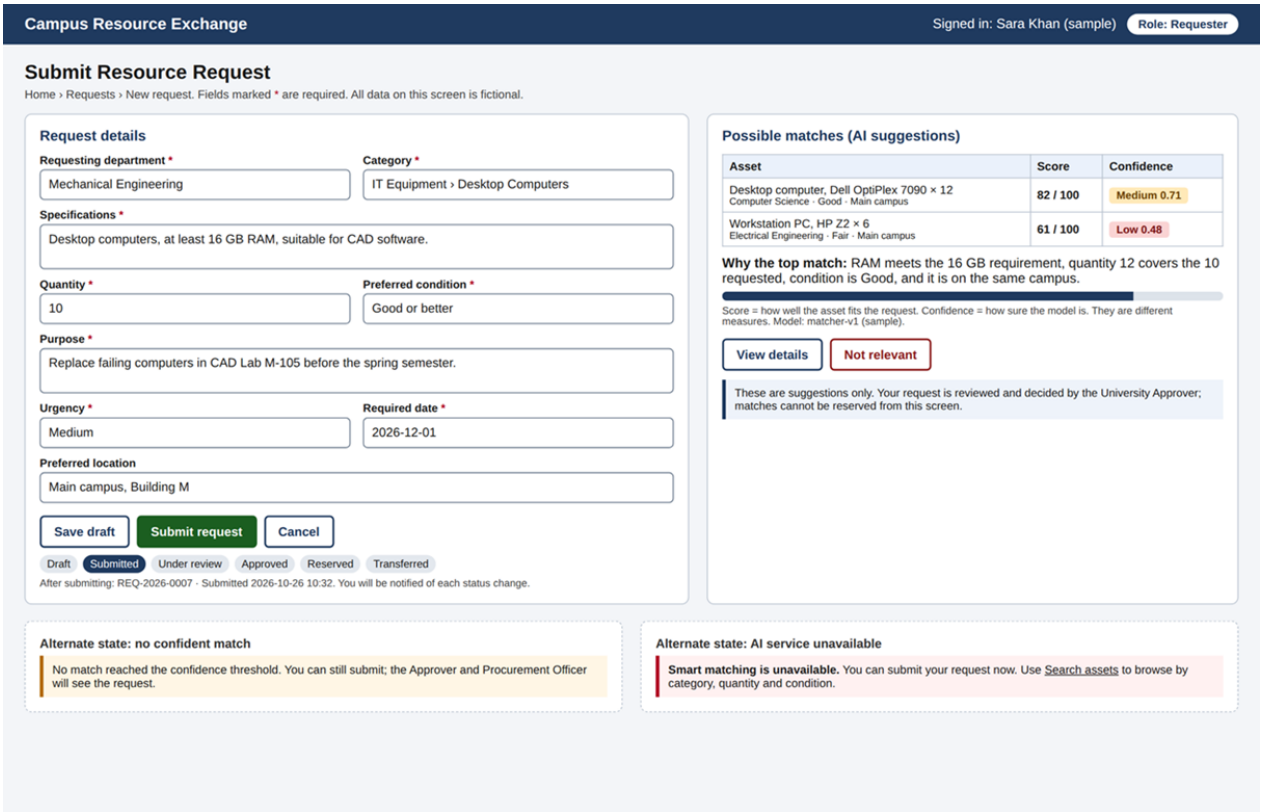


Figure 7.2: Resource-Request Submission (role: Requester). All data is fictional.

Screen element	What it shows / does	Phase 3 link
Role banner	Signed-in user and role (Requester).	FR-01
Request form	Requesting department, category, specifications, quantity, preferred condition, purpose, urgency, preferred location and required date. Required fields are marked.	FR-15
Submit / Save draft	The Requester submits the request. Status moves from Draft to Submitted and the tracker shows later steps. A request number and time are shown after submission.	FR-15, FR-16
Possible matches	Up to a few ranked assets with compatibility score, confidence, and a reason. Score (fit of asset to request) and confidence (how sure the model is) are labelled as different measures. Score and confidence rules are defined in 8.11.	FR-41–FR-46

Human decision	Matches are suggestions. The Requester may mark one as not relevant; reservation and approval are done by the University Approver.	FR-18, FR-48
Alternate: no confident match	The request can still be submitted; Approver and Procurement Officer can see it.	FR-47
Alternate: AI unavailable	Smart matching is switched off, the request is submitted normally, and keyword Search assets is offered.	FR-47

Accessibility: same rules as 7.1. Confidence colours are always paired with text and a number.

7.3 Recommendation Review

The Requester compares eligible assets using the compatibility score, confidence status, explanation and missing information. Selecting an item does not approve or transfer it; it only sends the choice to the normal reservation and approval workflow.

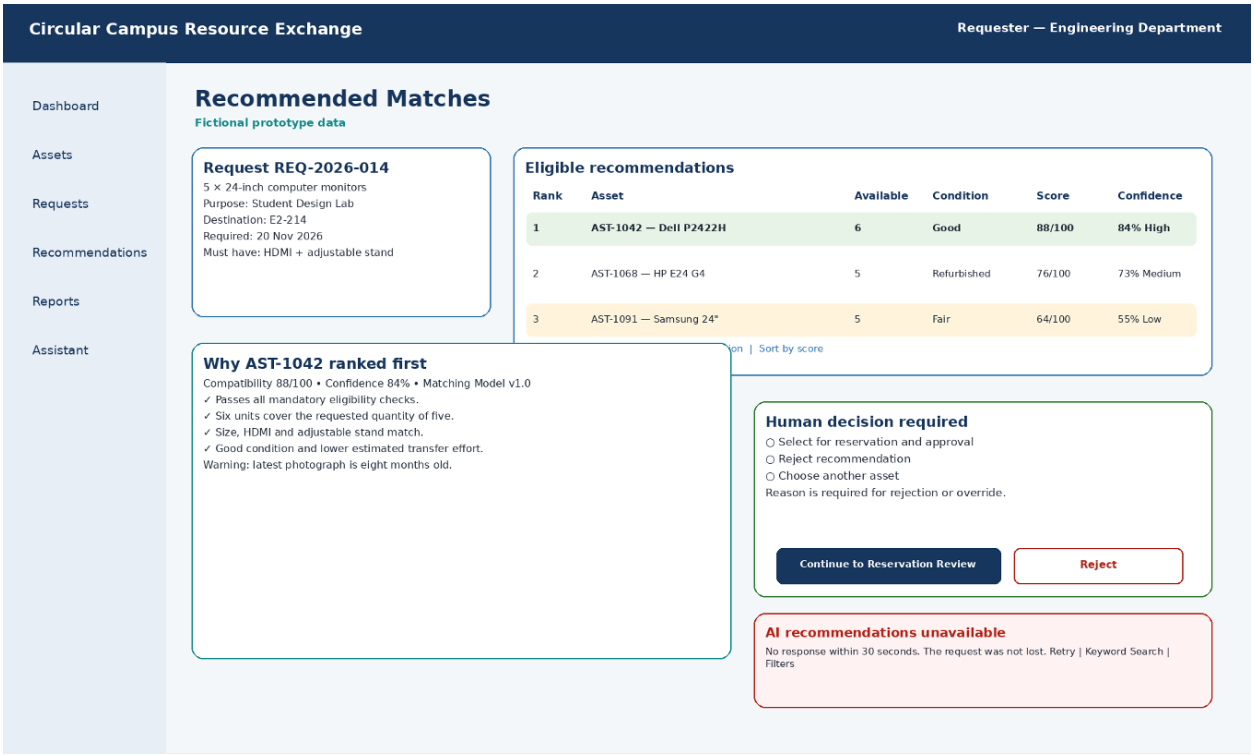


Figure 7.3
Recommendation Review Screen

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7.4 Transfer Approval

This screen allows the University Approver to review a proposed allocation, record a decision and reserve the approved quantity after approval.

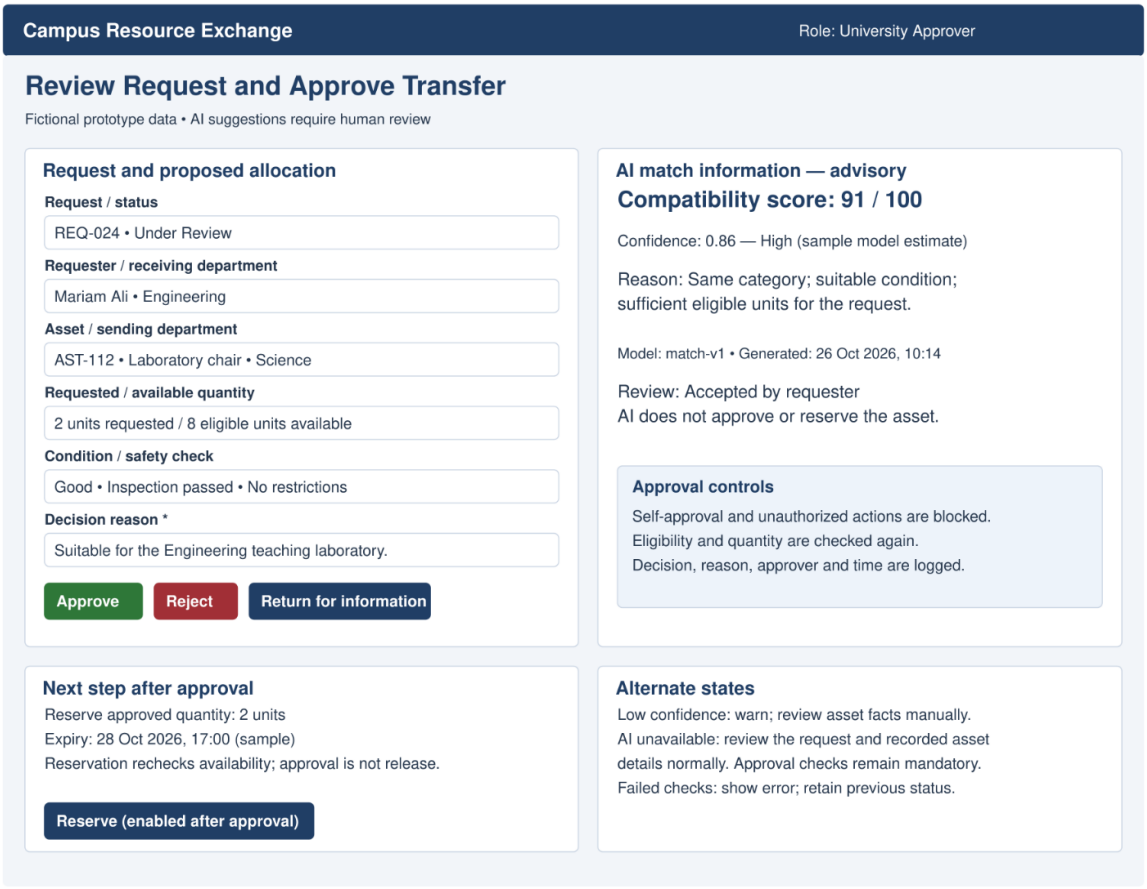


Figure 7.4: Transfer approval • FR-19–FR-24, FR-34, FR-47–FR-48, FR-53

Figure 7.4: Transfer Approval

7.5 Maintenance Tracking

This screen allows Maintenance Staff to record repair progress and costs, while keeping repair completion separate from inspection and custodian-confirmed return to availability.

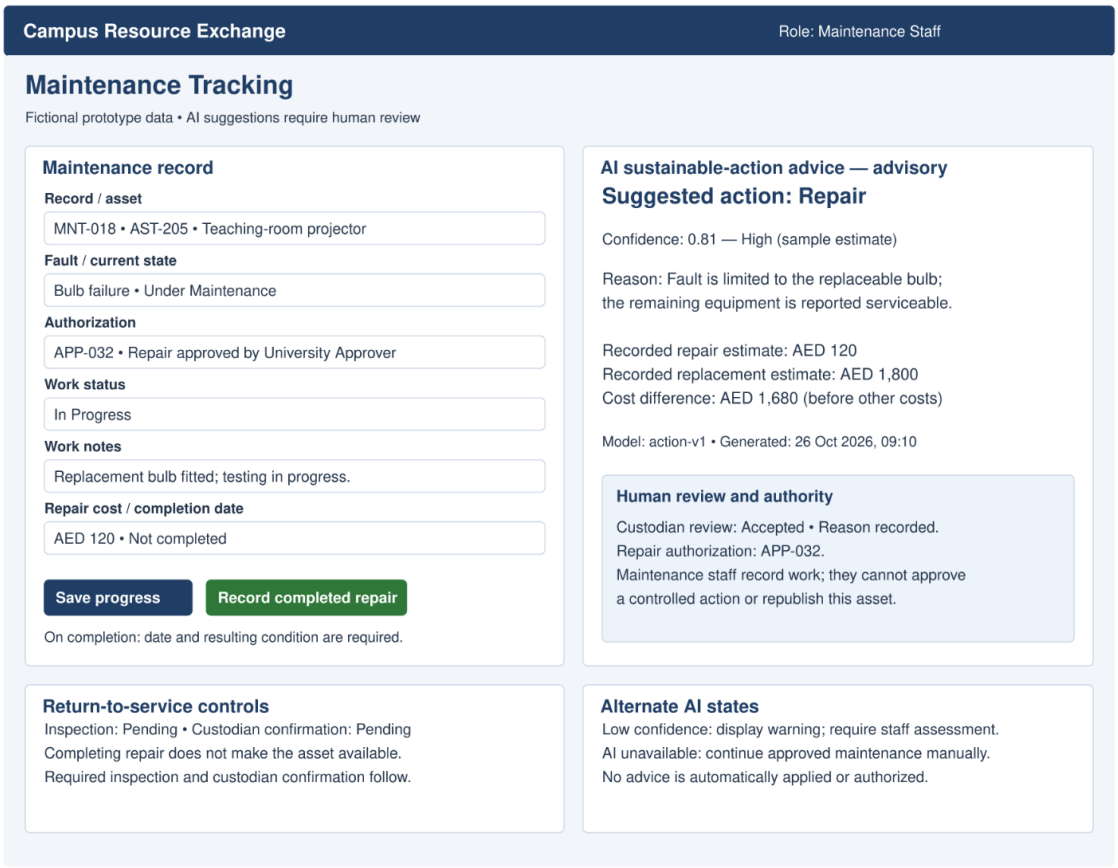


Figure 7.5: Maintenance Tracking Screen Mockup

7.6 Sustainability Dashboard

This screen presents calculated sustainability measures, their methods and sources, and an AI-generated interpretation for human review; calculated results remain available when AI is unavailable.

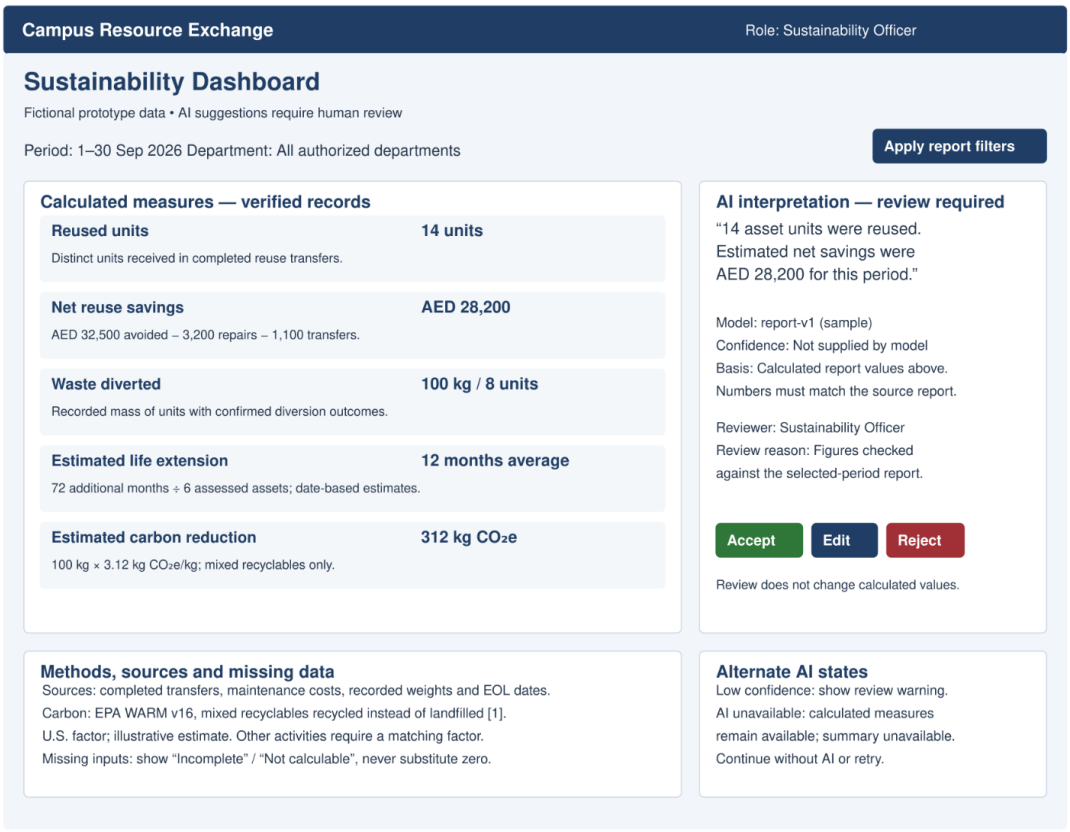


Figure 7.6: Sustainability dashboard • FR-36–FR-38, FR-47–FR-48, FR-51–FR-53

Figure 7.6: Sustainability Dashboard Screen Mockup

The eight user-interface mockups follow consistent design principles to support usability, accessibility, and human control. The screens use fictional prototype data and display AI-related information and fallback options where applicable. Table 7.1 summarizes the shared design rules applied across the mockups.

Table 7.1:
Shared User-Interface Mockup Design Rules

Rule	**How It Is Applied**
Consistent Layout	Screens use a consistent header, navigation menu, content layout, and visible user-role information.
Human Control	Authorized users can review, confirm, accept, edit, or reject relevant actions and AI suggestions. AI does not automatically approve or execute consequential actions.
AI Transparency	AI-assisted screens display relevant suggestions, explanations, confidence information, and model/version details where applicable.

AI Failure Handling	When AI is unavailable, users receive a clear message and can continue using the relevant manual functions or standard reports.
Accessibility	Screens use labelled fields, readable text, sufficient contrast, keyboard-accessible controls, and text labels alongside status colours or icons.
Sample Data	All displayed records and calculated examples are fictional prototype data, not actual university results.

7.7 Operational/Sustainability Reports and AI Summaries

An authorized officer generates a standard report from verified records, then optionally requests a draft AI summary. Missing values are not treated as zero, and the standard report remains available when AI fails.

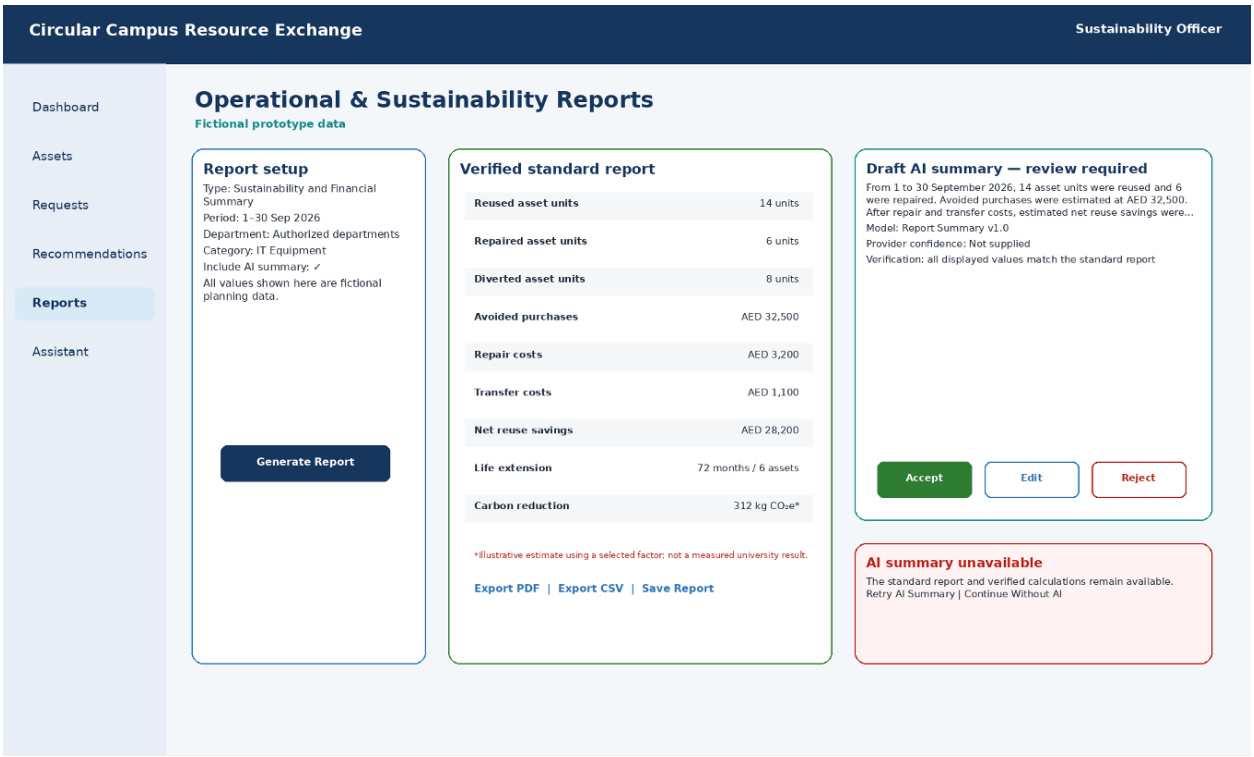


Figure 7.7

Operational and Sustainability Report and AI Summary Review Screen

7.8 Conversational Assistant

The assistant answers authorized questions and may prepare a draft request, but the user must review and confirm every field before submission. Permission checks occur before records are retrieved or displayed.

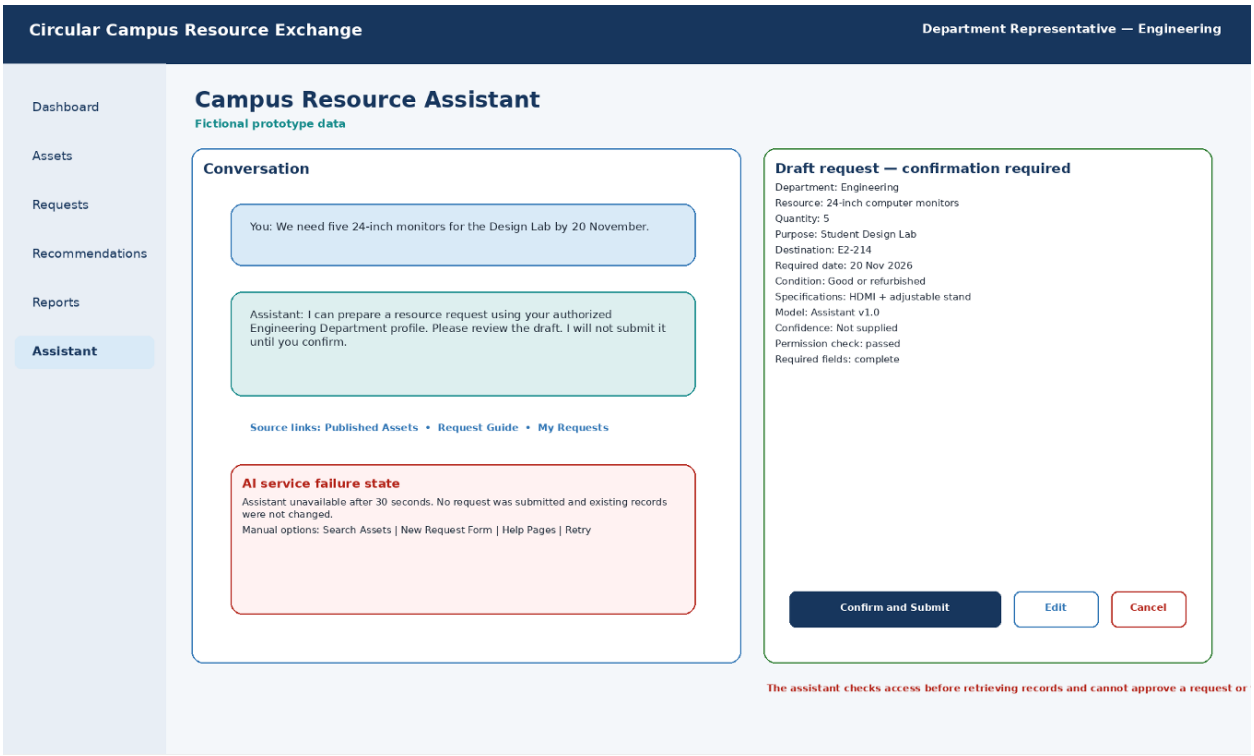


Figure 7.8

Conversational Assistant and Request Confirmation Screen

Source: *Screen content traces to FR-46–FR-48, FR-50, FR-52–FR-53, NFR-SE01–NFR-SE03, NFR-SE10 and NFR-AI05–NFR-AI10.*

8. Detailed Design and Modelling

8.1 Overall Use-Case Diagram

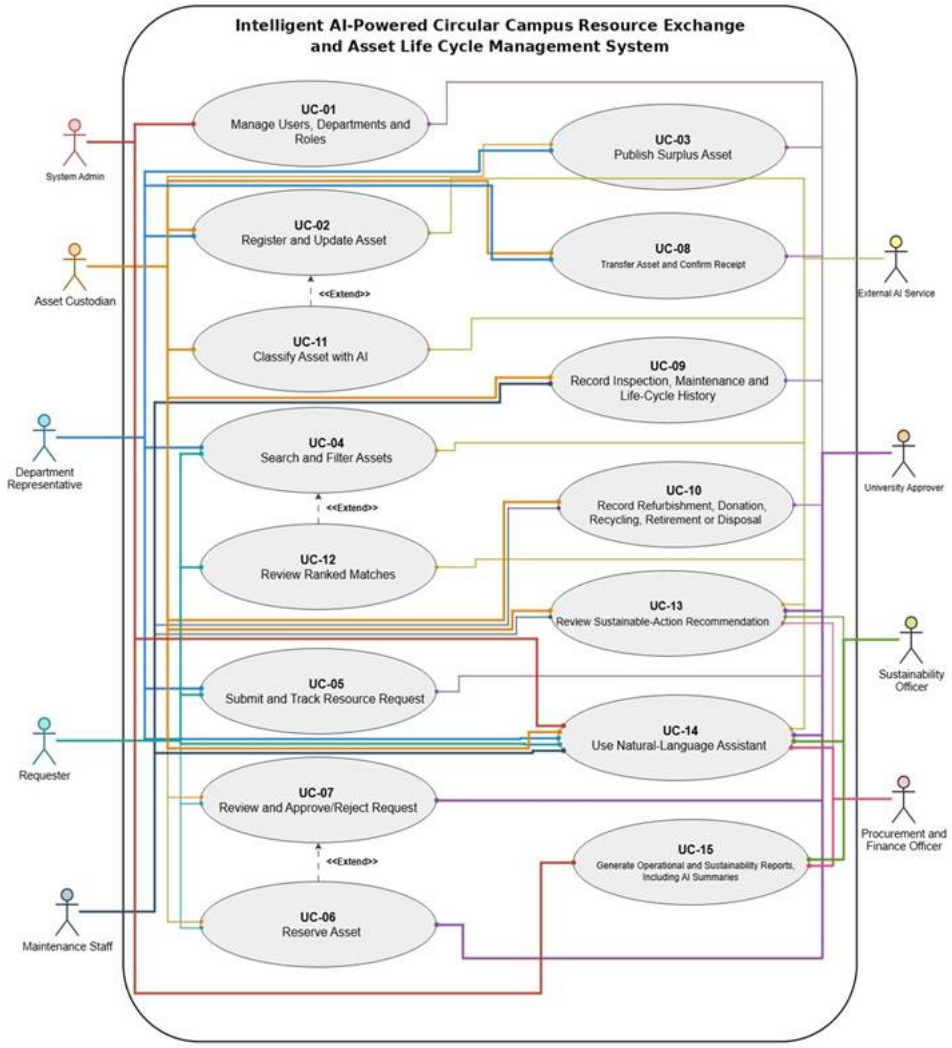


Figure 8.1: Overall use-case diagram (updated from Phase 3 Figure 1).

Table 8.1

Use cases, actors and changes from Phase 3

ID	Use case	Primary actor	Supporting actors
UC-01	Manage Users, Departments and Roles	System Administrator	University Approver
UC-02	Register and Update Asset	Asset Custodian	External AI Service

UC-03	Publish Surplus Asset	Department Representative	Asset Custodian, University Approver
UC-04	Search and Filter Assets	Requester	External AI Service
UC-05	Submit and Track Resource Request	Requester	Department Representative, University Approver
UC-06	Reserve Asset	University Approver	Requester, Asset Custodian
UC-07	Review and Approve/Reject Request	University Approver	Requester, Asset Custodian
UC-08	Transfer Asset and Confirm Receipt	Asset Custodian	Department Representative, University Approver
UC-09	Record Inspection, Maintenance and Life-Cycle History	Maintenance Staff	University Approver, Asset Custodian
UC-10	Record Refurbishment, Donation, Recycling, Retirement or Disposal	Asset Custodian	University Approver, Maintenance Staff
UC-11	Classify Asset with AI	Asset Custodian	External AI Service
UC-12	Review Ranked Matches	Requester	External AI Service
UC-13	Review Sustainable- Action Recommendation	Asset Custodian	Maintenance Staff, Procurement Officer, Finance Officer, Sustainability Officer, University Approver, External AI Service
UC-14	Use Natural- Language Assistant	Requester	Department Representative, Asset Custodian, University Approver, Sustainability Officer, External AI Service
UC-15	Generate Operational and Sustainability Reports, Including AI Summaries	Procurement Officer	Finance Officer, University Approver, Sustainability Officer, External AI Service

Changes made to the reviewed Phase 3 diagram:

- The Phase 3 actor “Procurement and Finance Officer” is split into Procurement Officer and Finance Officer (D1). Both are primary actors of UC-15; both support UC-13.
- The External AI Service is now connected to UC-15 (report summaries), which was missing in Phase 3. It remains a supporting actor of UC-02, UC-04, UC-11, UC-12, UC-13, UC-14 and UC-15 only.
- “Notification Service” and “Audit Log” are no longer listed as supporting actors, because they are parts of the system (D2). Each actor now appears once.
- UC-06 “Reserve Asset” is no longer drawn as an «extend» of UC-07. Reservation follows approval and is performed by the University Approver, so it is a separate use case. The other two «extend» relations are kept: UC-11 extends UC-02 and UC-12 extends UC-04.
- UC-08 shows both the sending Asset Custodian (release) and the Department Representative (receipt) as actors, matching the Phase 3 rule that release and receipt are separate responsibilities.

8.2 Class Diagram

The model is shown in five views that share the same class names. Collapsed grey boxes show a class from another view. “+” public, “-” private. Enumeration types (AccountStatus, AssetState, RequestStatus, TransferStatus, MaintenanceStatus, DecisionStatus, Decision) take their values from the Phase 3 state tables. Attribute names are the camelCase forms of the ERD columns in 8.9. Asset records its owner department (Department 1 to 0..* Asset), its optional custodian (AppUser 0..1 to 0..* Asset) and its location, as in the asset table of 8.9b. An Approval belongs to a request, an asset or a maintenance record depending on its type, so its request end is optional (0..1), as in 8.9c.

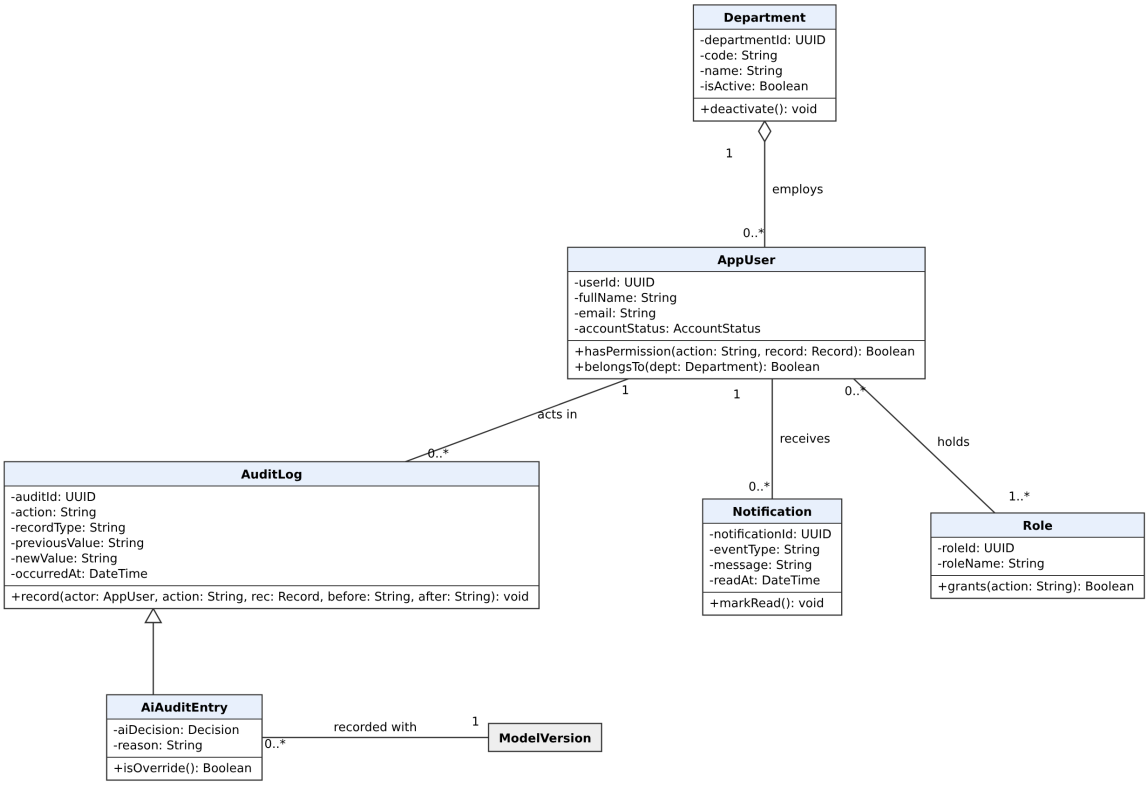


Figure 8.2a: Access, notification and audit. Department aggregates AppUser; AiAuditEntry specializes AuditLog.

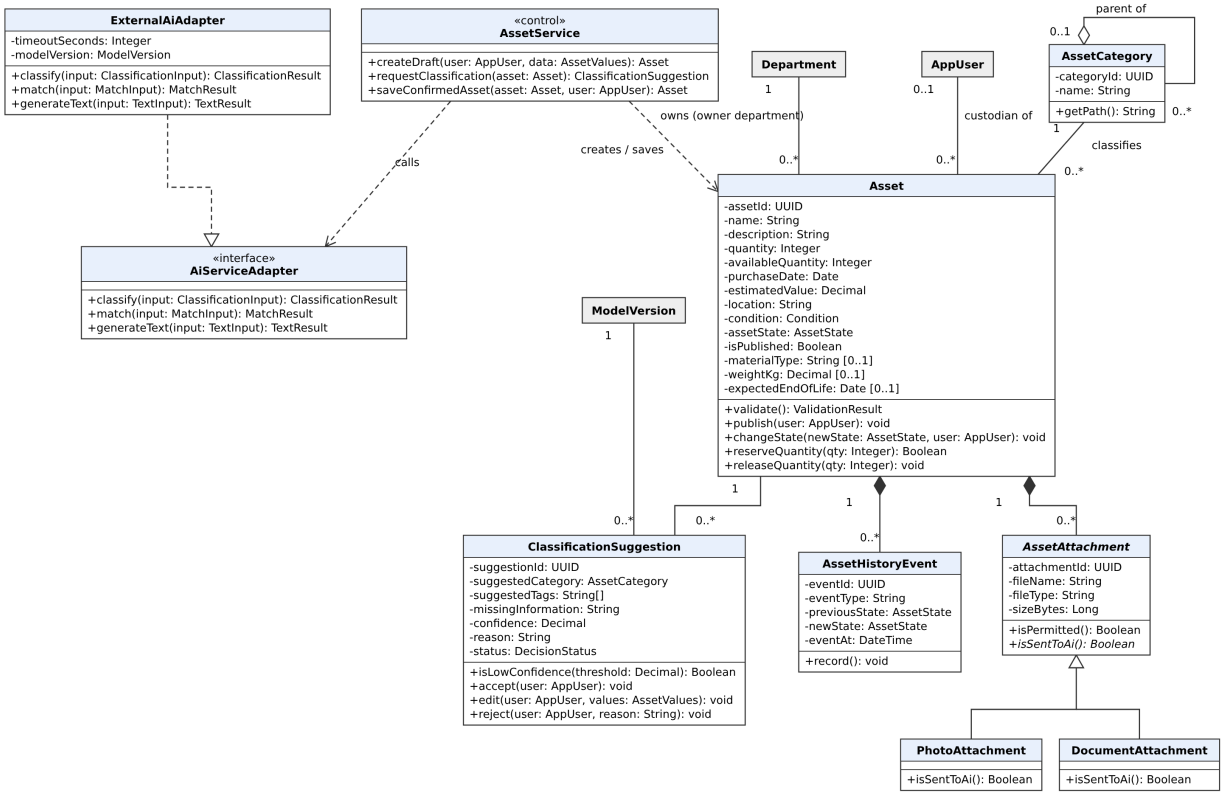


Figure 8.2b: Assets, AI classification and service classes (AssetService, AiServiceAdapter, ExternalAiAdapter). Asset is composed of history events and attachments; AssetCategory aggregates its sub-categories; PhotoAttachment and DocumentAttachment specialize AssetAttachment.

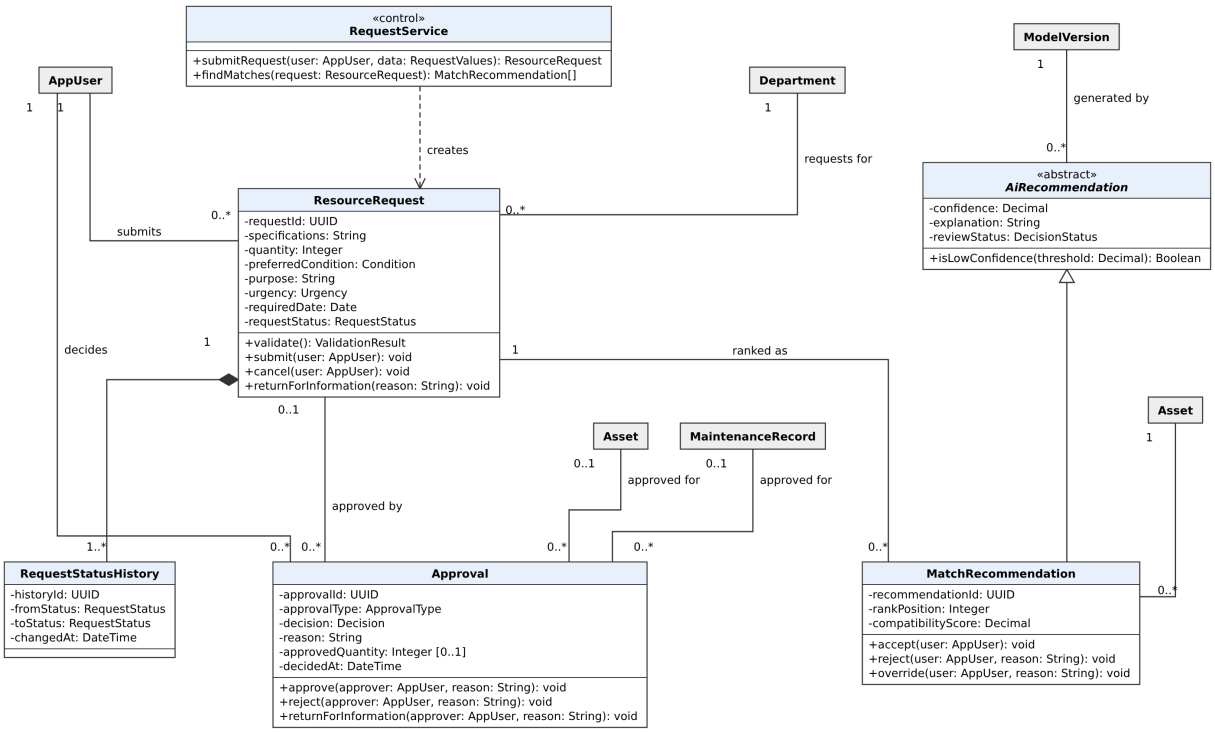


Figure 8.2c: Requests, AI matching and approval. An Approval may belong to a request, an asset or a maintenance record, so the request end is optional (0..1). ResourceRequest is composed of its status history; MatchRecommendation specializes the abstract AiRecommendation.

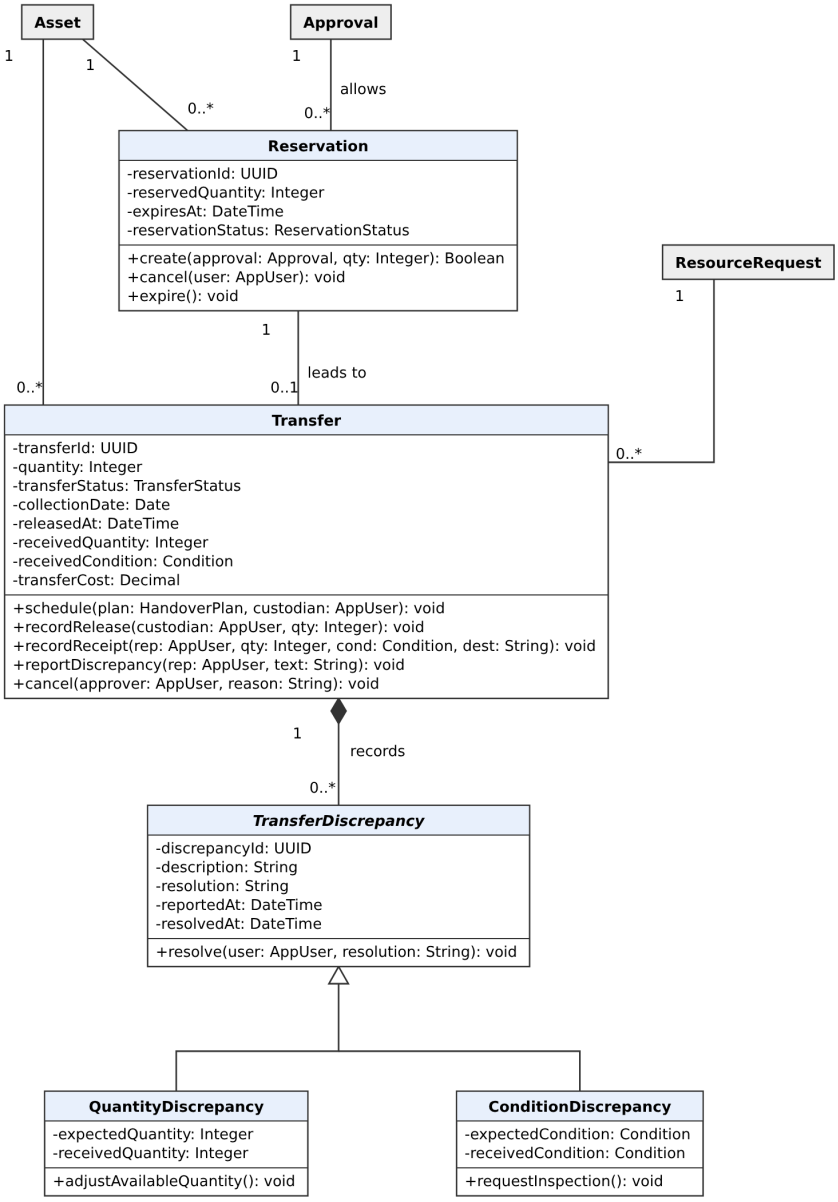


Figure 8.2d: Reservation, transfer and receipt discrepancies. A reservation leads to zero or one transfer. Transfer is composed of its discrepancies, which are specialized into quantity and condition discrepancies.

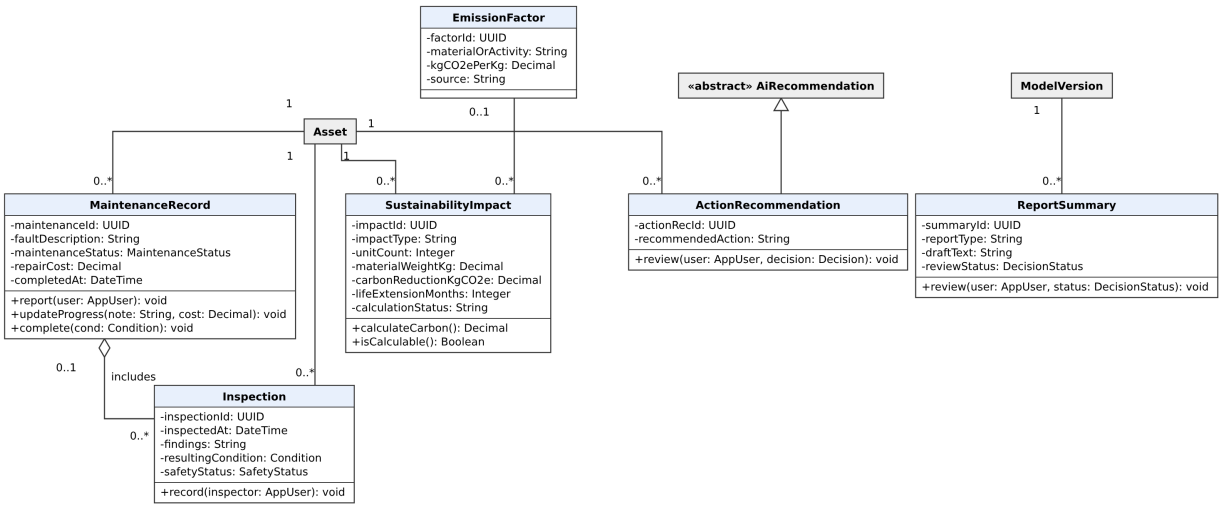


Figure 8.2e: Maintenance, sustainability and AI reporting. MaintenanceRecord aggregates inspections; ActionRecommendation specializes AiRecommendation.

Table 8.2: Whole–part and inheritance relationships in each view

View	Aggregation / composition	Inheritance	Why it is correct
8.2a	Aggregation: Department ◇— AppUser (1 to 0..*)	Generalization: AiAuditEntry ▷ AuditLog	A department groups its users, but deactivating a department does not delete user accounts; users can move between departments. An AI audit entry is an audit entry that also records the AI decision, the override reason and the model version (the model_version_id column of audit_log is filled only for these rows).
8.2b	Composition: Asset ◆— AssetHistoryEvent and Asset ◆— AssetAttachment (1 to 0..*). Aggregation: AssetCategory ◇— sub-category (0..1 to 0..*)	Generalization: PhotoAttachment, DocumentAttachment ▷ AssetAttachment (abstract)	History events and attachments have no meaning without their asset. A category groups sub-categories that exist on their own (parent_category_id in asset_category). Photos may be sent to the AI service and documents may not, so isSentToAi() is abstract and each subclass answers it; fileType tells them apart.
8.2c	Composition: ResourceRequest ◆— RequestStatusHistory (1 to 1..*)	Generalization: MatchRecommendation ▷ AiRecommendation (abstract)	A status history entry belongs to exactly one request. AiRecommendation holds what every AI recommendation shares (confidence, explanation, review status, model version) so these fields are defined once; MatchRecommendation adds rank and compatibility score.

View	Aggregation / composition	Inheritance	Why it is correct
8.2d	Composition: Transfer ◆— TransferDiscrepancy (1 to 0..*)	Generalization: QuantityDiscrepancy, ConditionDiscrepancy ▷ TransferDiscrepancy (abstract)	A discrepancy is recorded only as part of one transfer. A receipt can differ in quantity (available quantity must be corrected) or in condition (an inspection is requested); both share description and resolution. Stored in transfer_discrepancy with a type value.
8.2e	Aggregation: MaintenanceRecord ◇— Inspection (0..1 to 0..*)	Generalization: ActionRecommendation ▷ AiRecommendation (abstract, shown in 8.2c)	A maintenance job groups the inspections made for it, but routine inspections also exist without one (maintenance_id is optional). ActionRecommendation inherits confidence, explanation, review status and the model version link, and adds the recommended action.

Notation: a hollow diamond (◇) marks aggregation and a filled diamond (◆) marks composition, both drawn at the whole; a hollow triangle (▷) points from the subclass to the superclass. Abstract classes are shown in italics. All other classes, attributes, operations and associations are unchanged from the previous version.

Behaviour covered by operations:

- Submitting requests: ResourceRequest.validate() and submit(user); RequestService.submitRequest(user, data).
- Approving actions: Approval.approve(), reject() and returnForInformation(), each with the approver and a reason; Reservation.create(approval, qty) only after an approval exists.
- Release and receipt are separate: Transfer.recordRelease(custodian, qty) is called by the sending Asset Custodian; Transfer.recordReceipt(rep, qty, cond, dest) is called by the receiving Department Representative; Transfer.reportDiscrepancy() records a mismatch.
- AI use: AssetService.requestClassification(asset) calls AiServiceAdapter.classify(); ExternalAiAdapter is the only class that talks to the External AI Service, so the provider can change without changing business classes. Every AI result is stored with a ModelVersion.
- Audit: AuditLog.record(actor, action, record, before, after) is append-only; there is no update or delete operation.

Names used here (AssetService, createDraft, requestClassification, saveConfirmedAsset, AiServiceAdapter, classify, recordReceipt, AuditLog.record) match the sequence diagram in 8.3.1. The interface contracts in 8.10 should reuse AiServiceAdapter.classify, match and generateText.

8.3 Sequence Diagrams

8.3.1 Register Asset → AI Classification → Review → Save

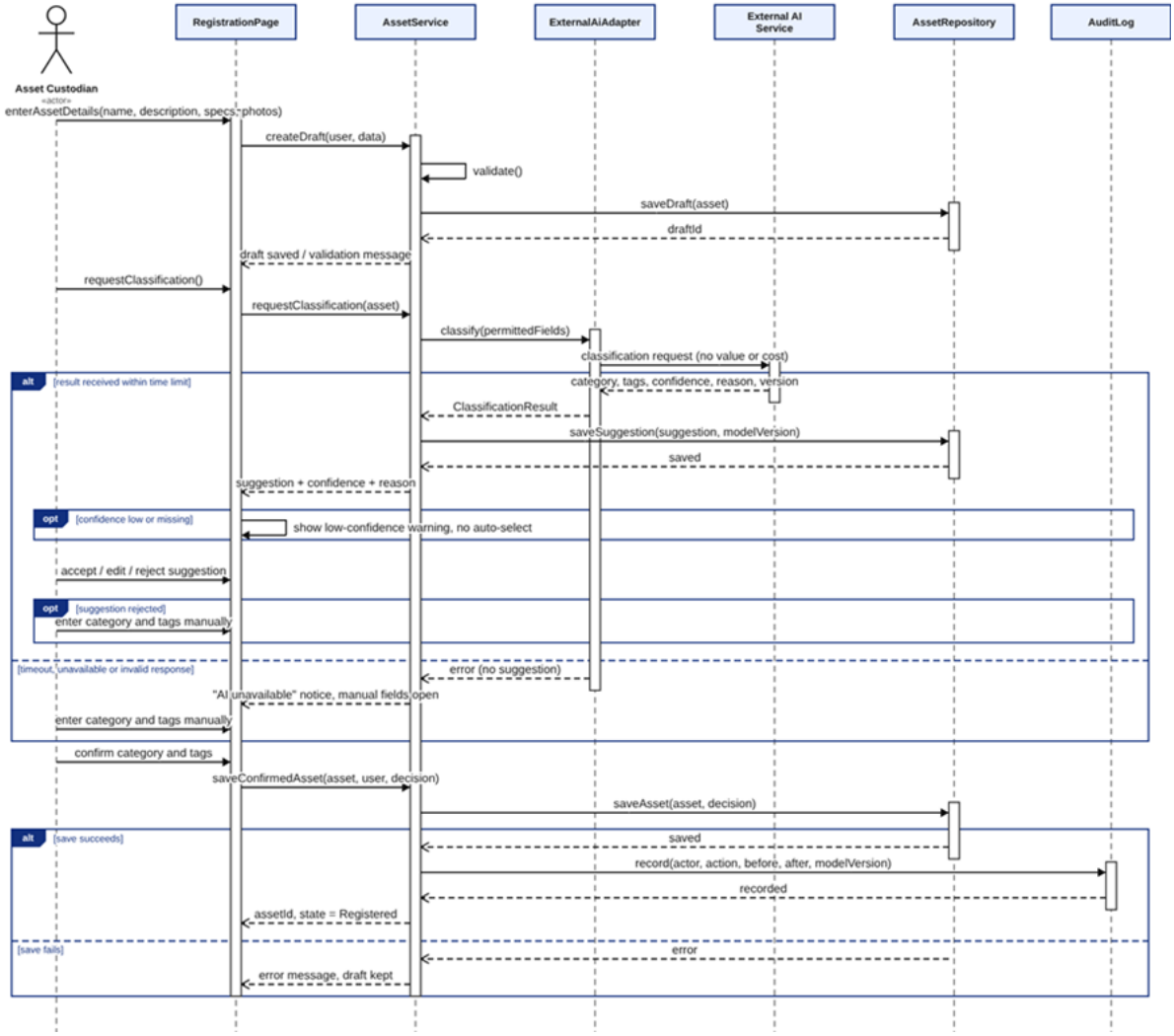


Figure 8.3.1: Register asset and review AI classification (UC-02 and UC-11). Solid arrows are calls, dashed arrows are replies; opt = only if confidence is low; alt = one branch runs.

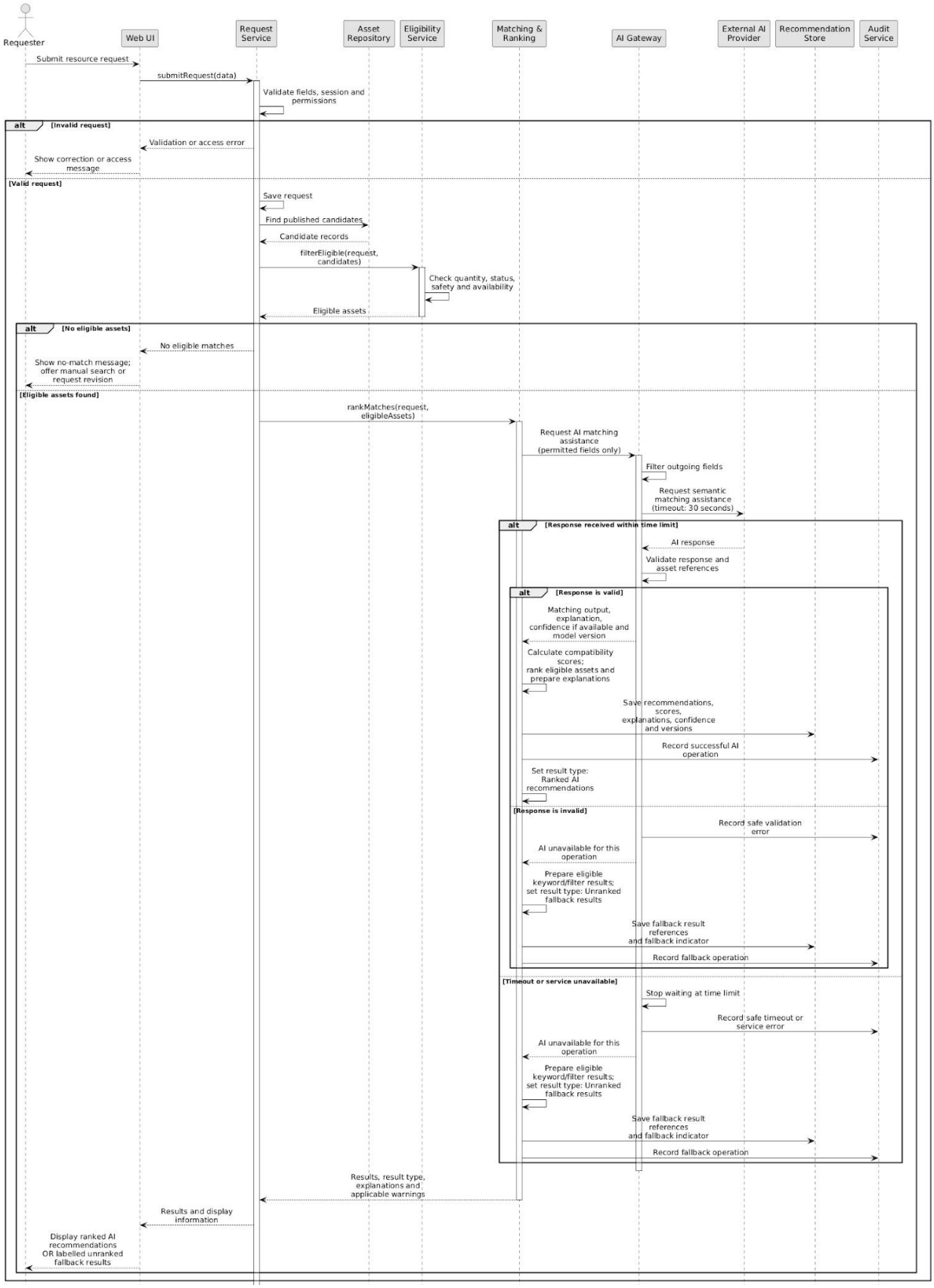
How to read the diagram:

- The Asset Custodian enters details; AssetService validates them and saves a draft, so work is not lost if AI fails.
- AssetService sends only permitted fields through ExternalAiAdapter. The adapter applies the time limit and retry rule defined in 8.10; if the limit is passed or the response is invalid, the failure branch runs.

- Success branch: the result (category, tags, confidence, reason, model version) is saved as a ClassificationSuggestion and shown. If confidence is below the threshold defined in Section 4, a warning is shown and nothing is pre-selected.
- Failure branch: an “AI unavailable” notice appears and the custodian enters the category and tags manually. All other functions continue.
- The custodian always decides: accept, edit or reject the suggestion (solid arrow). If the suggestion is rejected, the custodian enters the category and tags manually. In every case the custodian then confirms the values before saving; nothing is saved from the AI automatically. saveConfirmedAsset stores the asset and the decision; AuditLog.record stores who, what, when, before and after values, and the model version. If saving fails, the draft is kept and an error is shown.

8.3.2 Matching and Ranking Sequence Diagram

This sequence begins when an authorized Requester submits a complete resource request. The system first saves and validates the request, then removes ineligible assets before the matching service is called. The sequence includes invalid input, no-match, low/unknown confidence and AI-failure paths. A human decision is stored before the selected item enters the normal reservation or approval process.


Requester
Web UI
Request Service
Asset Repository
Eligibility Service
Matching & Ranking
AI Gateway
External AI Provider
Recommendation Store
Audit Service
Submit resource request
submitRequest(data)
Validate fields, session and permissions
alt [Invalid request]
Validation or access error
Show correction or access message
[Valid request]
Save request
Find published candidates
Candidate records
filterEligible(request, candidates)
Check quantity, status, safety and availability
Eligible assets
alt [No eligible assets]
No eligible matches
Show no-match message; offer manual search or request revision
[Eligible assets found]
rankMatches(request, eligibleAssets)
Request AI matching assistance (permitted fields only)
Filter outgoing fields
Request semantic matching assistance (timeout: 30 seconds)
alt [Response received within time limit]
AI response
Validate response and asset references
alt [Response is valid]
Matching output, explanation, confidence if available and model version
Calculate compatibility scores; rank eligible assets and prepare explanations
Save recommendations, scores, explanations, confidence and versions
Record successful AI operation
Set result type: Ranked AI recommendations
[Response is invalid]
Record safe validation error
AI unavailable for this operation
Prepare eligible keyword/filter results; set result type: Unranked fallback results
Save fallback result references and fallback indicator
Record fallback operation
[Timeout or service unavailable]
Stop waiting at time limit
Record safe timeout or service error
AI unavailable for this operation
Prepare eligible keyword/filter results; set result type: Unranked fallback results
Save fallback result references and fallback indicator
Record fallback operation
Results, result type, explanations and applicable warnings
Results and display information
Display ranked AI recommendations OR labelled unranked fallback results

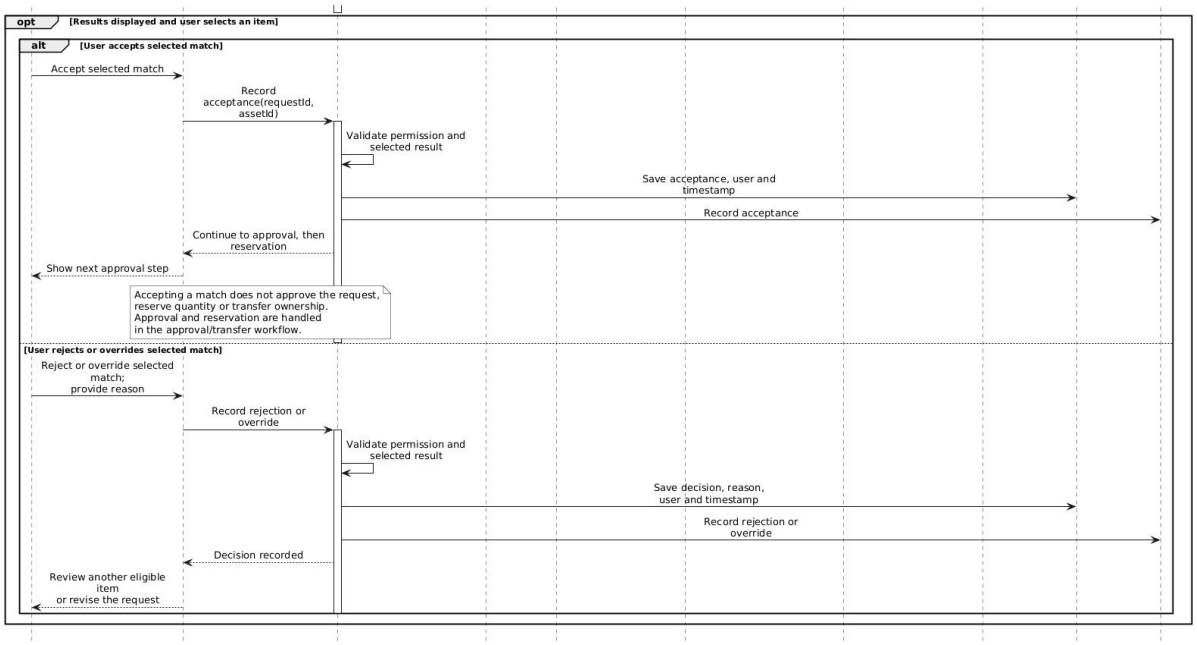


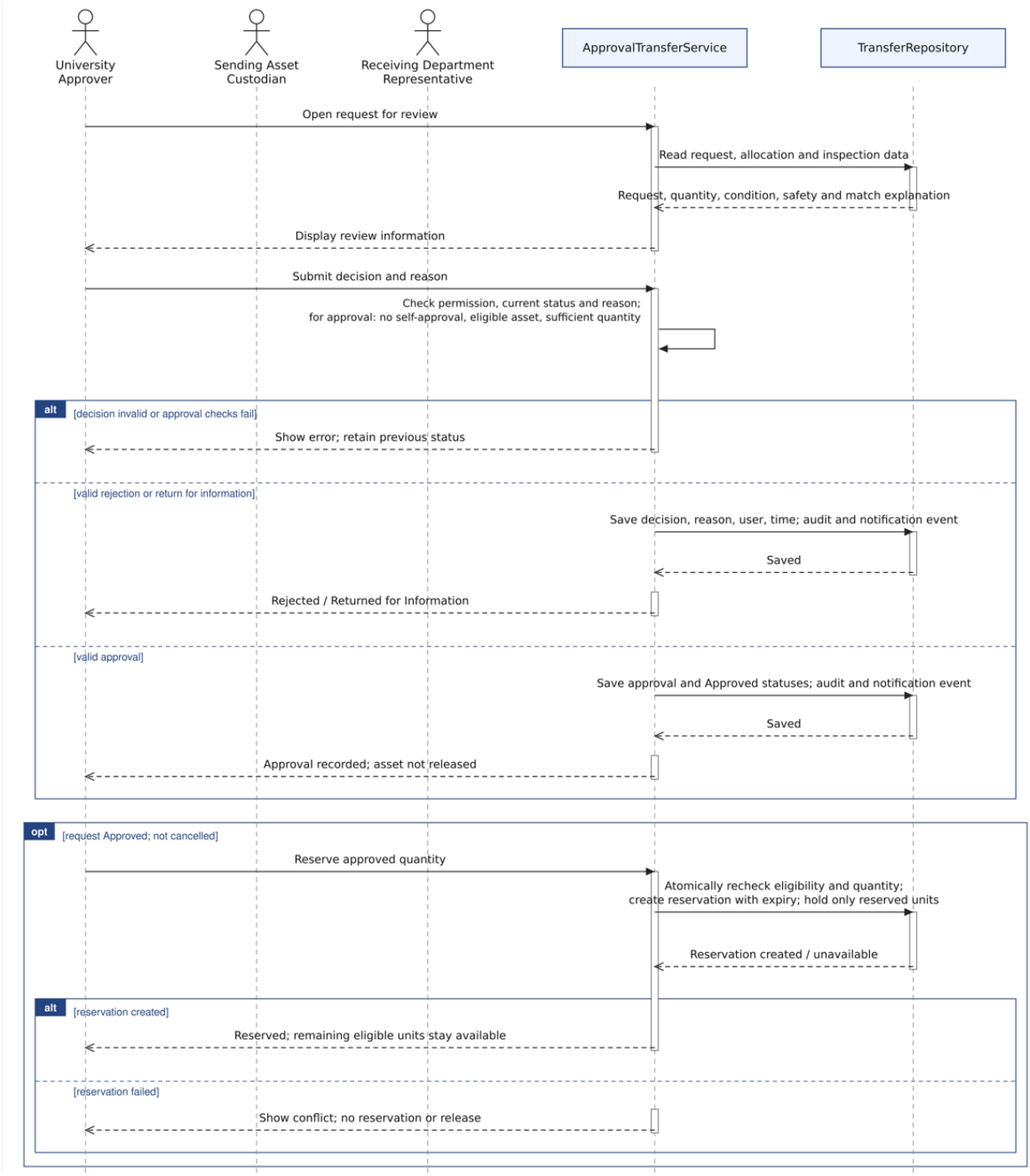
Figure 8.3.2

Submit Resource Request, Match, Rank and Review Recommendations

Source: *UC-12; FR-14–FR-16, FR-41–FR-48, FR-52–FR-53; NFR-05, NFR-AI03–NFR-AI10 and NFR-SE02–NFR-SE03.*

8.3.3 Approval and Transfer Sequence Diagram

This sequence begins after a request has been submitted with a proposed asset allocation. It shows approval, reservation, release and receipt confirmation, including failed checks and receipt discrepancies. Cancellation and reservation-expiry rules are described in Section 8.5.
Related requirements: FR-19–FR-26, FR-33 and FR-34.


University Approver
Sending Asset Custodian
Receiving Department Representative
ApprovalTransferService
TransferRepository
Open request for review
Read request, allocation and inspection data
Request, quantity, condition, safety and match explanation
Display review information
Submit decision and reason
Check permission, current status and reason;
for approval: no self-approval, eligible asset, sufficient quantity
alt
[decision invalid or approval checks fail]
Show error; retain previous status
[valid rejection or return for information]
Save decision, reason, user, time; audit and notification event
Saved
Rejected / Returned for Information
[valid approval]
Save approval and Approved statuses; audit and notification event
Saved
Approval recorded; asset not released
opt
[request Approved; not cancelled]
Reserve approved quantity
Atomically recheck eligibility and quantity;
create reservation with expiry; hold only reserved units
Reservation created / unavailable
alt
[reservation created]
Reserved; remaining eligible units stay available
[reservation failed]
Show conflict; no reservation or release

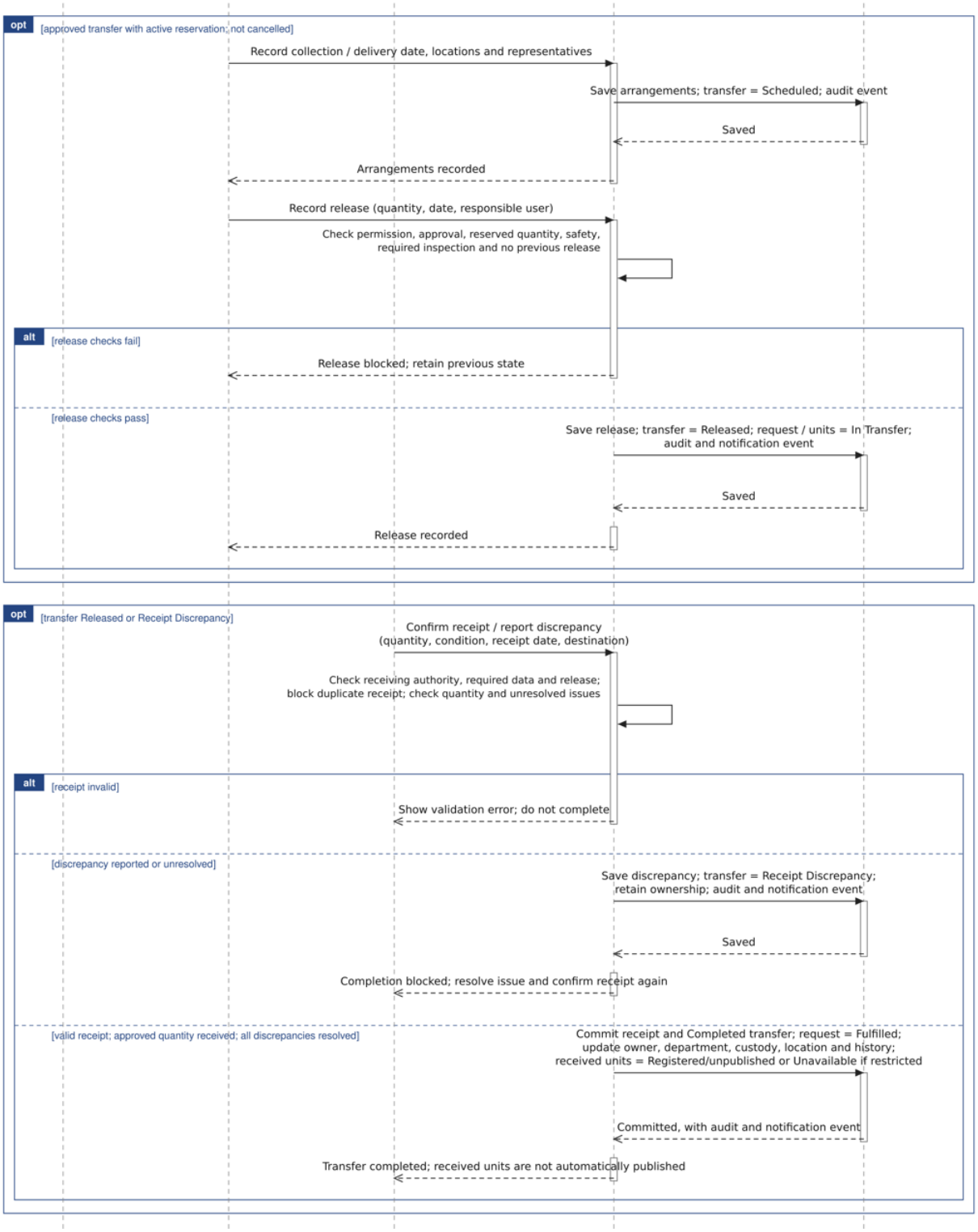


Figure 8.3.3

Submit Requests,Approvals, and Transfers

Precondition: a submitted request has a proposed asset allocation and user is authenticated.

Repository writes are transactional: a failed write rolls back, reports an error and retains the last valid state.
Solid arrows: calls, dashed arrows = replies. Service checks apply on every protected action. AI cannot authorize transfers

8.4 Activity Workflow

The exchange process starts by submitting a request. Eligible matches will appear for review or an alternative search can be done manually if AI is not available. The University Approver will review the proposed allocation and approve, decline or ask for more information. Once approved, the request can be reserved and the system will verify that items are still available.

The Asset Custodian (sending) will record the transfer details. They will release the asset once it has been approved, the quantity has been confirmed and safety checks passed. The Department Representative (receiving) will confirm they have received the item or report any discrepancies. Once they have received a valid confirmation and resolved any discrepancies the ownership, custodian, department, location and history can be updated. If the item is restricted then the units will remain unpublished/unavailable. See Sections 5.1 and 8.3.3 for examples of responsibilities and interactions during the exchange process. The rules around the lifecycle of the asset are described in Section 8.5.

8.5 Life-Cycle (State) Descriptions

8.5.1 Asset Lifecycle

An asset begins as **Draft** and becomes **Registered** when its record is saved with a unique identifier. Registered assets remain unpublished until an authorized user publishes them as **Available**.

Approved requests may reserve eligible asset quantities. Only the reserved units are held; any remaining eligible units remain available. Release moves the allocated units to **In Transfer**. After confirmed receipt, the receiving units become **Registered/unpublished**, or **Unavailable** if restrictions apply. Receipt does not automatically publish them for another exchange.

Assets requiring assessment or repair enter **Under Inspection** or **Under Maintenance**. Return to availability requires an acceptable condition, any required safety inspection and custodian confirmation. Assets that cannot be repaired become **Unrepairable**. Retirement, donation, recycling and disposal require authorized approval, and their records remain available for history and auditing.

The permitted asset transitions follow **Phase 3 SRS Section 3.3**. Invalid transitions are rejected without changing the record.

8.5.2 Request Lifecycle

The request life-cycle is shown below. Once a request has been prepared and submitted, it will go into review. If a request is returned then amendments need to be made and then the request re-submitted. When a request times out of reserved it will go back to approved so that availability can be re-checked. Any discrepancies in receipt will leave the request in Transfer until they have been resolved.

A request can only be cancelled in the stages indicated in the diagram above. This will also cancel any associated reservation. Requests which have been rejected cannot be reopened and a new request will need to be created.

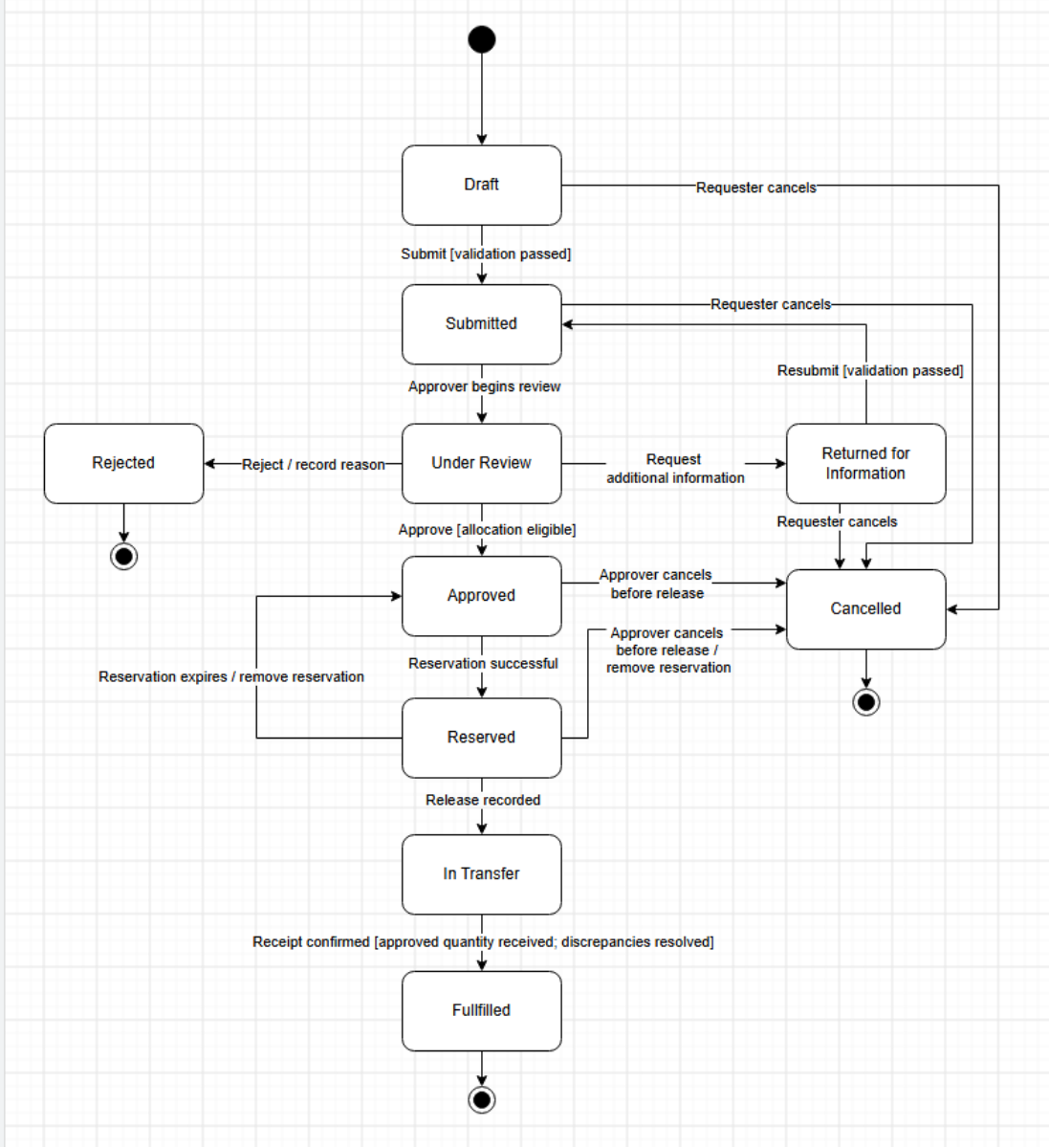


Figure 8.5.2: Request State Diagram

8.5.3 Approval Lifecycle

Below is a state diagram showing what happens when a request is approved, denied or asked for more info after being submitted. If a request is resubmitted it goes back into the review queue.

Only an authorised University Approver can approve/deny. You will not be allowed to approve yourself. In order to approve you must have an appropriate asset with enough stock available. Once approved/denied the reason for your decision, name of the approver and date/time of approval will be recorded. This does not mean that the asset has been booked, issued or received, just that the decision has been made to allow the allocation to go through.

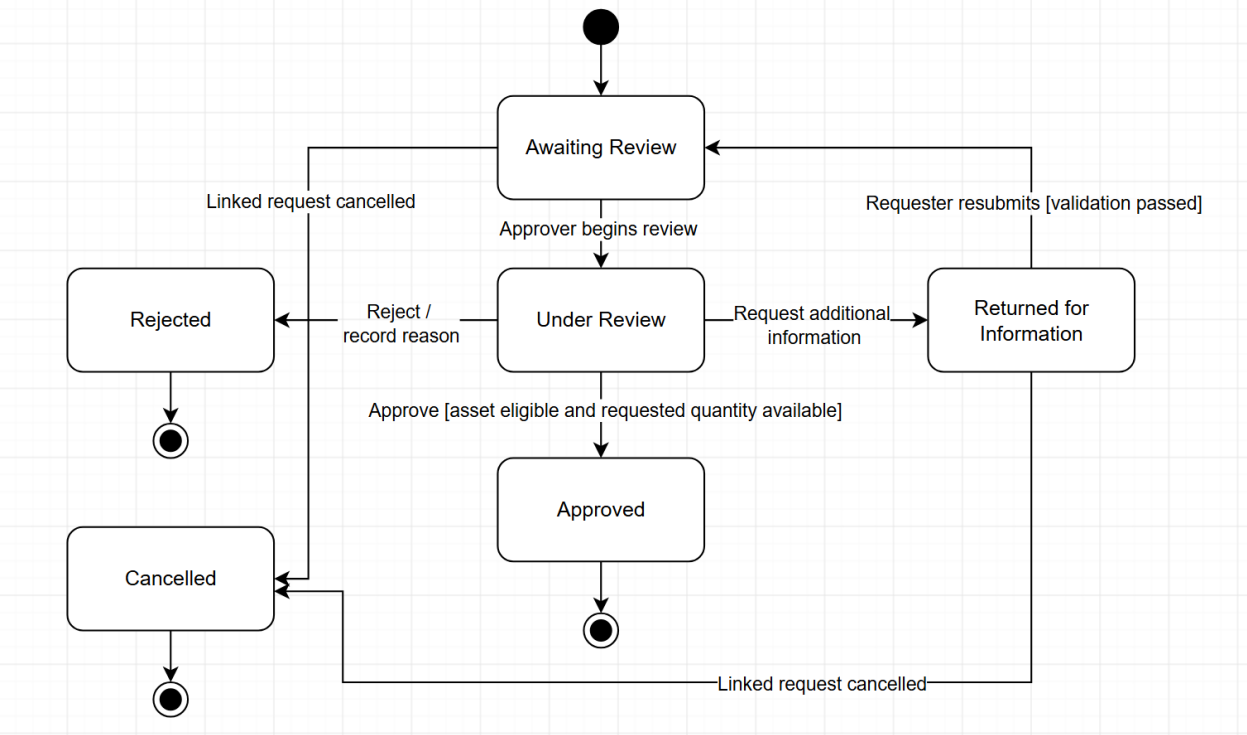


Figure 8.5.3: Approval State Diagram

8.5.4 Transfer Lifecycle

The following shows the state diagram of the transfer process. Once approved, the sending custodian will note any arrangements and release. The receiving Department Representative will confirm receipt or report discrepancies. The University Approver may cancel an approved or scheduled transfer before release, but once released any issues will follow the same process as confirmation and discrepancies.

Once complete (confirmed receipt, discrepancies resolved), the system will update the ownership, department, custody, location and asset history. Any assets received will be unpublished or restricted (as per section 8.5.1).

Related requirements: Phase 3 SRS Sections 3.2–3.5; FR-20–FR-26, FR-29, FR-32 and FR-34.

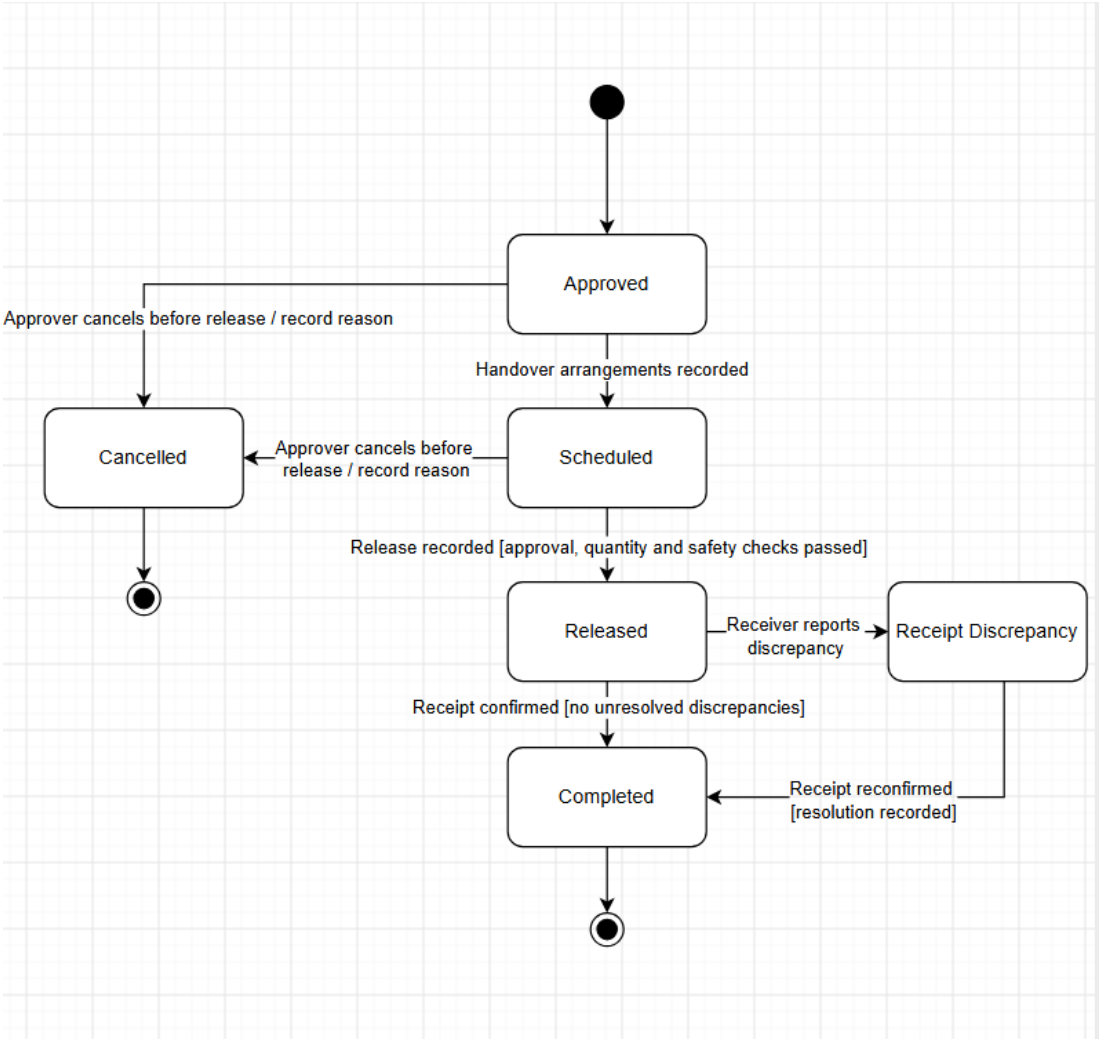


Figure 8.5.4: Transfer State Diagram

8.6 Component Diagram

The UML component diagram is Figure 3.1 in Section 3, so it is not repeated here. It shows the eight architecture areas, internal dependencies, the external AI boundary and the interfaces used by the main modules.

Table 8.6

Main Component Interfaces

Interface	Provider	Main Consumers	Purpose

IApplicationAPI	Application API	Web Screens	Provides secure JSON/HTTPS access to system functions.
IAssetManagement	Assets and Catalogue	Application API, Reporting, AI Gateway	Registers, updates, publishes and retrieves asset information.
IRequestManagement	Requests and Reservations	Application API, Approvals and Transfers, AI Gateway	Submits requests, checks eligible candidates and manages selections.
IApprovalTransfer	Approvals and Transfers	Application API, Requests and Reservations	Records approvals, asset releases, transfers and receipt confirmations.
IAIService	AI Gateway	Assets and Catalogue, Requests and Reservations, Maintenance and Inspection, Reporting	Provides controlled, provider-independent access to advisory AI functions.
IDataRepository	Repository Layer	Application modules, Reporting, AI monitoring	Provides controlled access to application records and file storage.
IAuditRecorder	Audit recording service	Protected application workflows, AI Gateway	Records append-only audit entries, user decisions, changes and AI-related actions.

8.7 Deployment Diagram

The prototype uses a browser client, a Vercel-hosted Next.js application, Supabase Auth/PostgreSQL/pgvector/Storage and an external AI provider. All AI calls originate from the server-side gateway. GitHub is not shown because it supports development and version control rather than runtime deployment.

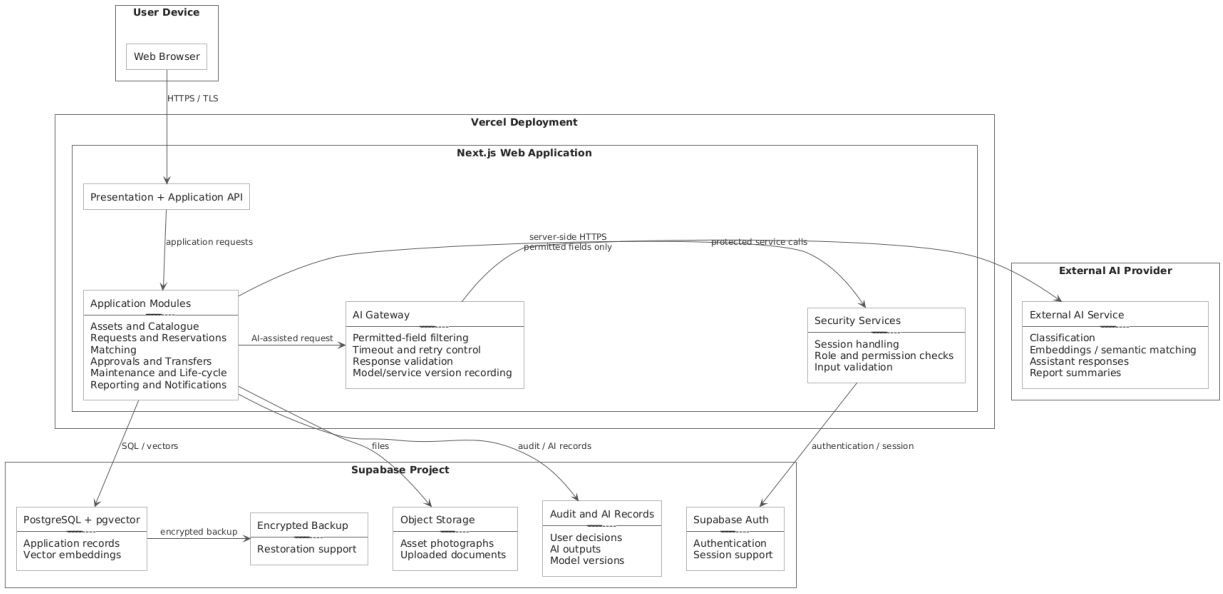


Figure 8.7
Deployment View of the Proposed Prototype

Source: *Deployment choices follow the Phase 2 feasibility study and Phase 3 security and scope constraints [1], [4].*

8.8 AI Recommendation Workflow

The workflow separates rule-based eligibility from AI-supported comparison and scoring. Only eligible assets reach the AI step. The human decision is recorded with the explanation and version, and unavailable or uncertain AI results return to manual tools.

Figure 8.8 — AI Recommendation Workflow

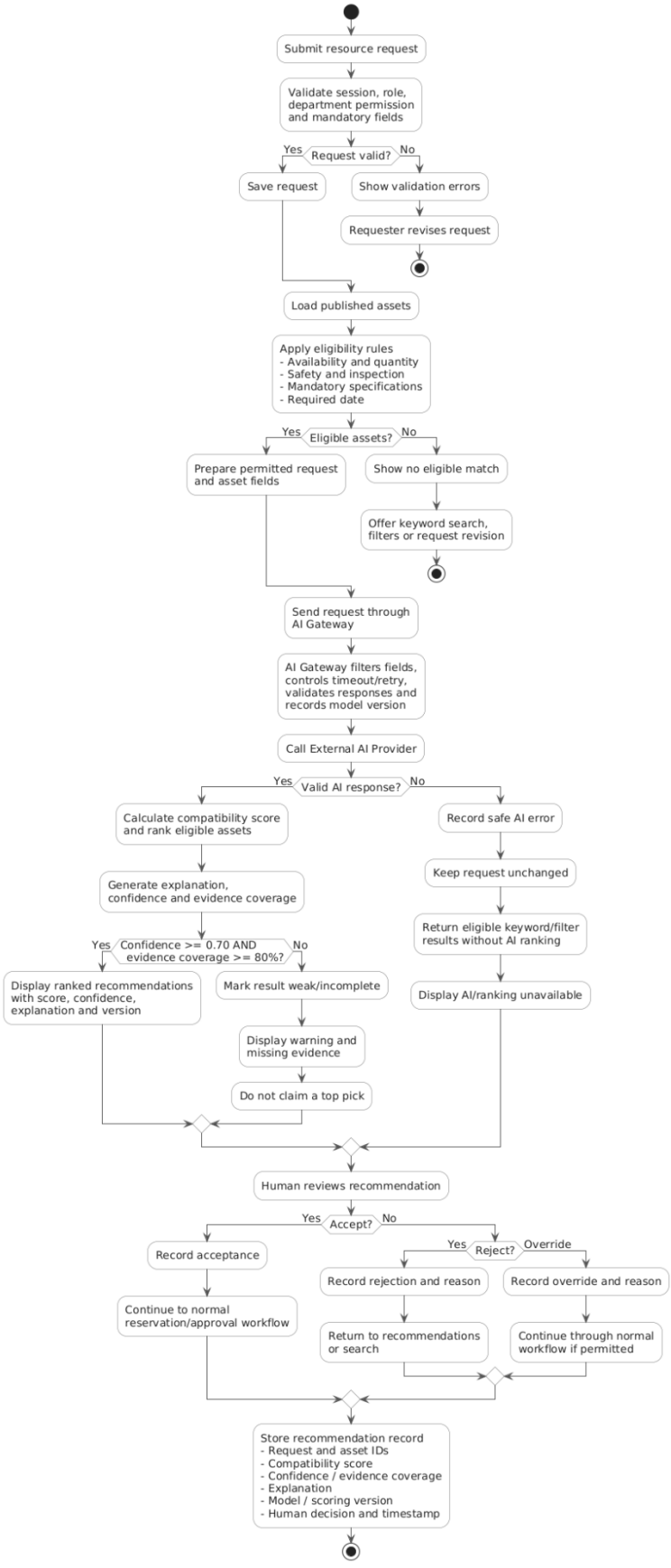


Figure 8.8

AI Matching, Ranking, Explanation and Human-Review Workflow

Source: *FR-41–FR-48, FR-52–FR-53; BR-AI-01, BR-AI-03–BR-AI-05 and BR-AI-09–BR-AI-10.*

8.9 Entity-Relationship Diagram

The database has 25 tables (PostgreSQL on Supabase, as chosen in Phase 2). It is shown in five views grouped by subject. Grey boxes are tables from another view (primary key only). PK = primary key, FK = foreign key (with target table), UQ = unique, (null) = optional. Every foreign key to app_user or department is many-to-one (1 : 0..*) and is listed in Table 8.9 rather than drawn, to avoid crossing lines. Cardinality symbols: tick = exactly one; circle + tick = zero or one; crow's foot with circle = zero or many.

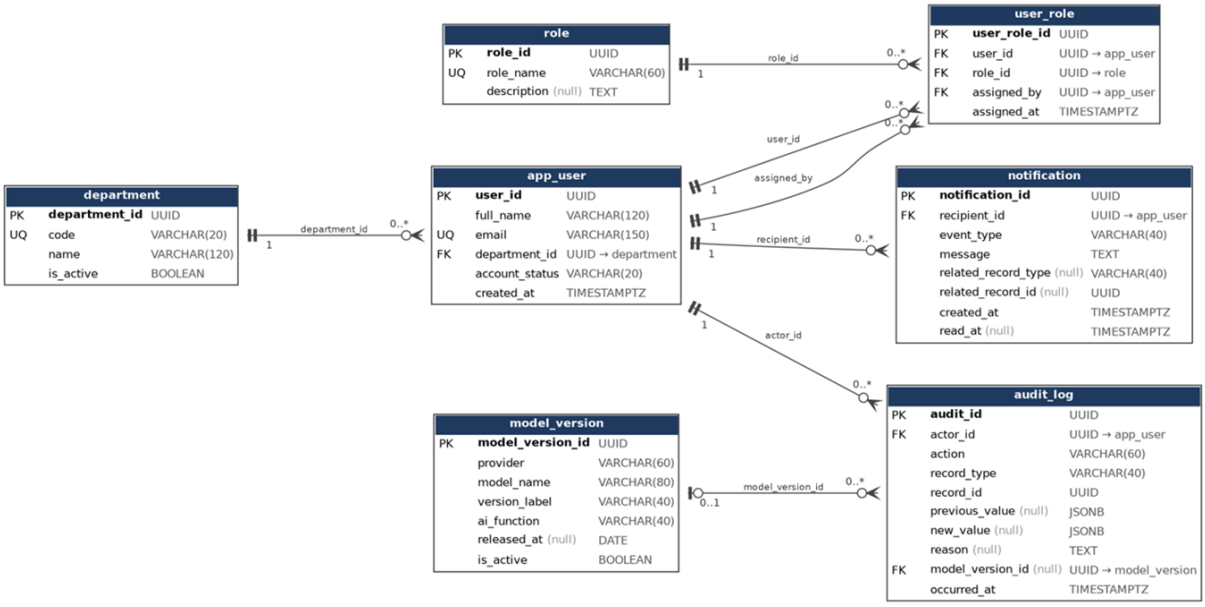


Figure 8.9a: Organisation, access, notification and audit.

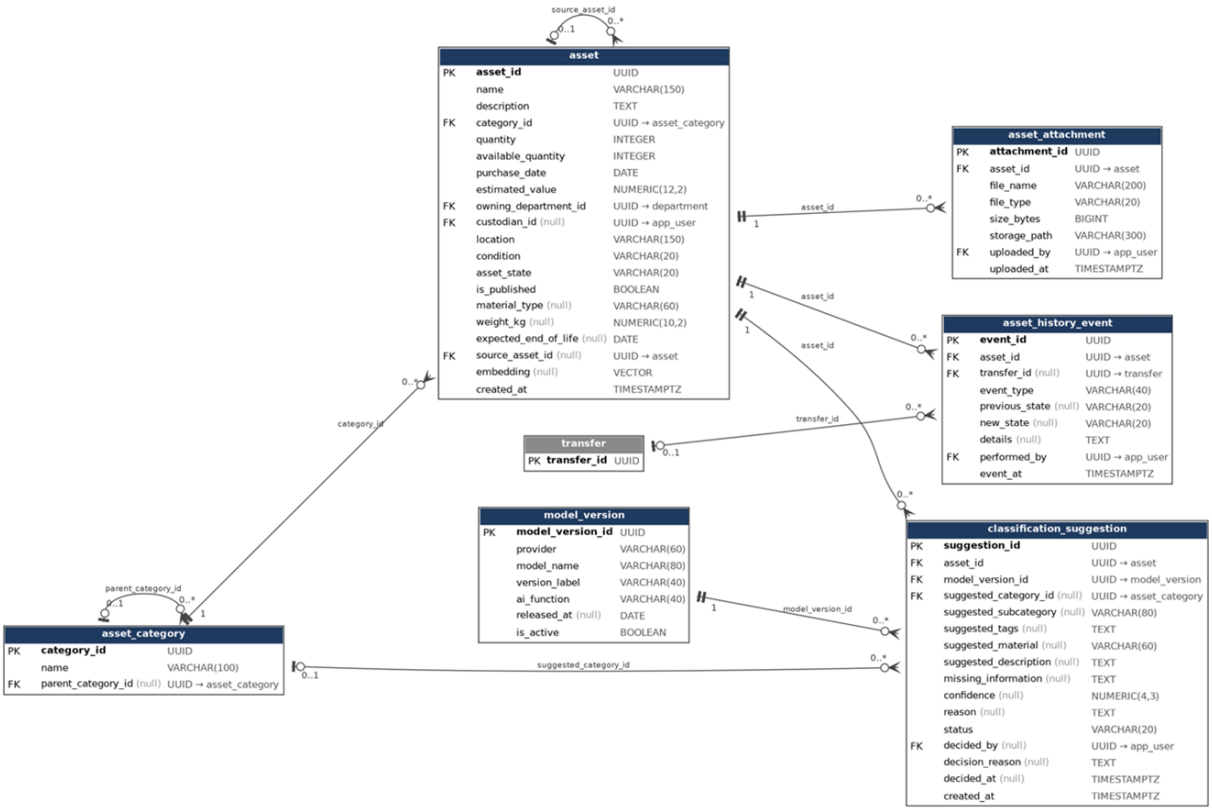


Figure 8.9b: Assets, attachments and AI classification.

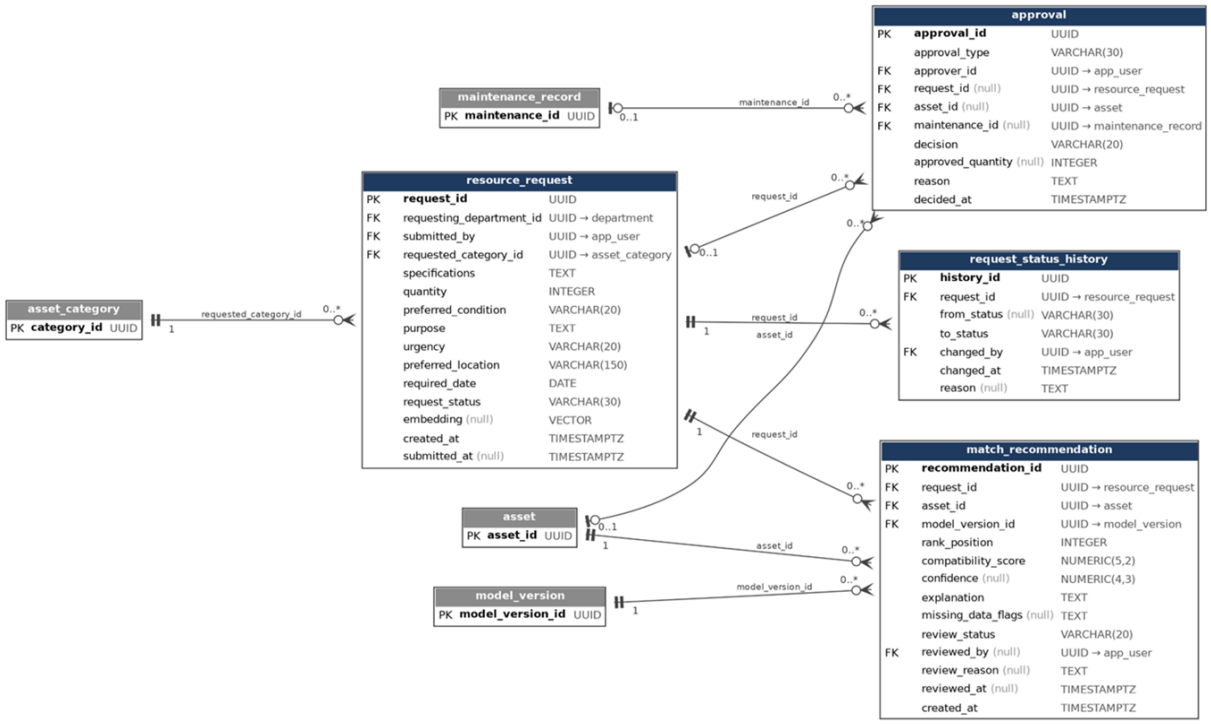


Figure 8.9c: Requests, AI matching (MatchRecommendation) and approval.

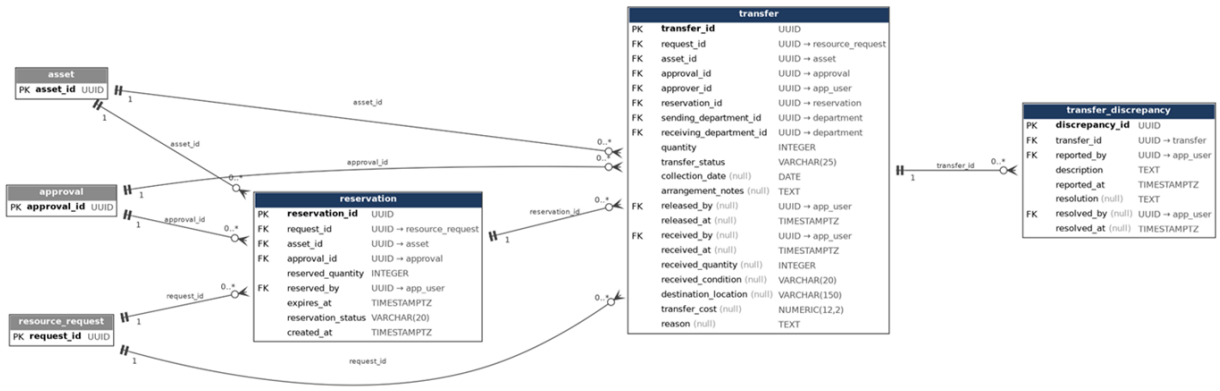


Figure 8.9d: Reservation, transfer and receipt discrepancies.

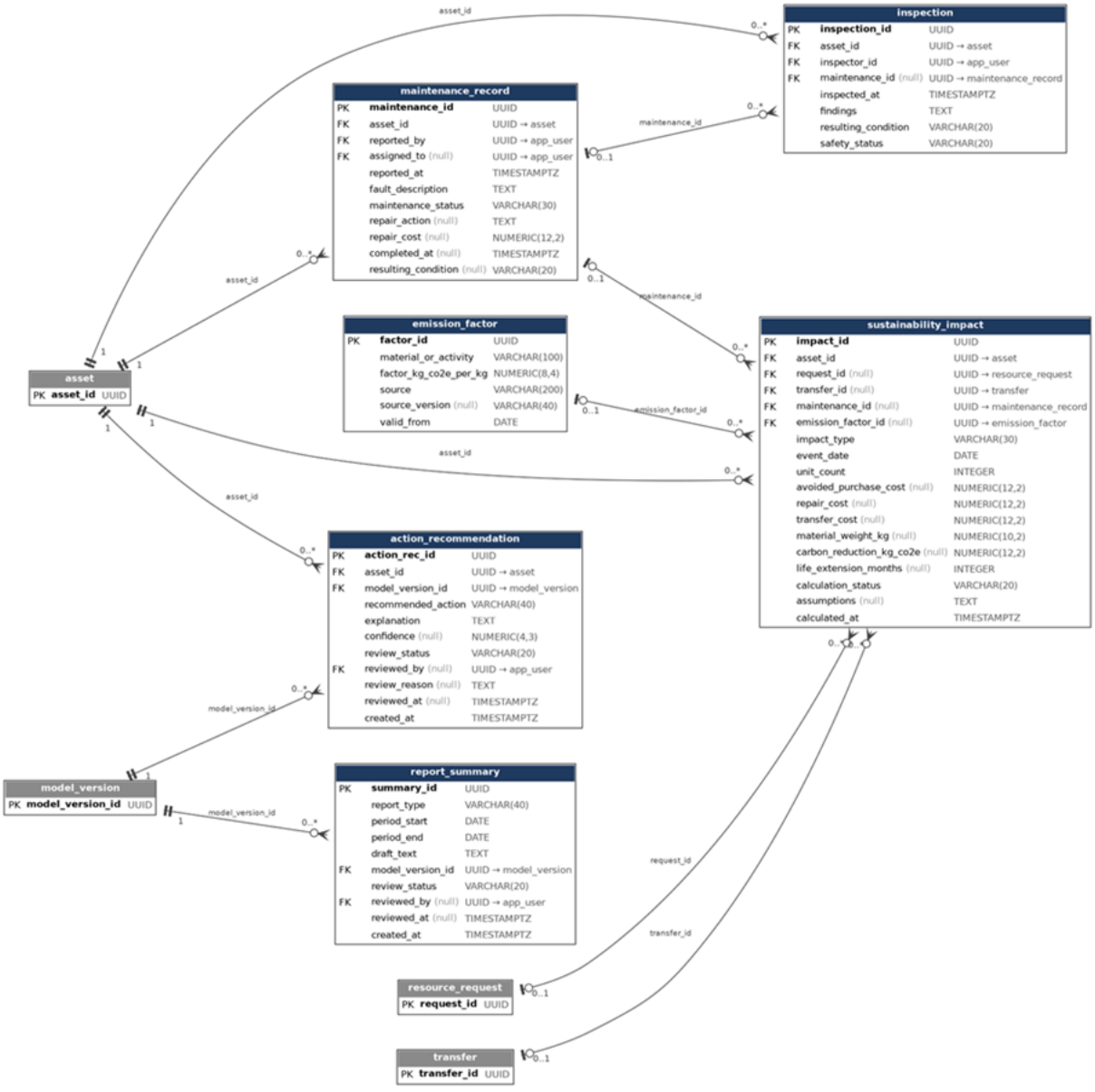


Figure 8.9e: Maintenance, sustainability and AI reporting.

Table 8.9a:

Required entities and where they are

Required item	Table(s)
Users	app_user, user_role
Roles	role
Departments	department
Assets	asset, asset_attachment, asset_history_event
Categories	asset_category
Requests	resource_request, request_status_history
Recommendations	match_recommendation (request–asset, with score, explanation, model version), classification_suggestion, action_recommendation
Reservations	reservation
Approvals	approval
Transfers	transfer, transfer_discrepancy
Inspections	inspection
Maintenance records	maintenance_record
Sustainability impacts	sustainability_impact, emission_factor
Notifications	notification
Model versions	model_version
Audit logs	audit_log
Added from Phase 3	report_summary (AI summaries need human review)

Key design points:

- MatchRecommendation resolves the many-to-many relation between requests and assets. It stores compatibility_score and confidence as separate fields, the explanation, the model_version_id, the rank, and the human review result.
- transfer links to one asset (asset_id), its request (request_id), its approval (approval_id) and the approver (approver_id). It keeps quantity, received_quantity and received_condition, so partial and discrepant receipts are recorded (decision D3: a new asset row with source_asset_id is created for received units).
- estimated_value is per unit. weight_kg, material_type and expected_end_of_life are optional and support the carbon and asset-life calculations; sustainability_impact stores assumptions and calculation_status so missing data is marked, not guessed.

- embedding columns (pgvector) on asset and resource_request support semantic search (Phase 2, Section 3.4). Their size follows the embedding model chosen in Section 4.
- audit_log has no update or delete path in the application and records previous_value, new_value, reason and model_version_id.
- No passwords are stored in these tables; sign-in credentials are handled by the authentication service (Phase 2).

Table 8.9b:
Relationships and cardinalities

Parent table	Child table	FK column in child	Parent end	Child end
department	app_user	department_id	1	0..*
app_user	user_role	user_id	1	0..*
role	user_role	role_id	1	0..*
app_user	user_role	assigned_by	1	0..*
asset_category	asset_category	parent_category_id	0..1	0..*
asset_category	asset	category_id	1	0..*
department	asset	owning_department_id	1	0..*
app_user	asset	custodian_id	0..1	0..*
asset	asset	source_asset_id	0..1	0..*
asset	asset_attachment	asset_id	1	0..*
app_user	asset_attachment	uploaded_by	1	0..*
asset	asset_history_event	asset_id	1	0..*
transfer	asset_history_event	transfer_id	0..1	0..*
app_user	asset_history_event	performed_by	1	0..*
asset	classification_suggestion	asset_id	1	0..*
model_version	classification_suggestion	model_version_id	1	0..*
asset_category	classification_suggestion	suggested_category_id	0..1	0..*
app_user	classification_suggestion	decided_by	0..1	0..*
department	resource_request	requesting_department_id	1	0..*
app_user	resource_request	submitted_by	1	0..*
asset_category	resource_request	requested_category_id	1	0..*
resource_request	request_status_history	request_id	1	0..*

app_user	request_status_history	changed_by	1	0..*
resource_request	match_recommendation	request_id	1	0..*
asset	match_recommendation	asset_id	1	0..*
model_version	match_recommendation	model_version_id	1	0..*
app_user	match_recommendation	reviewed_by	0..1	0..*
app_user	approval	approver_id	1	0..*
resource_request	approval	request_id	0..1	0..*
asset	approval	asset_id	0..1	0..*
maintenance_record	approval	maintenance_id	0..1	0..*
resource_request	reservation	request_id	1	0..*
asset	reservation	asset_id	1	0..*
approval	reservation	approval_id	1	0..*
app_user	reservation	reserved_by	1	0..*
resource_request	transfer	request_id	1	0..*
asset	transfer	asset_id	1	0..*
approval	transfer	approval_id	1	0..*
app_user	transfer	approver_id	1	0..*
reservation	transfer	reservation_id	1	0..*
department	transfer	sending_department_id	1	0..*
department	transfer	receiving_department_id	1	0..*
app_user	transfer	released_by	0..1	0..*
app_user	transfer	received_by	0..1	0..*
transfer	transfer_discrepancy	transfer_id	1	0..*
app_user	transfer_discrepancy	reported_by	1	0..*
app_user	transfer_discrepancy	resolved_by	0..1	0..*
asset	maintenance_record	asset_id	1	0..*
app_user	maintenance_record	reported_by	1	0..*
app_user	maintenance_record	assigned_to	0..1	0..*
asset	inspection	asset_id	1	0..*

app_user	inspection	inspector_id	1	0..*
maintenance_record	inspection	maintenance_id	0..1	0..*
asset	sustainability_impact	asset_id	1	0..*
resource_request	sustainability_impact	request_id	0..1	0..*
transfer	sustainability_impact	transfer_id	0..1	0..*
maintenance_record	sustainability_impact	maintenance_id	0..1	0..*
emission_factor	sustainability_impact	emission_factor_id	0..1	0..*
asset	action_recommendation	asset_id	1	0..*
model_version	action_recommendation	model_version_id	1	0..*
app_user	action_recommendation	reviewed_by	0..1	0..*
model_version	report_summary	model_version_id	1	0..*
app_user	report_summary	reviewed_by	0..1	0..*
app_user	notification	recipient_id	1	0..*
app_user	audit_log	actor_id	1	0..*
model_version	audit_log	model_version_id	0..1	0..*

8.10 Interface Contracts

All AI calls include a trace ID, operation name, caller identity, permitted record references and schema version. Responses return the same trace ID, status, output or safe error code, model/service version and timing information. Provider-specific stack traces and credentials are never returned to the user.

Table 8.10

Application-to-AI Interface Contracts

Operation	Caller → receiver	Inputs / outputs	Rules	Errors and fallback

suggestAssetClassification	Asset module → AI Gateway → Classification Service	Asset draft fields → category, tags, material, description, missing fields, confidence/version	Custodian update access; classification allow-list; response schema check	VALIDATION_ERROR, FORBIDDEN, AI_TIMEOUT, INVALID_RESPONSE → manual fields
buildAssetRepresentation	Asset module / background worker → AI Gateway → Semantic Matching Service	Published text/specifications → representation ID/version/time	Asset ready for publication; exclude identity/unrelated cost	Failure marks representation unavailable; keyword search continues; one queued retry
findSemanticCandidates	Matching coordinator → AI Gateway → Semantic Matching Service	Request profile + eligible representation IDs → candidate IDs, similarity/evidence/version	Request access; visibility and basic eligibility checked first	NO_CANDIDATES, LOW_CONFIDENCE, AI_TIMEOUT → rule/keyword list
rankMatches	Matching coordinator → AI Gateway → Ranking Service	Eligible candidates + factor data → order, score, contributions, coverage, explanation	Recheck quantity/status/safety/specifications; no unrelated personal data	NO_ELIGIBLE_ASSETS, STALE_ASSET, SCORING_ERROR → unranked eligible results

recommendSustainableAction	Life-cycle module → AI Gateway → Sustainable-Action Recommender	Condition/inspection/history/cost/material → ordered actions/reasons/approvals	Life-cycle permission; missing safety check blocks action	INSPECTION_REQUIRED, MISSING_DATA, AI_TIMEOUT → human checklist
answerAssistantQuestion	Assistant module → AI Gateway → LLM Assistant	Role/question/authorized references → answer, sources, clarification/confirmation flag	Check access before retrieval/display; cannot approve; confirmation before submission	FORBIDDEN, UNVERIFIED_ANSWER, AI_TIMEOUT → help/search/forms
draftReportSummary	Reporting module → AI Gateway → Report Summary Generator	Verified metrics/units/period/assumptions → draft + metric references/version	Report access; every stated value must match standard report	REPORT_INCOMPLETE, VALUE_MISMATCH, AI_TIMEOUT → standard report only

8.11 Compatibility Scoring and Ranking

8.11.1 Eligibility Before Scoring

An asset is removed before scoring if it is not published and available, has insufficient quantity, has a safety restriction, requires a missing inspection, fails a mandatory specification or cannot meet the required date. This step is rule based. AI cannot restore an excluded asset.

8.11.2 Compatibility Factors

Table 8.11

Proposed Prototype Scoring Factors

Factor	Weight	Meaning
Required and preferred specifications	35%	Degree to which the recorded specifications satisfy the request; mandatory failure is already excluded.
Intended-purpose similarity	25%	Meaning similarity between the request purpose and the approved asset description/use.
Preferred condition	10%	Suitability of the recorded condition.
Required date and urgency	10%	Availability by the required date.
Location and transfer effort	10%	Relative campus distance or approved transfer effort.
Financial effect	5%	Relative transfer and repair cost among eligible candidates.
Environmental effect	5%	Available approved reuse, repair or diversion evidence.

$$\textbf{Compatibility score} = 100 \times [\Sigma(w_i \times s_i) \div \Sigma(w_i \text{ for factors with available evidence})]$$

$$\textbf{Evidence coverage} = 100 \times [\Sigma(\text{weights with available evidence}) \div \Sigma(\text{all weights})]$$

Mandatory data must be present before scoring. Missing optional information is removed from the denominator and displayed; it is not silently treated as zero. Evidence coverage below 80% is labelled incomplete and prevents a top-pick claim. Ties are resolved by lower estimated transfer/repair cost, then asset ID for a repeatable order.

8.11.3 Illustrative Calculation

Factor	Weight	Candidate value	Contribution
Specifications	0.35	0.90	0.315
Intended purpose	0.25	0.84	0.210
Condition	0.10	0.80	0.080
Required date/urgency	0.10	1.00	0.100
Location/transfer effort	0.10	0.70	0.070
Financial effect	0.05	0.60	0.030
Environmental effect	0.05	0.80	0.040
Total	1.00	—	0.845

The compatibility score is 0.845 × 100 = 84.5/100 and evidence coverage is 100%. If the provider reports confidence of 0.78, the interface shows Compatibility: 84.5/100 and AI confidence: 78% as separate values. This is an illustrative design example, not a measured university result.

8.11.4 Compatibility, Confidence and Explanation

Measure	Meaning	Use
Compatibility score	How well an eligible asset fits the request under the versioned factor rules.	Ranks eligible candidates.
AI confidence	How certain the provider reports that it is about its semantic/generated output.	Displays a warning below 0.70; never approves a request.
Evidence coverage	How much of the factor weight is supported by recorded data.	Labels a result incomplete below 80%.

When confidence is not meaningful or not returned, the database stores null and the screen displays Confidence not supplied. Every recommendation stores the request/asset IDs, rank, factor contributions, warnings, model/scoring version, time and the user's acceptance, rejection, edit or override. A rejection or override requires a reason.

8.12 Sustainability Calculations

Financial and sustainability figures are calculated based on the data captured. The AI may summarise the figures but will not calculate them. Each figure will include the unit of measurement, inputs, source, calculation method and assumptions.

Table 8.12:
Sustainability Calculations

Measure	Calculation and counting rule	Unit
Reused assets	Count the number of unique asset units that were successfully reused via a completed transfer. Excludes incomplete, cancelled and unresolved transfers.	Asset units
Repaired assets	Count the number of unique asset units that have had a completed maintenance log showing they were repaired.	Asset units
Assets returned to service	Count the number of unique asset units that were repaired or refurbished and had an acceptable inspection result and were confirmed by the custodian to be back in service. Completed repairs are not enough. Asset Units	Asset units

Waste diversion	Count the number of unique asset units that had an outcome logged to divert it from being disposed of. Sum up the total weight of these items. Excludes normal disposal. Not all transfers will be waste diversion.	Asset units; kg
Estimated asset-life extension	Revised expected end-of-life date after reuse, repair or refurbishment minus the previously recorded expected end-of-life date. Convert the difference to months using days ÷ 30.44. Report per asset.	Months
Avoided purchase costs	Sum documented replacement unit costs × quantities for completed reuse events confirmed to replace a planned purchase.	AED
Net reuse savings	`Avoided purchase costs − associated repair costs − associated transfer costs.`	AED
Estimated carbon reduction	Eligible material weight × an avoided-emission factor matching the material and activity.	kg CO_2e
Reuse rate	(Number of unique asset units successfully reused through completed transfers during the reporting period ÷ Total number of eligible asset units made available for reuse during the same period) × 100. Exclude cancelled or incomplete transfers from the numerator.	%

Reuse rate is calculated for a defined reporting period using the number of eligible asset units made available for reuse as the denominator. If the denominator is zero or unavailable, the reuse rate is reported as “Not calculable” rather than 0%.

Carbon and factor source

Estimated carbon reduction (Kg CO2e) = Material weight (Kg) x Relevant avoided emissions factor (Kg CO2e/Kg)

According to the U.S. Environmental Protection Agency (EPA), the Waste Reduction Model (WARM), Version 16, estimates an avoided emission of 2.83 metric tons CO_2e per US short ton of mixed recyclables recycled instead of landfilled [1]. Converting this factor to kilograms gives approximately 3.12 kg CO_2e per kg of mixed recyclables.

Illustrative calculation:

Estimated carbon reduction = Material weight × Avoided-emission factor

= 100 kg × 3.12 kg CO_2e/kg

≈ **312 kg CO_2e avoided**

This example applies only to mixed recyclables recycled instead of landfilled. It does not automatically apply to reuse, repair, or other materials. The system must use an appropriate documented factor for each material and activity. If no suitable factor is available, the result is marked "Not calculable." This is an illustrative estimate, not a measured university result.

Calculation controls

- Count each asset unit only once within the same count-based measure and reporting period.
- Keep links to the contributing transfer, maintenance or lifecycle records.
- Identify missing inputs and mark affected results **“Incomplete”** or **“Not calculable”**; do not substitute zero.
- Record the source and version of emission factors, and label carbon and life-extension results as estimates.
- Ensure AI-generated summaries reproduce the calculated report values.

Related requirements: FR-36–FR-38, FR-51 and NFR-AI09.

8.13 Model-Monitoring Records

Monitoring records are linked by IDs and are visible only to authorized staff. They are append-only: a correction creates a new entry rather than changing the earlier record. Passwords, API tokens and unnecessary personal data are excluded.

8.13.1 Model Service Version Record

Field	Type	Purpose
modelVersionId	UUID, PK	Internal version identifier
serviceName	String	Classification, semantic matching, ranking, sustainable action or LLM
providerName / providerModelVersion	String	Exact provider and configured/returned model version
promptTemplateVersion	String, nullable	Instruction/template version
scoringConfigVersion	String, nullable	Factor, weight and threshold version
confidenceThreshold	Decimal, nullable	Applied low-confidence threshold, normally 0.70
activatedAt / retiredAt	DateTime	Active period
status	Enum	Test, active, suspended or retired

8.13.2 AI Interaction Record

Field	Type	Purpose
interactionId / traceId	UUID	Unique attempt and cross-module trace
featureType	Enum	Classification, matching, ranking, sustainable action, assistant or report summary
modelVersionId	UUID, FK	Exact model/service configuration
relatedRecordType / relatedRecordId	String / UUID	Asset, request, recommendation or report reference
inputReferenceIds / inputHash	JSON / String	Approved source references and change-detection hash
startedAt / completedAt / durationMs	DateTime / Integer	Timing and performance
status / retryNumber	Enum / Integer	Success, low confidence, timeout, unavailable, invalid or failed; attempt number
outputReferenceId	UUID, nullable	Stored output reference

compatibilityScore / confidence / evidenceCoverage	Decimal, nullable	Separate score, provider confidence and evidence coverage
fallbackUsed / errorCode	Boolean / String	Manual/rule fallback and safe error code

8.13.3 Human Review and Feedback Record

Field	**Type**	**Purpose**
reviewId / interactionId	UUID, PK / FK	Review and related AI attempt
reviewerUserId	UUID, FK	Authorized reviewer
decision	Enum	Accepted, edited, rejected, overridden or no action
reason	String, nullable	Required for rejection or override
editedOutputReference	UUID, nullable	User-approved correction
decidedAt	DateTime	Review time
relevanceLabel / referenceLabel	Boolean / String, nullable	Evaluation label for matching or classification
feedbackNotes	String, nullable	Authorized evaluation note

8.13.4 Monitoring Measures

- classification accuracy against 20 agreed reference-labelled assets, with the Phase 3 proposed target of at least 16 correct primary categories;
- match relevance using up to 30 reviewed recommendations, with a proposed target of at least 75%;
- the percentage of low-confidence results that displayed the required warning;
- acceptance, edit, rejection and override counts by feature and model version;
- response-time distribution, timeout rate, unavailable rate and fallback use;
- report-summary value mismatches, which must be zero before approval; and
- results by asset category and request type to support fairness review without unrelated personal characteristics.

Evaluation feedback does not retrain or update a model automatically. The team reviews the records and approves any later change to the model, prompt, thresholds or scoring configuration.

9. Use of Artificial Intelligence

Throughout Phases 4 and 5, the team used ChatGPT (OpenAI) as an aid in producing the Architecture and Detailed Design Document. The team were responsible for making the plans, decisions and requirements for the system. The AI was primarily used to help produce the report structure, organise the topics and paragraphs and improve the explanations. Some diagrams were also produced by the AI.

All content produced using the AI was checked by the team to ensure it fit with the requirements, project description and lecture slides from Phase 3. Any necessary changes were made and the team were fully responsible for any decisions made, diagrams, documentation and any work they submitted.

10. References

[1] U.S. Environmental Protection Agency, “Greenhouse Gas Equivalencies Calculator—Calculations and References,” sec. “Tons of waste recycled instead of landfilled.” Accessed: Oct. 28, 2026. [Online]. Available: https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator-calculations-and-references

[2] Khalifa University, Department of Computer Science, “Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System,” *COSC 336: Introduction to Software Engineering, Project-September2026*, Fall 2026. [Course project description].

[3] Khalifa University, Department of Computer Science, “Phase 4: Architecture Design,” *COSC 336: Introduction to Software Engineering*, Fall 2026. [Lecture slides].

[4] Khalifa University, Department of Computer Science, “Circular Campus Resource Exchange and Asset Life Cycle Management—Phase 4 & 5: Design Document, Sequence, ER and Class Diagrams,” *COSC 336: Introduction to Software Engineering*, Fall 2026. [Lecture slides].

[5] Group 2, “Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System—Phase 3 Requirements Document,” *COSC 336: Introduction to Software Engineering*, Khalifa University, Fall 2026. [Unpublished course project report].